

Indian Ship Recycling Industry

Rudra Green Ship Recycling Limited

September 29, 2026

Contents

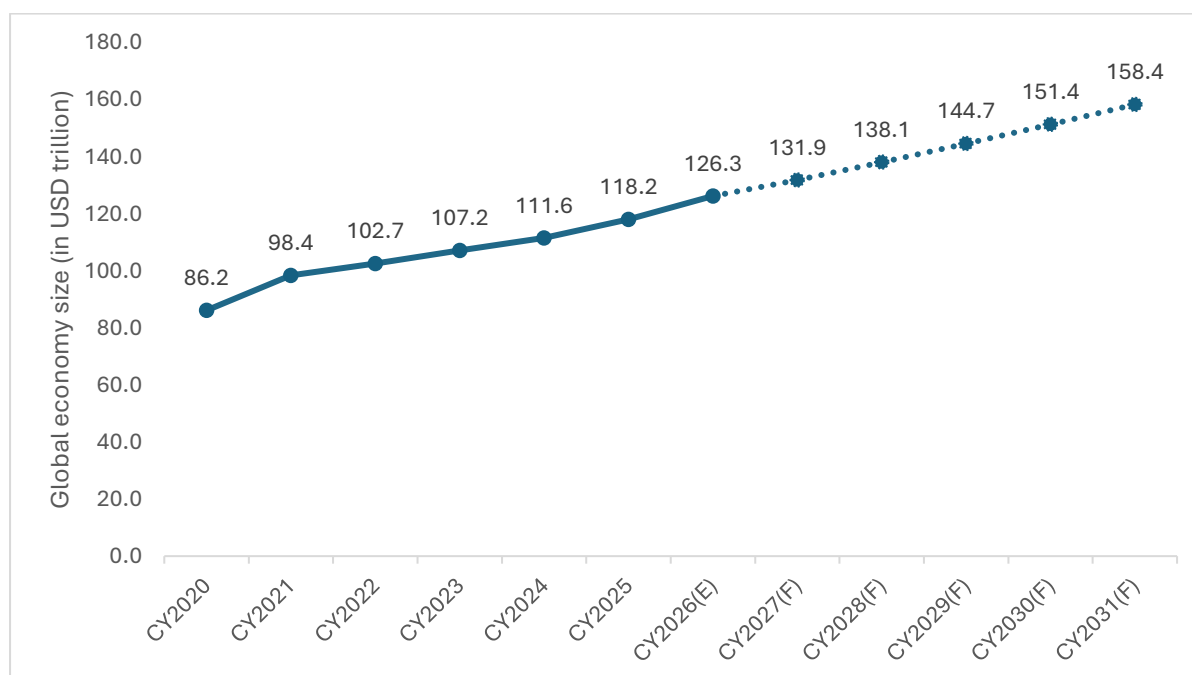
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1. Global macroeconomic overview

The International Monetary Fund (IMF) projects that the global economy will reach a nominal size of USD 126.3 trillion in CY2026 and grow to USD 158.4 trillion by CY2031, representing a CAGR of about 4.6%. According to the IMF's April 2026 World Economic Outlook, global economic momentum, which had been supported by strong labour markets, steady incomes, and resilient household consumption in major economies, is beginning to moderate due to rising geopolitical tensions and disruptions in energy supply. While disinflation had been progressing and inflation had been easing and moving closer to target levels across many advanced economies, renewed supply side pressures, particularly from increased energy prices, have slowed this progress. In response, central banks are continuing to take a cautious approach, implementing only gradual and limited monetary easing, depending on sustained inflation stability and improving financial conditions.

Chart: Global economy growth and prediction till CY2031 (in USD trillion- nominal terms)



Source: IMF (WEO April 2026), ICRA Analytics

CY2026 (E)- E is estimated

Note: F-Forecasted; data from CY2027-2031 are forecasted.

According to the IMF's World Economic Outlook of April 2026, global growth is expected to ease from an estimated 3.4% in CY2025 to 3.1% in CY2026, before recovering slightly to 3.2% in CY2027. This outlook is weaker than the January 2026 projections, with the CY2026 forecast revised downward from earlier expectations of about 3.3% to 3.4%.

The revisions are broad based across both advanced and emerging economies and reflect a noticeable loss of growth momentum, driven by rising geopolitical tensions and disruptions to trade, particularly those linked to energy supply shocks and uncertainty around global trade routes.

The downward revisions are mainly attributed to direct trade and supply side disruptions, including interruptions in energy flows and higher logistics costs, along with indirect effects such as tighter financial conditions, elevated uncertainty, and weaker business and consumer confidence. While ongoing uncertainty related to trade policies and tariff driven fragmentation continues to weigh on economic activity, the degree of impact varies across countries depending on factors such as trade exposure, industrial structure, macroeconomic resilience, and the ability to reroute trade or diversify export markets. In some economies, especially China and parts of the euro area, focused fiscal support and increased public investment are expected to partly offset these growth pressures.

The global slowdown carries important implications for the container shipping and container manufacturing industries, given their strong dependence on global trade volumes. Containerized cargo represents a significant share of seaborne trade, and weaker economic growth typically results in lower exports and imports of manufactured goods, especially in trade intensive sectors such as electronics, machinery, textiles, and automotive

components. In addition, the IMF notes that disruptions in trade and logistics tend to have a more pronounced impact on economies and industries that are deeply integrated into global value chains, thereby increasing volatility in container traffic volumes.

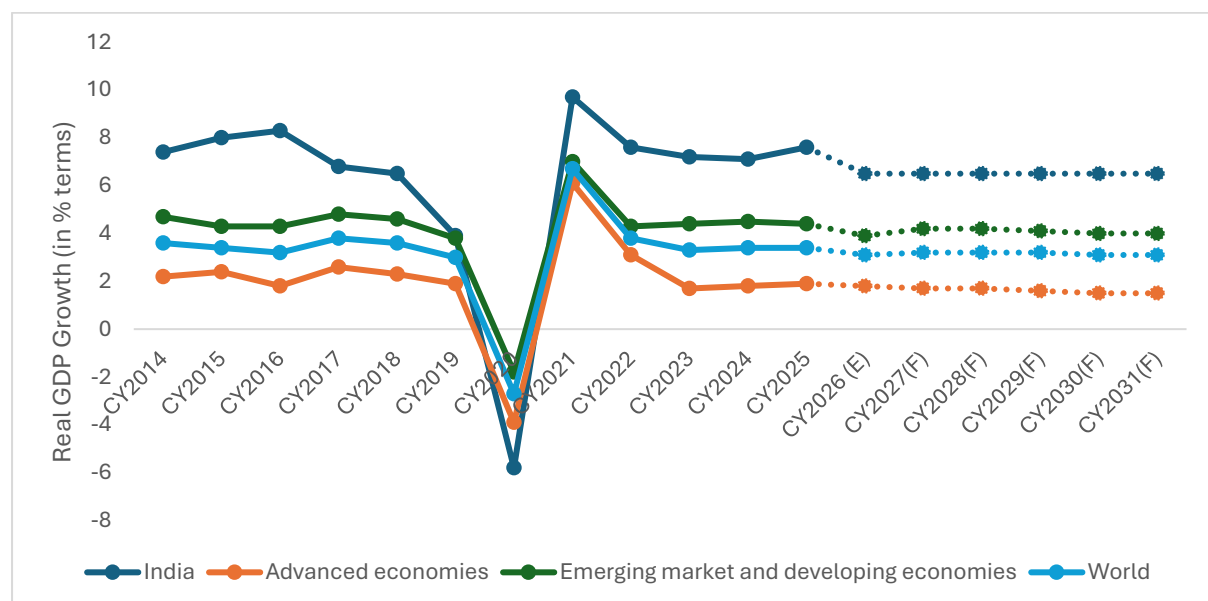
The global economy is navigating an increasingly complex landscape shaped by interconnected challenges such as geopolitical tensions, evolving monetary policy conditions, high debt-to-GDP ratios, and constrained fiscal expansion space. Although overall economic activity remains relatively resilient, growth is moderating, with global GDP expected to be in the range of 3.1%-3.3% in CY2026. Taken together, these factors continue to exert pressure on global growth prospects. The key underlying dynamics include:

- **Inflationary Pressures:** Global headline inflation is expected to ease to around 4.4% in CY2026, continuing the broader disinflation trend seen in recent years, though short term volatility cannot be ruled out. Inflation in advanced economies is likely to return to target levels of approximately 2% sooner, whereas emerging markets may continue to face relatively higher inflation due to structural challenges and currency related pressures. This disinflation is supported by improving supply chain conditions, reduced tightness in labour markets, and the delayed impact of earlier monetary tightening. However, risks persist, including potential commodity price shocks arising from geopolitical tensions, financial market volatility, and supply disruptions, particularly linked to ongoing conflicts in the Middle East and other regions.
- **Monetary and Fiscal Policies:** After a phase of aggressive monetary tightening between 2022 and 2024, most central banks have begun shifting toward a gradual easing cycle, with policy rates expected to move closer to neutral levels by CY2026. However, the trajectory of monetary policy is becoming increasingly uneven across economies, leading to fluctuations in capital flows and exchange rates. On the fiscal front, while some economies are providing targeted support to sustain growth, their ability to do so remains limited due to elevated public debt levels and higher borrowing costs. As a result, policymakers are balancing the need to support near term economic activity with the imperative of maintaining long term fiscal sustainability and rebuilding fiscal buffers.
- **Financial Market Volatility:** Financial markets in CY2026 continue to experience elevated volatility, driven by persistent policy uncertainty, geopolitical developments, and ongoing trade tensions. Aggressive tariff actions and shifts in global trade policies, particularly by the United States, have led to fluctuations in asset prices, disrupted trade flows, and weakened investor sentiment. Additionally, differing monetary policy approaches across major economies are contributing to increased volatility in global bond and currency markets.
- **Geopolitical Crises:** Geopolitical risks have intensified and are emerging as a key threat to global economic stability. Ongoing conflicts, particularly in the Middle East, along with continued tensions in Ukraine and broader geoeconomic fragmentation, are disrupting trade flows, financial markets, and supply chains. At the same time, rising inflation, increasing cost of living, and widening inequality are fuelling social unrest across regions, which could hinder structural reforms and further weaken economic growth prospects.
- **Commodity Price Increases:** Commodity prices have risen substantially in CY2026, driven by supply disruptions, geopolitical tensions, and climate related factors. Global prices are expected to see a notable increase, with energy costs rising sharply due to supply constraints. Disruptions to critical trade routes and energy infrastructure, particularly in the Middle East, have further intensified price volatility and raised the cost of essential goods such as food, fuel, and industrial inputs, thereby complicating efforts to manage inflation and placing additional strain on global supply chains.

- Sovereign Debt Crisis and Fiscal Vulnerability in Emerging Markets:** Based on insights from the World Bank's *Global Economic Prospects*, global credit markets are facing increasing strain as soaring debt levels significantly reduce the flexibility of sovereign borrowing. The rapid accumulation of public debt has sharply raised debt-servicing burdens worldwide, placing considerable pressure on government finances. This challenge is particularly acute for highly leveraged emerging market and developing economies (EMDEs), where rising borrowing costs are driving higher sovereign risk premiums. In response to these mounting fiscal pressures and the threat of capital flow volatility, vulnerable countries are increasingly compelled to shift toward stronger domestic revenue generation, implement comprehensive tax reforms, and enforce disciplined fiscal frameworks to maintain economic stability.
- Geoeconomic Realignment of Foreign Direct Investment (FDI) Flows:** The global landscape of foreign direct investment (FDI) is undergoing a fundamental shift, as companies increasingly prioritize geopolitical stability and regulatory certainty over cost advantages. Cross-border capital flows are becoming more fragmented, with a decline in greenfield investments in countries that are politically or strategically misaligned. In response, firms are redirecting investments through “friend-shoring” and “near-shoring” strategies, channelling capital into trusted, stable, and aligned economic blocs. This realignment is fostering the emergence of new regional manufacturing and financial hubs, while simultaneously reducing capital inflows to less-aligned developing economies.

1.1 Global Economies and Growth Trend

Chart: Real GDP growth rate (annual % change) of India and other economies



Source: IMF, ICRA Analytics

Note: E-Estimated, CY2026 is estimated; F-Forecasted; data from CY2027-2031 are forecasted

Table: India v/s Other Economies (Real GDP, Y-o-Y % change)

Real GDP growth (Annual % change)	CY 2020	CY 2021	CY 2022	CY 2023	CY 2024	CY 2025	CY 2026(E)	CY 2027(F)	CY 2028(F)	CY 2029(F)	CY 2030(F)	CY 2031(F)
India*	-5.8	9.7	7.6	7.2	7.1	7.6	6.5	6.5	6.5	6.5	6.5	6.5
Advanced economies	-3.9	6.1	3.1	1.7	1.8	1.9	1.8	1.7	1.7	1.6	1.5	1.5
Emerging market and developing economies	-1.8	7.0	4.3	4.4	4.5	4.4	3.9	4.2	4.2	4.1	4.0	4.0
World	-2.7	6.7	3.8	3.3	3.4	3.4	3.1	3.2	3.2	3.2	3.1	3.1

Source: IMF, ICRA Analytics

Note: E-Estimated, CY2026 is estimated; F- Forecasted; data from CY2027-2031 are forecasted, Advanced Economies includes United States, Germany, France, Japan, United Kingdoms, Canada and other developed countries. Emerging market and developing economies includes India, China, Saudia Arabia, Mexico, Vietnam and other developing economies.

*India's value in IMF document is given in terms of fiscal year. Eg- CY2020 is FY2021 for India

Global real GDP growth stood at around 3.4% in CY2025 and is expected to moderate to 3.1% in CY2026, before edging up slightly to 3.2% in CY2027. The sharp rise in trade tensions and unusually high levels of policy uncertainty is likely to weigh significantly on global economic activity. Beyond this period, growth is projected to remain broadly stable at approximately 3.1% through CY2031, supported mainly by the easing of monetary policy and continued strength in private consumption.

• Growth trend in Advanced Economies

Real GDP growth in advanced economies stood at approximately 1.9% in CY2025 and is projected to moderate slightly to around 1.8% in CY2026, followed by a further easing to about 1.7% in CY2027, and gradually stabilizing at close to 1.5% by CY2031. In the United States, real GDP growth was around 2.1% in CY2025 and is expected to improve to about 2.3% in CY2026, reflecting steady expansion despite ongoing policy and trade uncertainties. Growth is then projected to ease to approximately 2.1% in CY2027 and stabilize at around 1.8% through CY2031.

In Europe, growth recovered from 0.9% in CY2024 to 1.4% in CY2025 but is expected to slow to 1.1% in CY2026 before edging up to 1.2% in CY2027. Within the euro area, France, Italy, and Spain recorded growth rates of 0.9%, 0.5%, and 2.8% respectively in CY2025, and are expected to maintain similar growth levels in CY2026, while Germany, which grew by 0.2% in CY2025, is projected to rise to 0.8% in CY2026.

Among other advanced economies, the United Kingdom registered real GDP growth of around 1.3% in CY2025, which is expected to decline to approximately 0.8% in CY2026 before stabilizing at about 1.3% in CY2027.

Real GDP growth in the Euro area stood at approximately 1.4% in CY2025 and is projected to decline to around 1.1% in CY2026, showing a more pronounced slowdown compared to broader Europe. Growth recovers slightly to about 1.2% in CY2027 and reaches 1.4% in CY2028 before easing again and stabilizing at around 1.1-1.2% through CY2031. The pattern indicates persistent weakness and limited growth momentum, reflecting structural economic challenges.

• Growth trend in emerging markets and developing economies

Real GDP growth in emerging and developing economies stood at around 4.4% in CY2025 and is projected to moderate to 3.9% in CY2026, before recovering to 4.2% in CY2027, with notable downward revisions for countries most affected by recent trade measures, particularly China. Growth in emerging and developing Asia is expected to decline from approximately 5.5% in CY2025 to about 4.9% by CY2026, with Association of Southeast Asian Nations (ASEAN) countries among the most impacted due to recent tariff changes.

In Sub-Saharan Africa, GDP expanded at a strong 4.5% in CY2025 and is projected to grow at 4.3% in CY2026, with South Africa and Nigeria recording Real GDP growth of 1.1% and 4.0% respectively in CY2025 and expected to grow at around 1.0% and 4.1% in CY2026.

Real GDP growth in Africa region was approximately 4.5% in CY2025 and is expected to ease to around 4.2% in CY2026, indicating a short-term moderation. However, growth rebounds thereafter to about 4.4% in CY2027 and peaks at approximately 4.7% in CY2028, before stabilizing around 4.5-4.6% through CY2031. This pattern reflects strong underlying growth fundamentals despite periodic fluctuations, with the region maintaining one of the highest growth trajectories globally.

Meanwhile, growth in emerging and developing Europe remained at about 2.0% in CY2025 and is expected to stay broadly stable in CY2026. In Latin America and the Caribbean, real GDP growth is projected to ease slightly from around 2.4% in CY2025 to 2.3% in CY2026, before rising again to approximately 2.7% thereafter.

Real GDP growth in Turkey was about 3.6% in CY2025 and is projected to soften slightly to around 3.4% in CY2026, indicating a mild slowdown. Growth is expected to gradually recover thereafter, reaching approximately 3.5% in CY2027 and strengthening further to about 3.8% in CY2028. From CY2029 onwards, growth stabilizes at a higher level of around 4.0% through CY2031.

Real GDP growth in Bangladesh stood at approximately 3.5% in CY2025 and is projected to rise sharply to around 4.7% in CY2026, marking the most significant year-on-year improvement among the compared economies. Growth moderates slightly to about 4.3% in CY2027 and remains steady around 4.5% in CY2028, before accelerating strongly to approximately 5.7% in CY2029 and continuing an upward trajectory to about 6.1% by CY2031. This reflects a robust and sustained growth acceleration over the medium term following an initial recovery phase.

Real GDP growth in Pakistan was around 3.1% in CY2025 and is expected to improve to approximately 3.6% in CY2026, indicating a recovery in economic activity. Growth remains relatively stable at about 3.5% in CY2027 before accelerating to around 4.3% in CY2028. From CY2029 onwards, growth stabilizes at approximately 4.5% through CY2031. The trajectory reflects a gradual strengthening followed by stabilization at a higher growth level, suggesting improving macroeconomic conditions.

- **Growth trend in India**

India continues to stand out as the fastest-growing major economy, with real GDP growth remaining robust despite minor fluctuations. Growth moderated slightly from approximately 7.2% in CY2023 to around 7.1% in CY2024, before accelerating to a strong 7.6% in CY2025, reflecting resilient economic fundamentals and strong domestic demand. According to the IMF, this pace is expected to normalize, with growth easing to about 6.5% in both CY2026 and CY2027 as the economy transitions toward a more sustainable growth trajectory. This expansion is primarily supported by robust private consumption, particularly in rural areas, alongside improving investment activity and ongoing structural reforms. Over the medium term, the growth momentum is projected to remain stable, with forecasts indicating consistent expansion at around 6.5% annually through CY2031. Overall, India's growth outlook highlights a balance between short-term moderation and long-term resilience, underpinned by favourable demographics, rising income levels, and a strong consumption-driven economic base.

2. Domestic Economic overview

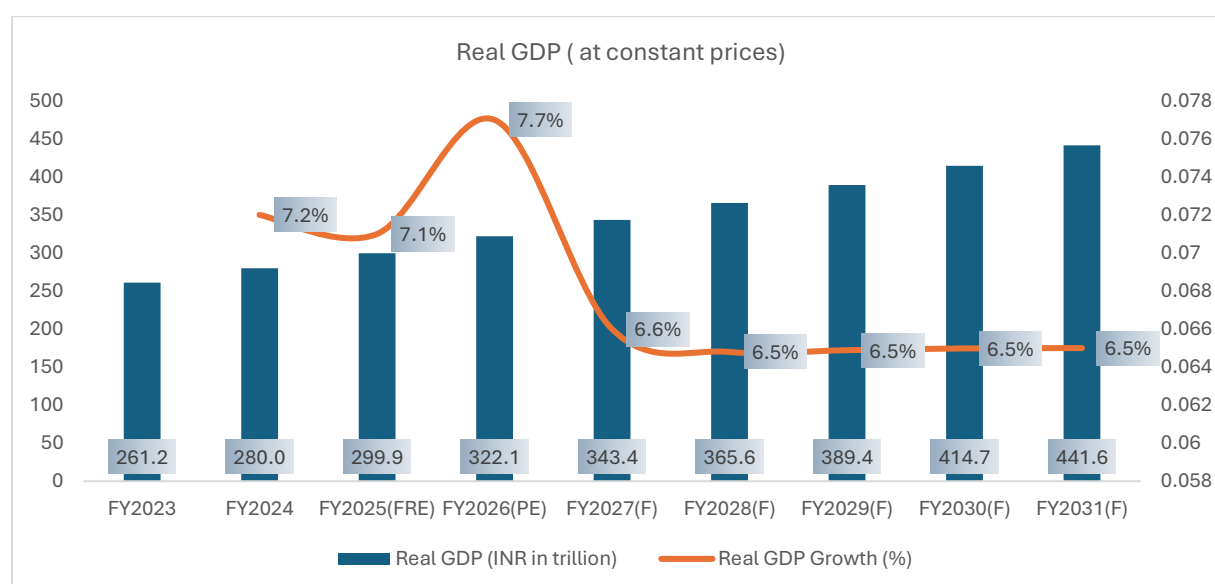
2.1 Trend in GDP growth in India and its Outlook

India's real GDP is projected to follow a strong upward trajectory, increasing from INR 261.2 trillion in FY2023 to INR 441.6 trillion by FY2031 according to the Provisional Estimates (PE) released by the National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI) and the projections given by the International Monetary Fund (IMF). This reflects a sustained expansion in economic output, with consistent year on year increases across the entire period. The pace of increase in absolute terms becomes more pronounced over time, as gains between later years such as FY2029 and FY2031 are significantly larger than those observed in the earlier years.

India's Real GDP growth moderates slightly from 7.2 percent in FY2024 to 7.1 percent in FY2025, before accelerating to a peak of 7.7 percent in FY2026, suggesting a phase of strong expansion. This moderation reflects the typical progression of a growing economy, where maintaining very high growth rates becomes more challenging as the overall size increases. This continued momentum was primarily driven by robust domestic demand, a favourable demographic profile, and ongoing economic reforms. India's growing role in global trade, investment, and innovation, along with the government's continued focus on infrastructure and economic development, also supported this growth trajectory.

Further, the International Monetary Fund (IMF) has projected that India will remain the world's fastest-growing major economy, with annual growth expected at 6.5% from FY2028 to FY2031.

Chart: Historical trend and projection of Real GDP of India (INR Trillion) - Base year (2022-23)



Source: RBI, IMF, ICRA Analytics

Note: F-Forecasted.

Data from FY2027-2031F are forecasted from IMF

FY2025 (FRE) is the First Revised Estimates, FY2026(PE) is the Provisional Estimates

The Monetary Policy Committee (MPC) of the Reserve Bank of India (RBI) has revised its forecast for India's real GDP growth for FY2027 to 6.6%, as announced by Governor Sanjay Malhotra in the policy statement on 5th June 2026. Although this marks a moderation from the earlier projection of 6.9% made in April, the RBI highlighted that domestic demand conditions continue to remain resilient, supported by urban consumption, improving rural activity, and sustained investment momentum.

The quarterly real GDP growth pattern for FY2027 (P) reflects a steady but slightly uneven trajectory across the fiscal year. Growth is recorded at 6.6% in Q1 (P), followed by a moderation to 6.3% in Q2 (P), indicating a brief softening in economic momentum during the first half of the year. This is followed by a recovery trend, with growth improving to 6.5% in Q3 (P) and further strengthening to 6.8% in Q4 (P). The gradual pickup in the latter half suggests improving economic conditions, potentially driven by stronger demand or sectoral performance.

These revisions indicate a more realistic and slightly moderated growth trajectory, reflecting a balance between strong domestic fundamentals and evolving global risks.

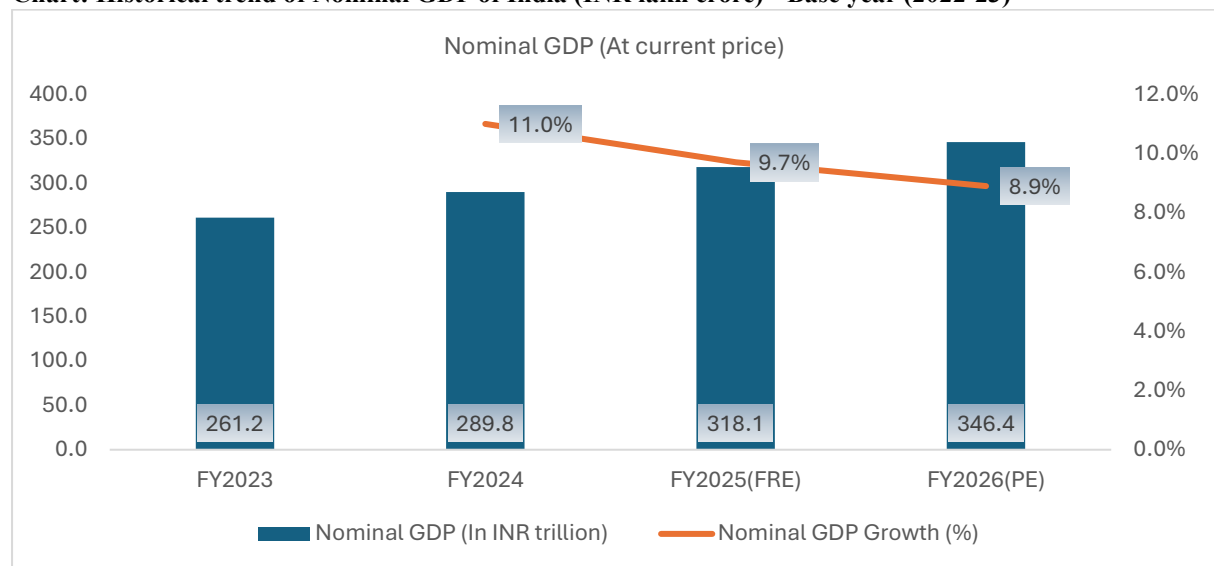
Table: Real GDP growth forecasted by Reserve Bank of India

Real GDP Growth (at constant 2022-23 prices)	FY2027 (P)	FY2027 (P)			
Quarter		Q1 (P)	Q2 (P)	Q3(P)	Q4 (P)

% Growth	6.6	6.6	6.3	6.5	6.8
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P- Projected.
Source: RBI, ICRA Analytics

Chart: Historical trend of Nominal GDP of India (INR lakh crore) - Base year (2022-23)



Source: RBI, ICRA Analytics

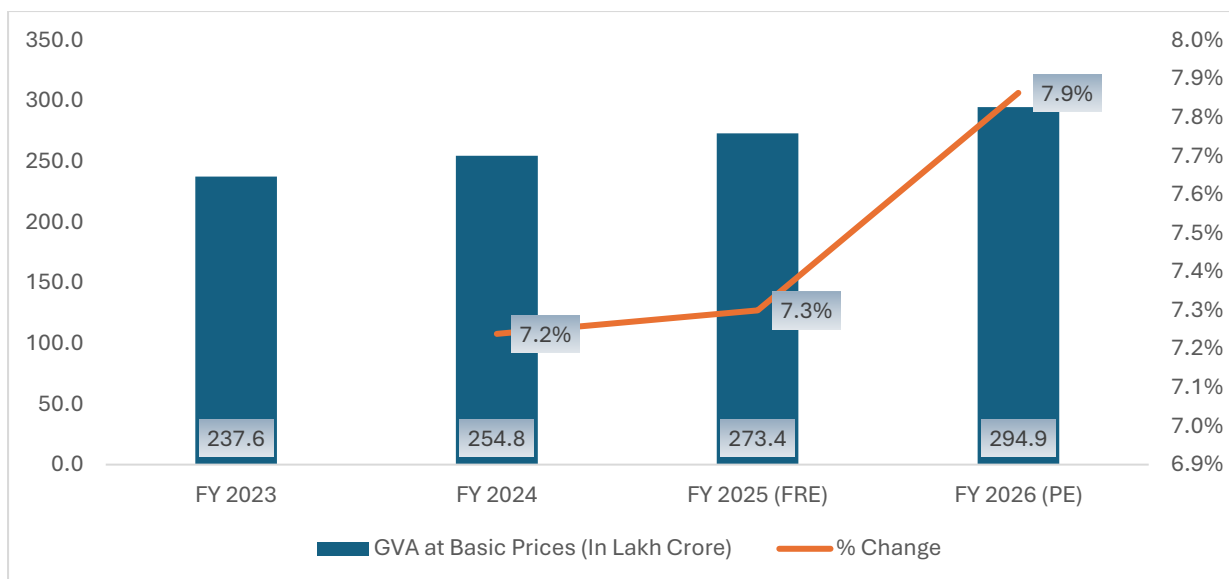
Note: FY2025 (FRE) is the First Revised Estimates, FY2026 (PE) is the Provisional Estimates released by the National Statistical Office (NSO)

India's nominal GDP has increased steadily from approximately INR 261.2 trillion in FY2023 to INR 289.8 trillion in FY2024, followed by INR 318.1 trillion in FY2025 (FRE) and INR 346.4 trillion in FY2026 (P). This sustained rise reflects continued expansion in the size of the economy in absolute terms, suggesting resilience and stable growth drivers such as consumption, investment, and price-level effects. The progression remains smooth and incremental, with no evident disruptions, indicating a relatively predictable economic environment.

In contrast, the nominal GDP growth rate shows a gradual moderation over the same period. After registering 11.0% in FY2024, growth declines to 9.7% in FY2025 (FRE) and further to 8.9% in FY2026 (P). This trend highlights that while the economy continues to expand in size, the pace of growth is decelerating. Such a pattern is often associated with base effects or a transition from a high-growth phase to a more stable and mature growth cycle. The divergence between rising GDP levels and easing growth rates underscores the importance of focusing on sustaining long-term growth momentum.

2.2 Gross Value Added growth in India

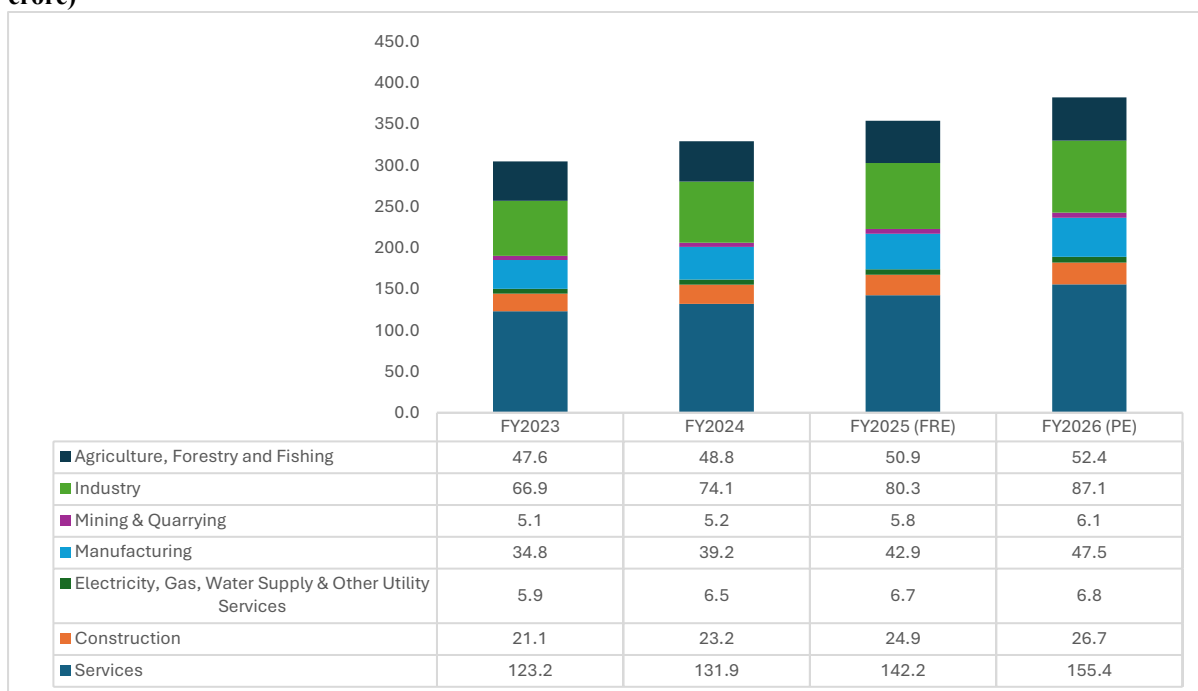
Chart: GVA at Basic Prices (Base Year: 2022-23) Constant Prices (in INR lakh crore) and y-o-y growth (in %) FY2023-FY2026



Source: RBI, ICRA Analytics

Note: FY2025 (FRE) is the First Revised Estimates, FY2026 (PE) is the Provisional Estimates

Chart: Gross Value Added at Constant Basic Prices: By Economic Activity: Base Year 2022-23 (in Rs lakh crore)



Source: RBI, ICRA Analytics

Note: FY2025 (FRE) is the First Revised Estimates, FY2026(PE) is the Provisional Estimates

As per the latest Provisional Estimates (PE) released by the National Statistical Office (NSO) for FY2026, India's Gross Value Added (GVA) is expected to witness broad-based growth across all key sectors of the economy.

The industry sector is estimated to generate a GVA of INR 87.1 lakh crore, accounting for nearly 29.5% of India's total GVA of INR 294.9 lakh crore. The services sector is projected to remain the largest contributor to the economy, with a GVA of INR 155.4 lakh crore, representing approximately 52.7% of the total. Meanwhile, the agriculture, forestry and fishing sector is expected to contribute INR 52.4 lakh crore, constituting around 17.8% of the country's total GVA.

Between FY2025 and FY2026, the agriculture sector is estimated to expand by 3.0%, while the industry and services sectors are projected to grow at 8.5% and 9.3%, respectively.

Within the industry sector, manufacturing is estimated to contribute 16.1% of India's total GVA in FY2026. The sector's GVA is expected to increase from INR 42.9 lakh crore in FY2025 to INR 47.5 lakh crore in FY2026, reflecting a growth rate of approximately 10.7%. The electricity, gas, water supply and other utility services segment is projected to maintain a GVA of INR 6.8 lakh crore, broadly in line with the previous year.

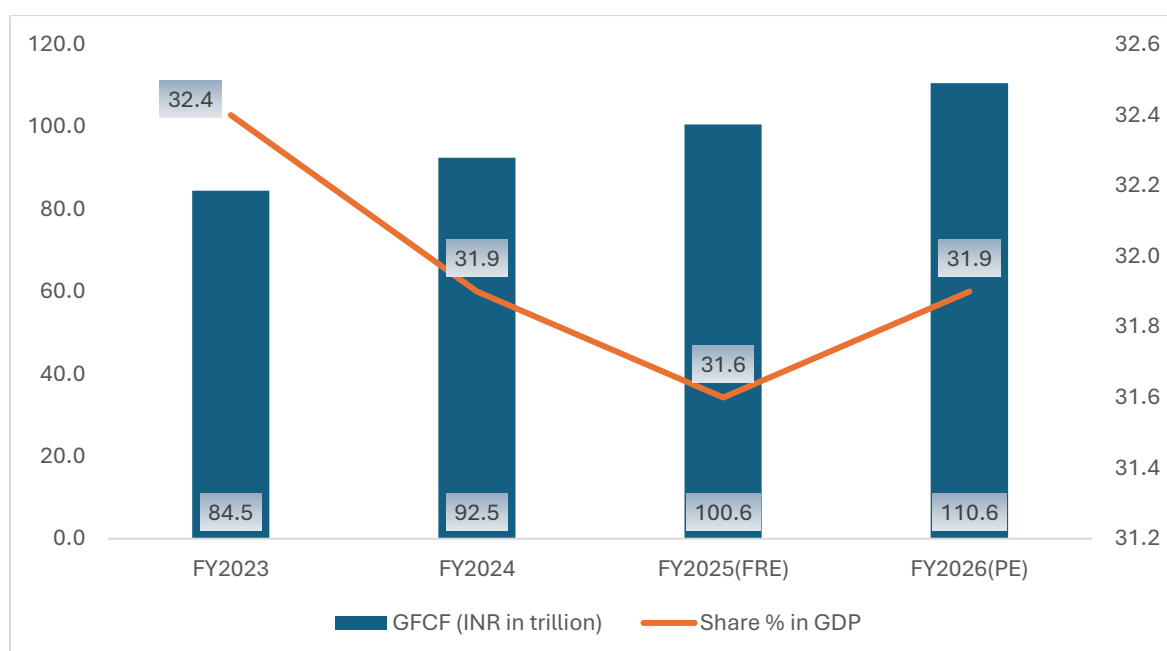
Construction activity is also expected to remain strong, with GVA rising from INR 24.9 lakh crore in FY2025 to INR 26.7 lakh crore in FY2026, supported by continued infrastructure development and investment activity. At the same time, the mining and quarrying sector is estimated to remain relatively stable at INR 6.1 lakh crore.

The FY2026 estimates indicate that the services sector will continue to be the primary driver of economic growth, supported by sustained expansion across a wide range of sub-sectors. Simultaneously, the industry sector is expected to strengthen further, aided by robust growth in manufacturing and construction activities. The agriculture sector is projected to maintain steady, albeit moderate, growth in line with long-term structural trends in the economy.

2.3 Growth Trend in Investment & Consumption Demand

2.3.1 Gross Fixed Capital Formation (GFCF) Growth in India

Chart: India's Gross Fixed Capital Formation (GFCF) (in INR Lakh crores) and share of GDP (in %) FY2023-FY2026 - - Base year (2022-23)



Source: MOSPI, RBI, ICRA Analytics

Note: FRE: First Revised Estimates; PE: Provisional Estimates

India's Gross Fixed Capital Formation shows a clear and consistent upward trend over the period, reflecting strengthening investment activity in the economy. GFCF increases gradually from around INR 82.5 lakh crores in FY2023 to INR 92.5 lakh crores in FY2024, followed by a further rise to approximately INR 100.6 lakh crores in FY2025 and to about INR 110.6 lakh crores in FY2026. The acceleration indicates growing momentum in capital expenditure, likely driven by infrastructure expansion, private sector investment, and supportive policy measures, highlighting an increasingly investment led growth trajectory.

The corresponding GFCF to GDP ratio for FY2024 and FY2025 stood at 31.9% and 31.6% respectively, indicating stable investment intensity in the economy. For FY2026, GFCF to GDP ratio is projected 31.9% of GDP, suggesting continued strengthening of investment demand.

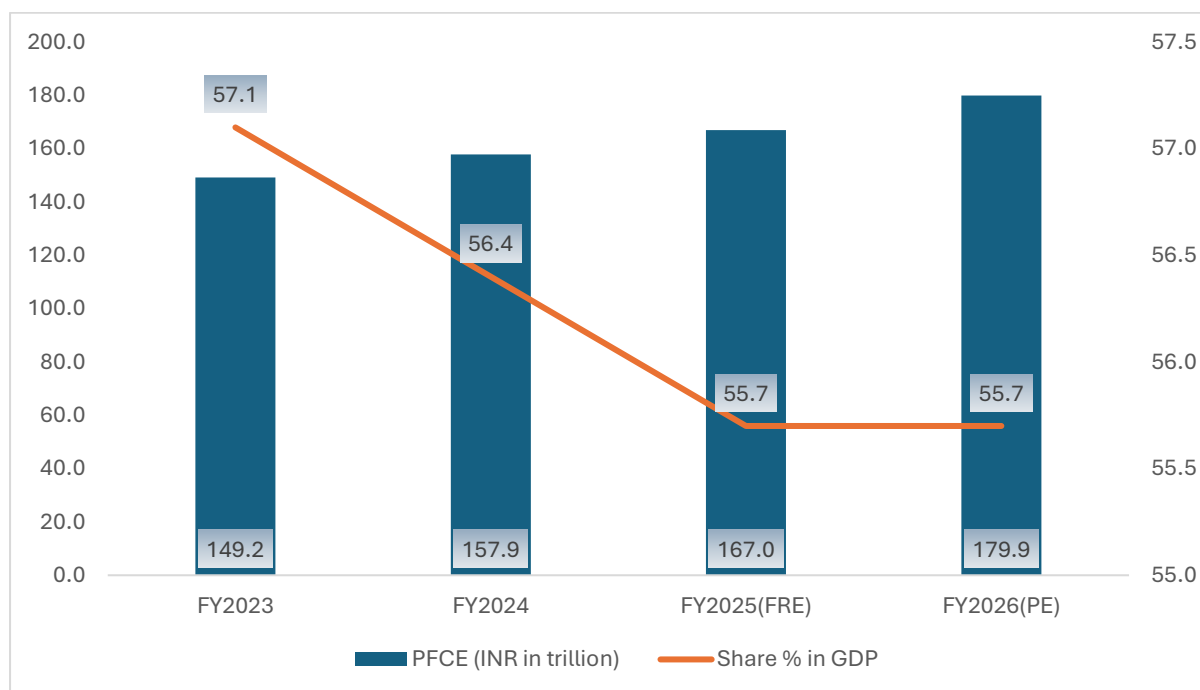
The rising trend in GFCF is supported by multiple factors such as increased government capital expenditure on infrastructure, expansion in private corporate investments, growth in real estate and construction activities, and policy initiatives like National Infrastructure Pipeline (NIP) and Production Linked Incentive (PLI) schemes.

Additionally, improving credit availability, stronger balance sheets of corporates, and increased public-private partnerships have contributed to sustained capital formation.

Investment plays a crucial role in enhancing productive capacity, generating employment, and supporting long-term economic growth. The steady rise in GFCF highlights strengthening investment momentum in the economy. The growth trend remains stable, with GFCF expanding from 9-10% range in recent years, reflecting an incredible improvement in capital formation and economic expansion.

2.3.2 Private Final Consumption Growth in India

Chart: India's Private Final Consumption Expenditure (PFCE) (in INR Lakh crores) and share of GDP (in %) FY2023-FY2026 - - Base year (2022-23)



Source: MOSPI, RBI, ICRA Analytics

Note: FRE: First Revised Estimates; PE: Provisional Estimates

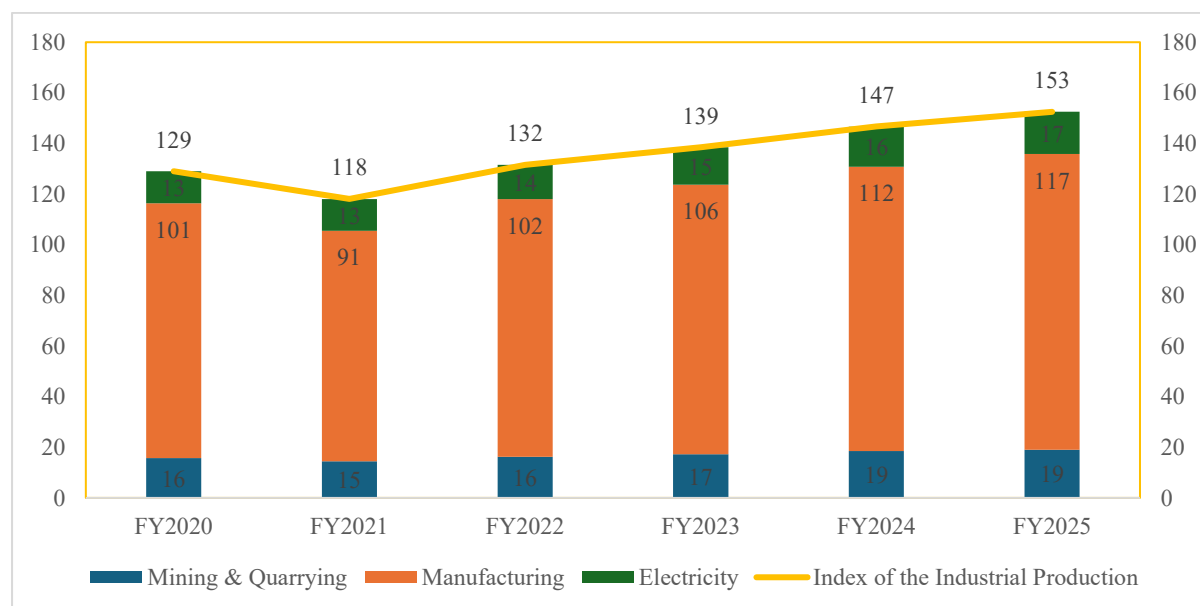
India's Private Final Consumption Expenditure demonstrates a steady and sustained expansion in absolute terms over the period from FY2023 to FY2026. The chart shows PFCE increasing from approximately INR 149.2 lakh crores in FY2023 to about INR 157.9 lakh crores in FY2024, followed by a further rise to INR 167.0 lakh crores in FY2025 and reaching nearly INR 179.9 lakh crores in FY2026. The rising PFCE is contributing by various factors such as expanding middle class population, increasing disposable income which enables consumers to spend more, easy access to credit and financing options which enables consumers to make purchases and invest in assets, various governmental policies (schemes such as Pradhan Mantri Awas Yojana, Ayushman Bharat Yojana) that supports consumer spending, advancements in technology, digital payments and other innovative developments.

However, despite the strong growth in absolute consumption, the share of PFCE in GDP exhibits a mild but noticeable decline over the same period. The contribution falls from 57.1 percent in FY2023 to 56.4 percent in FY2024 and stabilizes at around 55.7 percent in both FY2025 and FY2026. This divergence indicates that while consumption is growing, other components of GDP such as investment and government expenditure are expanding at a relatively faster pace. The plateauing of the consumption share in later years suggests a gradual rebalancing of economic drivers, with a greater emphasis on capital formation and production-led growth alongside consumption.

PFCE growth remained steady at 5.8 percent across two consecutive years, indicating a phase of stable and moderate expansion in consumption demand. This consistency reflects gradual strengthening without significant acceleration. The growth rate then rises to 7.7 percent in FY2026, signalling a clear pickup in consumption momentum driven by improving economic conditions and stronger consumer spending.

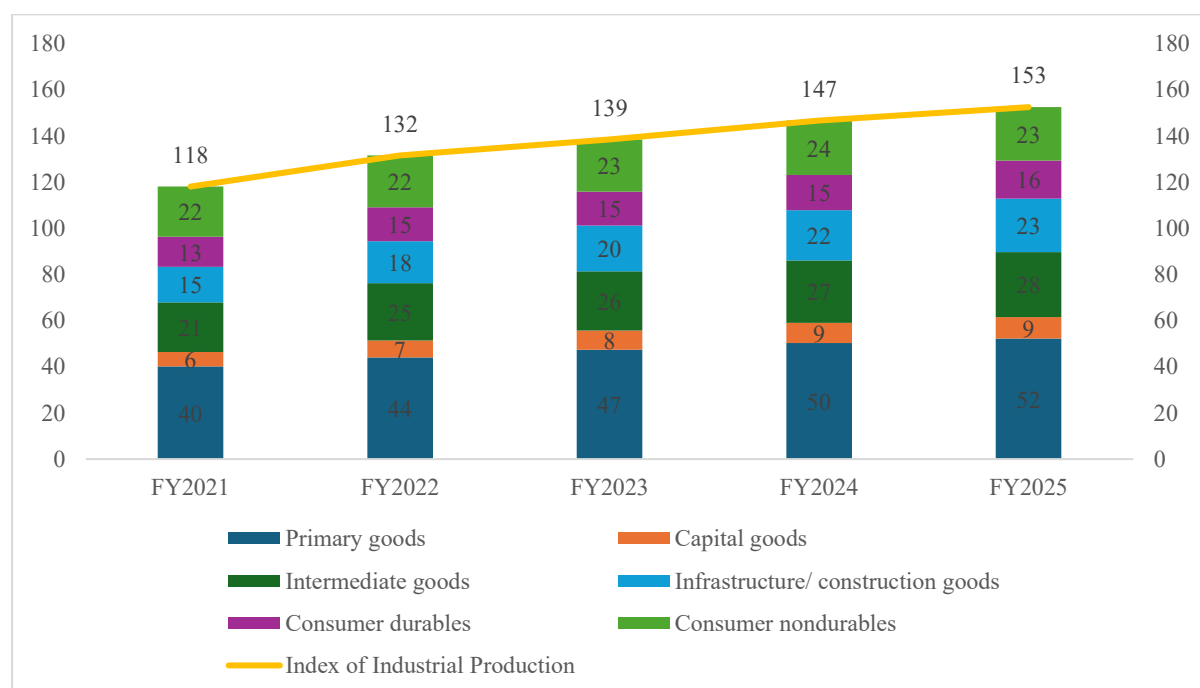
2.4 Trends in Industrial growth

Chart: Movement in Index of Industrial Production and its Components (Base Year 2011-12)



Source: PIB, RBI, ICRA Analytics

Chart: Index of Industrial Production - Use-Based Classification (Base Year 2011-12)



Source: RBI, ICRA Analytics

In FY2026, India's Index of Industrial Production (IIP) recorded year-on-year growth of 4.3%, moderating from 6.4% in FY2025. The slowdown was largely attributable to weaker performance in the manufacturing and electricity sectors, along with softer growth in mining, lower capital expenditure, and subdued domestic demand.

Despite this moderation, all three major sectors remained in expansion territory during FY2026. Mining and quarrying grew by 3.7%, compared with 2.8% in FY2025; manufacturing expanded by 4.9% versus 6.3%; and electricity & gas supply generation increased by 1.1%, down from 9.5%.

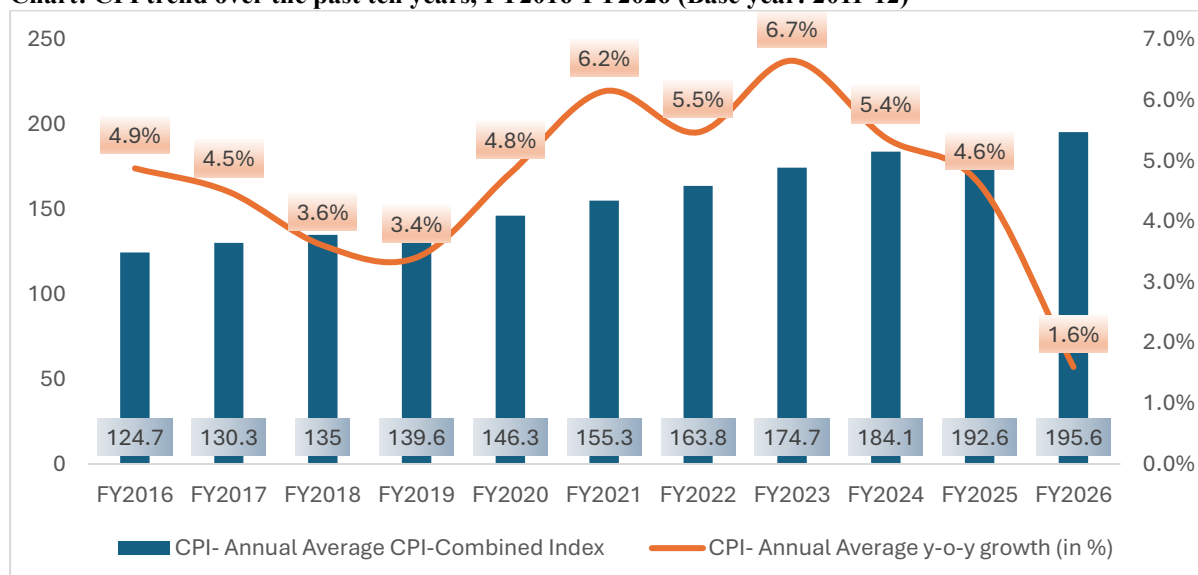
In May 2026, the IIP grew by 5.1% year-on-year, improving from the 4.9% growth recorded in April 2026. The increase was supported by healthy growth in the manufacturing and electricity sectors, indicating continued resilience in industrial activity despite variations across individual segments.

The composition of annual IIP data by use-based classification shows broad-based and consistent expansion across all categories from FY2024 to FY2026, indicating strengthening industrial activity. Primary goods remained the largest contributor, rising steadily from 32.7 to 35.5, reflecting stable growth in core sectors. Intermediate goods increased from 23.8 to 26.3, highlighting robust supply chain activity and higher input demand for production. Capital goods rose from 9.0 to 11.0, suggesting improving investment sentiment and capacity expansion, while infrastructure and construction goods grew from 12.0 to 14.1, signalling sustained momentum in infrastructure development. Consumer durables increased from 11.9 to 13.5, indicating improving discretionary demand, and consumer non-durables remained relatively stable with a slight increase from 17.3 to 18.0, reflecting steady essential consumption.

Overall, the data points to a balanced and broad-based industrial growth trajectory, driven by both consumption- and investment-linked sectors.

2.5 Review of inflation in India

Chart: CPI trend over the past ten years, FY2016-FY2026 (Base year: 2011-12)



Source: RBI, MOSPI, ICRA Analytics

Retail inflation in India, as measured by the Consumer Price Index (CPI), which reflects the cost of daily goods and services, has shown a consistent downward trend over the past four financial years. Inflation declined from 6.7% in FY2023 to 5.4% in FY2024 and further to 4.6% in FY2025, before moderating sharply to 1.6% in FY2026. This sustained reduction highlights the combined effectiveness of the Reserve Bank of India's calibrated monetary policy and the Government of India's targeted interventions to address supply-side constraints and stabilize prices of essential commodities. The declining trajectory has significantly eased cost-of-living pressures and contributed to a more stable and supportive environment for economic growth.

The Monetary Policy Committee (MPC) of the Reserve Bank of India made revision to its inflation forecast for FY2026, lowering the Consumer Price Index (CPI) estimate to 2.1% from the previous 2%. This decision was revealed at the end of the MPC meeting on 6th Feb 2026, led by Sanjay Malhotra, along with the monetary policy statement from the central bank.

The committee further estimated the CPI inflation for FY 2026-27 at 5.1%. This decision was revealed at the end of the MPC meeting on 5th Jun 2026, led by Sanjay Malhotra, along with the latest monetary policy statement from the central bank.

Furthermore, the MPC has given its inflation forecasts for the four quarters of fiscal year 2027, now anticipating 4.2% inflation in Q1, 5.1% in Q2, 5.9% in Q3 and 5.4% in Q4. According to the revised assessment made in June 2026 from April 2026, the MPC has outlined the following inflation trajectory:

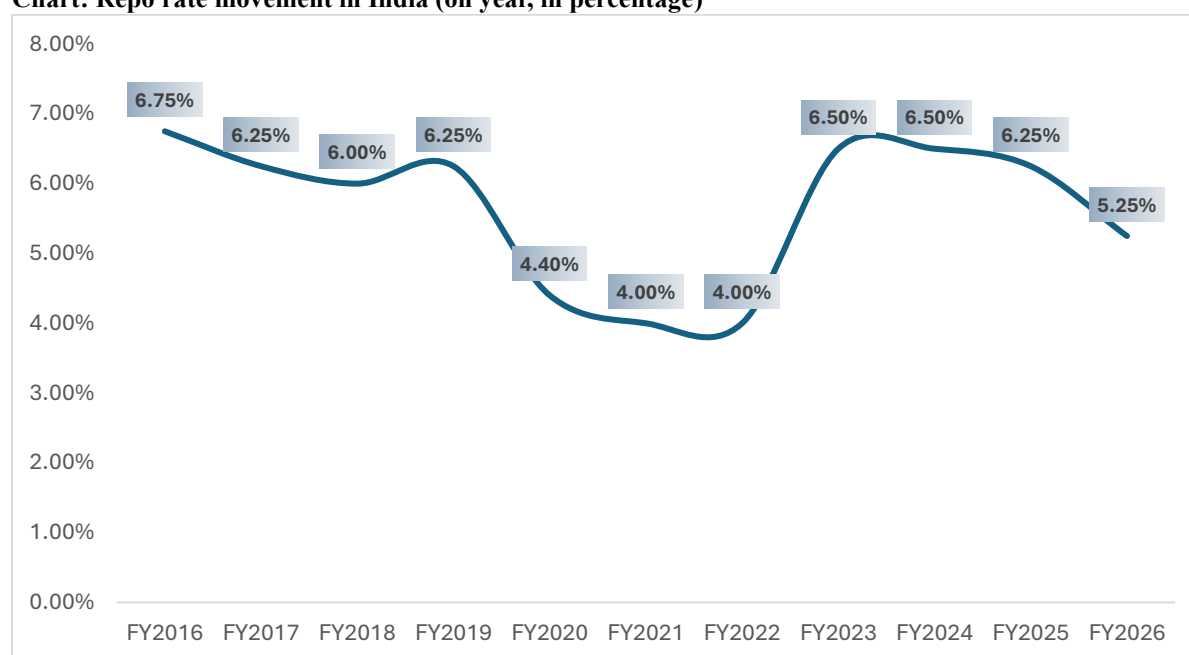
- Q1 FY27: Adjusted to 4.2% from 4.0%
- Q2 FY27: Adjusted to 5.1% from 4.4%
- Q3 FY27: Adjusted to 5.9% from 5.2%.
- Q4 FY27: Adjusted to 5.4% from 4.7%.

The central bank indicated that, excluding precious metals, the underlying inflationary pressures remain subdued, and the risks are currently balanced. The committee noted that the near-term food supply outlook is positive, bolstered by robust kharif production, sufficient foodgrain buffer stocks, promising rabi sowing, and adequate reservoir levels. Core inflation, which excludes potential fluctuations from precious metals, is anticipated to remain stable, with underlying pressures continuing to be contained.

The RBI warned that geopolitical uncertainties, fluctuations in energy prices, and adverse weather conditions present upward risks to the inflation forecast. Additionally, it pointed out that negative base effects, following a significant drop in prices in Q4 of 2024-25, are likely to elevate year-on-year inflation in Q4 of the current year, despite overall momentum remaining subdued.

2.6 Interest rate movement

Chart: Repo rate movement in India (on year, in percentage)



Source: CMIE, ICRA Analytics

Table: Recent Repo Rate Maintained by the Reserve Bank of India

Date	Repo Rate
5-Apr-2024	6.50%
7-Jun-2024	6.50%
8-Aug-2024	6.50%

9-Oct-2024	6.50%
6-Dec-2024	6.50%
7-Feb-2025	6.25%
9-Apr-2025	6.00%
6-Jun-2025	5.50%
1-Oct-2025	5.50%
6-Feb-2026	5.25%
8-Apr-2026	5.25%
5-Jun-2026	5.25%

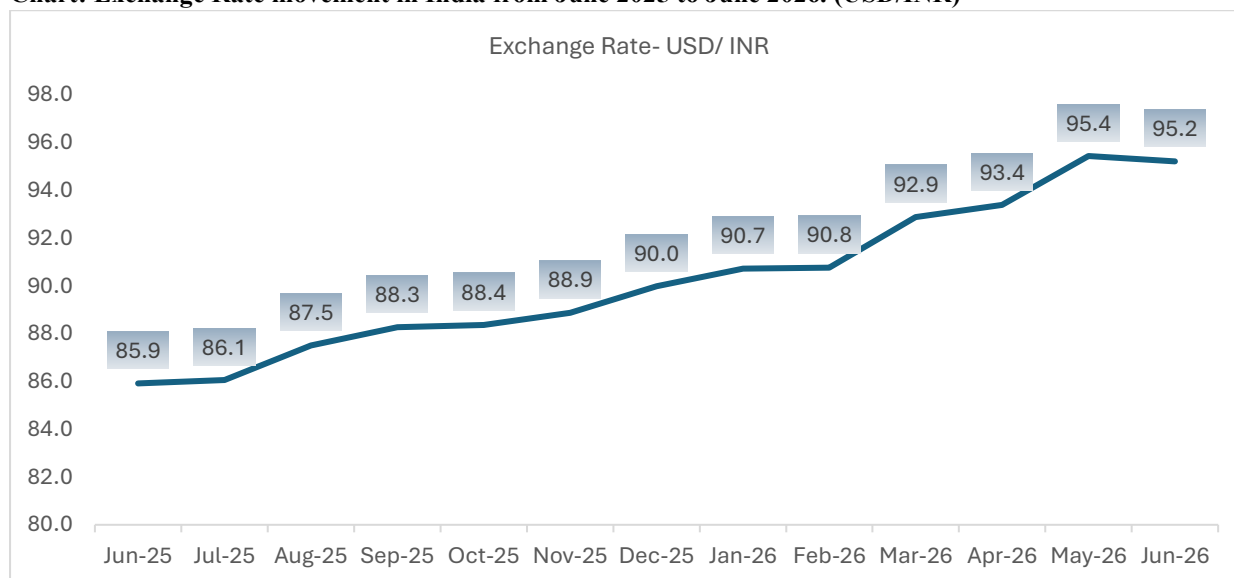
Source: RBI, ICRA Analytics

In the current landscape, India's repo rate trajectory continues to illustrate the RBI's prudent and data-driven approach to balancing inflation control with growth support. Following a series of rate hikes since 2022 to rein in high inflation, the central bank's decision to maintain the repo rate at 5.25% in the February 2026 meeting, along with a neutral stance, signals a continued "wait-and-watch" posture. With CPI inflation remaining below the RBI's tolerance band and economic activity showing resilience across key sectors, the conditions for future policy easing are gradually strengthening. However, any rate adjustment will remain highly contingent on incoming data, particularly relating to inflation durability, fiscal dynamics, global energy prices, and evolving geopolitical and trade risks.

Overall, India's evolving interest-rate path reaffirms the RBI's unwavering commitment to preserving financial stability, anchoring inflation expectations, and supporting sustainable long-term economic growth, while remaining responsive to a rapidly shifting global macroeconomic environment.

2.7 Review of Exchange Rates in India

Chart: Exchange Rate movement in India from June 2025 to June 2026. (USD/INR)



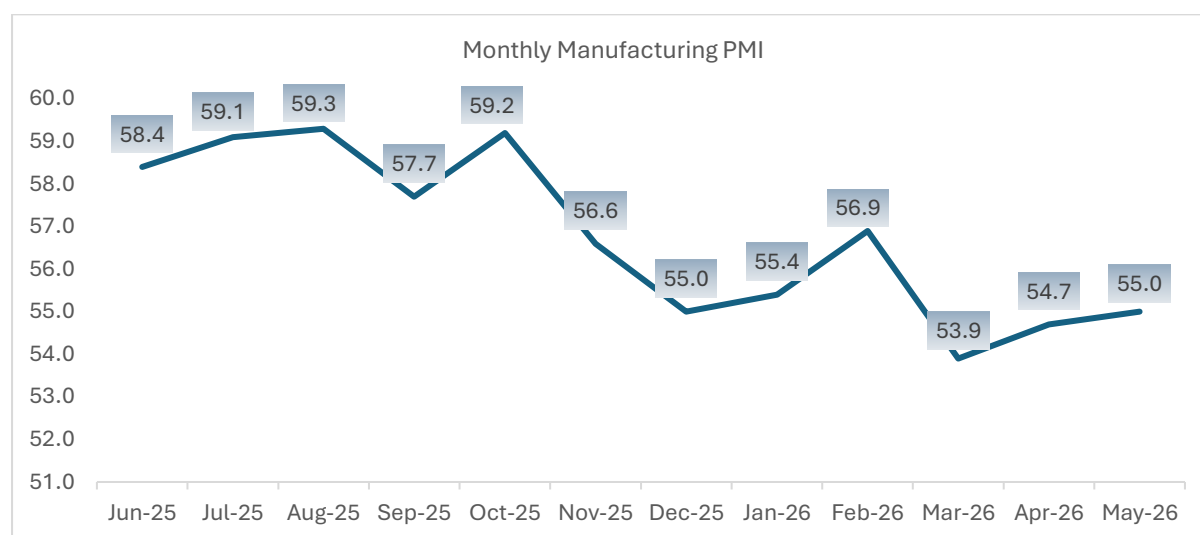
Source: X-Rates, ICRA Analytics.

India's exchange rate, as shown in the chart, exhibits a steady depreciation of the rupee against the US dollar over the one-year period from June 2025 to June 2026. The rate rises from INR 85.9 per USD in June 2025 to INR 95.2 per USD in June 2026, indicating that the rupee loses value consistently across the timeline. The increase is gradual in the early months, moving from INR 85.9 in June to INR 88.9 by November 2025, which suggests relatively controlled currency pressure. However, once the exchange rate crosses the INR 90 mark in December 2025, the upward trend becomes more pronounced, reflecting a faster pace of depreciation.

From January to May 2026, the data highlights more noticeable fluctuations, including a short plateau between January and February at INR 90.7 to INR 90.8, followed by sharper increases to INR 92.9 in March and INR 95.4 in May, which is the peak value in the chart. The slight dip to INR 95.2 in June 2026 does little to offset the overall upward movement. This sustained weakening of the rupee suggests underlying macroeconomic factors such as rising import demand, global dollar strength, inflationary pressures, and capital outflows. The trend has important implications, as it can make exports more competitive internationally while also making imports more expensive, especially commodities like oil, potentially adding pressure to India's trade balance and inflation.

2.8 Review of PMI trends across manufacturing sector

Chart: PMI trends in manufacturing sector in India from June 2025 to May 2026.



Source: Trading Economics, ICRA Analytics.

India's Monthly Manufacturing PMI remains consistently above the neutral 50 mark throughout the period from June 2025 to May 2026, indicating continuous expansion in manufacturing activity, though with clear fluctuations in momentum. The index begins at 58.4 in June and rises to a peak of 59.3 in August, reflecting strong growth in mid-2025, before dipping to 57.7 in September and rebounding to 59.2 in October. After this, a noticeable slowdown sets in, with the PMI falling to 56.6 in November and further to 55.0 in December, suggesting easing demand or production activity. Early 2026 shows a mild recovery as the index improves to 55.4 in January and 56.9 in February, but this is followed by a sharper decline to the lowest point of 53.9 in March, indicating a temporary soft patch. The PMI then stabilizes with a gradual recovery to 54.7 in April and 55.0 in May, signalling modest improvement but still below earlier highs. Overall, the trend reflects strong expansion initially followed by moderation and volatility, pointing to a gradual cooling of manufacturing growth while remaining in positive territory.

3. Global Ship Recycling Industry Overview

3.1 Market Overview

The global ship recycling industry is an integral part of the maritime value chain, facilitating the safe dismantling of end-of-life vessels and the recovery of valuable materials such as steel, non-ferrous metals, machinery, equipment, and other reusable components. The industry recycles a wide range of vessels, including bulk carriers, oil tankers, container ships, offshore vessels, passenger ships, and general cargo vessels that have reached the end of their economic or operational life. As steel generally accounts for a significant proportion of a vessel's lightweight, ship recycling serves as an important source of secondary steel and reusable marine equipment for industries including construction, infrastructure, engineering, manufacturing, and ship repair. By supplying recycled materials back into the economy, ship recycling supports resource efficiency, reduces dependence on virgin raw materials, lowers energy consumption, and contributes to lower greenhouse gas emissions compared with primary steel production, thereby supporting circular economy objectives.

Globally, ship recycling is undertaken through various dismantling methods, including beaching, dry dock, pier-side, and alongside recycling, with the choice of method depending on regulatory requirements, available infrastructure, environmental considerations, and economic viability. The industry is largely concentrated in South Asia, with India, Bangladesh, and Pakistan accounting for a substantial share of global ship recycling activity due to their established recycling ecosystems, competitive operating costs, favorable coastal infrastructure, and strong domestic demand for scrap steel.

The global regulatory landscape entered a new phase with the implementation of the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC) on 26 June 2025, which established globally harmonized standards for the safe and environmentally sound recycling of ships. Together with the European Union Ship Recycling Regulation (EU SRR), these regulations have strengthened requirements relating to hazardous material management, environmental protection, worker health and safety, operational procedures, and facility authorization. The evolving regulatory framework is encouraging shipowners to recycle vessels at internationally compliant recycling facilities and is driving investments in modern infrastructure, environmental management systems, mechanized operations, and internationally recognized certifications.

The long-term outlook for the global ship recycling industry remains positive, supported by an aging global fleet, ongoing fleet renewal and modernization, increasingly stringent environmental regulations, decarbonization initiatives within the shipping industry, and continued demand for recycled steel. According to the Baltic and International Maritime Council (BIMCO), more than 16,000 vessels are expected to be recycled globally over the next decade, creating significant opportunities for compliant ship recycling facilities. At the same time, increasing adoption of mechanized dismantling technologies, digital monitoring systems, advanced hazardous waste management practices, and internationally recognized management standards is enhancing operational efficiency, environmental performance, and occupational safety. Growing emphasis on ESG compliance and sustainable supply chains is also expected to support a gradual shift of recycling volumes towards authorized and environmentally compliant recycling facilities.

Despite its positive long-term outlook, ship recycling activity remains cyclical and is influenced by factors such as global shipping market conditions, freight rates, vessel earnings, scrap steel prices, ship acquisition prices, steel demand, and broader macroeconomic conditions, which collectively affect the timing and volume of vessels being retired for recycling.

Table: Global: Ship Recycling Market: Key Industry Highlights, CY2025 and CY2030

Particulars	CY 2025	CY 2030	CAGR (CY2025-CY2030)
Global Ship Recycling Market: Volume Trends (in Million Gross Tonnage (GT))	8.36	11.86	7.20%

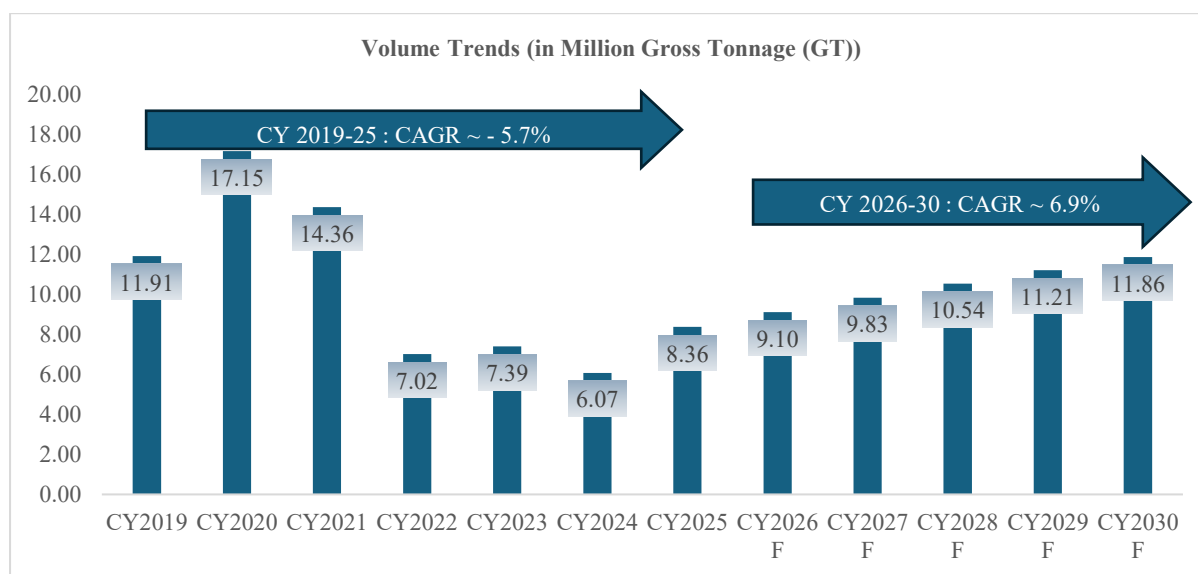
Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

Market by Region	CY2025 (in Thousand Gross Tonnage (GT))	CY2030F (in Thousand Gross Tonnage (GT))	CAGR (CY2025-CY2030F)
Asia-Pacific	7,136	10,298	7.60%
Middle East & Africa	1,036	1,317	4.90%
Europe	171.8	225.4	5.60%
North America	13.6	16.6	4.10%
Latin America	6.7	7.7	2.70%

Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

3.2 Market Performance & Forecast

Chart: Global: Ship Recycling Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The global ship recycling market reached 8.36 million Gross Tonnage (GT) in CY2025, despite registering a negative CAGR of 5.7% during CY2019-CY2025. Market activity during the period was influenced by fluctuations in vessel availability, freight market cycles, and delays in scrapping decisions as shipowners extended the operational lives of vessels amid periods of favorable shipping demand.

Despite the overall decline, the market continued to be supported by the growing demand for recycled steel, which remains a critical feedstock for the construction, infrastructure, automotive, and manufacturing sectors. Steel recovered from end-of-life vessels offers a cost-effective and environmentally sustainable alternative to primary steel production, helping reduce raw material consumption and supporting circular economy initiatives.

In addition, an aging global fleet and ongoing fleet modernization programs increased the number of vessels approaching the end of their economic lifespan. Rising maintenance expenditures, compliance costs, and stricter environmental regulations particularly related to emissions and fuel efficiency further encouraged shipowners to retire and recycle older vessels.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the global ship recycling market is projected to grow at a CAGR of 6.9% during CY2026F-CY2030F, reaching approximately 11.86 million GT by CY2030F. Growth is expected to be driven by a combination of regulatory, environmental, and economic factors that are accelerating the retirement of older vessels and promoting responsible recycling practices.

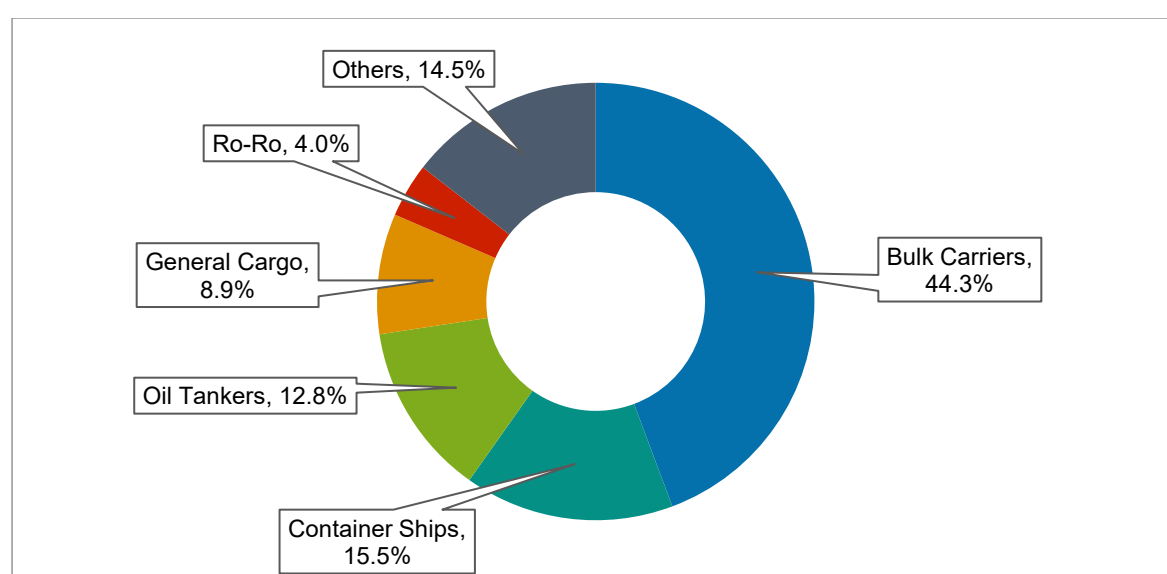
A major catalyst for future growth is the entry into force of the Hong Kong International Convention (HKC) in June 2025, alongside regional regulations such as the European Union Ship Recycling Regulation (EU SRR).

These frameworks are strengthening demand for environmentally compliant recycling facilities by encouraging shipowners to dismantle vessels at approved yards that adhere to internationally recognized environmental, health and safety standards. Furthermore, the shipping industry's ongoing decarbonization efforts, including the transition toward more fuel-efficient and low-emission vessels, are expected to accelerate fleet renewal and increase the supply of end-of-life ships available for recycling.

The market is also expected to benefit from continued investments in modern recycling infrastructure and advanced dismantling technologies across major recycling hubs. The adoption of mechanized equipment, digital monitoring systems, improved hazardous waste management practices, and internationally recognized management standards is enhancing operational efficiency, worker safety, and environmental performance. Additionally, increasing global steel demand, growing emphasis on resource efficiency, and sustainability-focused policies are expected to reinforce the strategic importance of ship recycling as a key contributor to circular economy objectives and sustainable materials sourcing.

3.3 Market Breakup by Type

Chart: Global: Ship Recycling Market: Breakup by Type (in %), CY2025



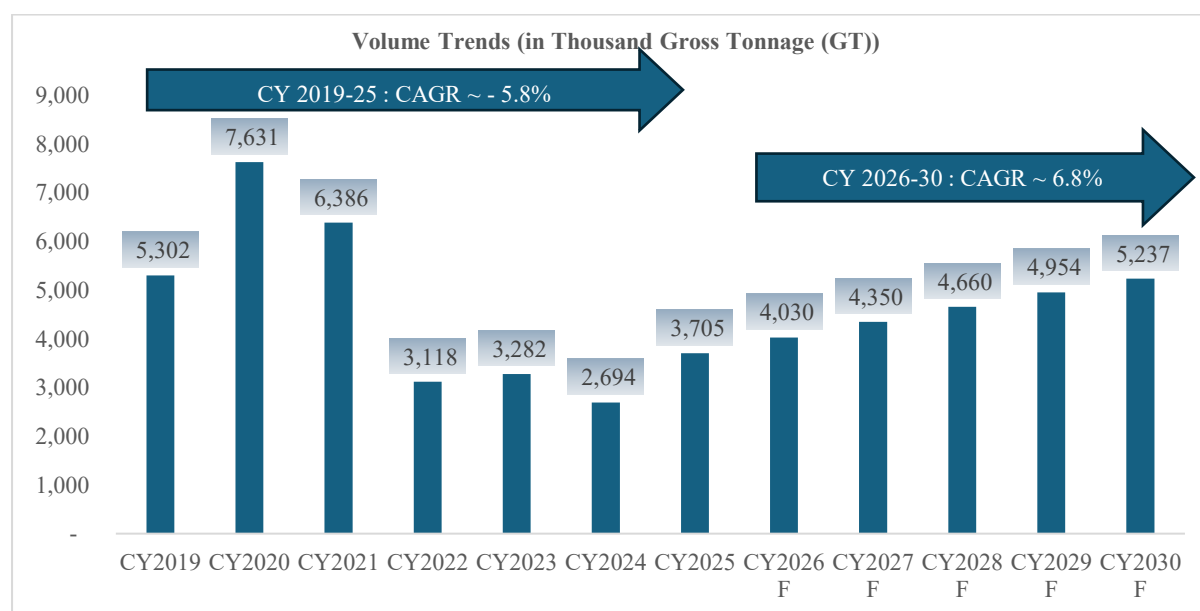
Source: Expert Interactions, IMARC, ICRA Analytics

In CY2025, bulk carriers accounted for the largest share of the global ship recycling market, representing 44.3% of total recycling volume. The segment was followed by container ships, which contributed 15.5% of the market, while oil tankers and general cargo vessels accounted for 12.8% and 8.9%, respectively. Ro-Ro (Roll-on/Roll-off) vessels represented 4.0% of total recycling activity, while the remaining 14.5% was attributed to other vessel types.

Recycling activity across these vessel categories was supported by increasing compliance costs, fleet modernization initiatives, decarbonization efforts, and the growing adoption of technologically advanced and fuel-efficient vessels. These factors encouraged shipowners to retire older vessels and replace them with more efficient and environmentally compliant assets, thereby driving recycling demand across the global shipping industry

3.3.1 Bulk Carriers

Chart: Global: Ship Recycling (Bulk Carriers) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

In CY2025, bulk carriers accounted for 44.3% of the global ship recycling market, making them the largest vessel segment by recycling volume. The segment reached a recycling volume of 3,705 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -5.8% during CY2019-CY2025. The trend was primarily influenced by fluctuations in scrapping activity, freight market cycles, and periods of extended vessel utilization.

The recycling of bulk carriers is largely driven by the aging global dry bulk fleet, particularly vessels engaged in the transportation of iron ore, coal, grains, and other commodities. As vessels age, increasing maintenance expenditure, declining fuel efficiency, and lower operational reliability reduce their economic viability.

In addition, the implementation of stricter environmental regulations and energy-efficiency standards has accelerated fleet renewal, encouraging shipowners to replace older vessels with modern bulk carriers that offer lower emissions, improved fuel economy, and enhanced operating performance.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the bulk carrier ship recycling segment is projected to reach a volume of 5,237 thousand GT by CY2030F, expanding at a CAGR of 6.8% during CY2026F-CY2030F. Growth is expected to be supported by a combination of fleet modernization initiatives, regulatory compliance requirements, and evolving market dynamics within the dry bulk shipping industry.

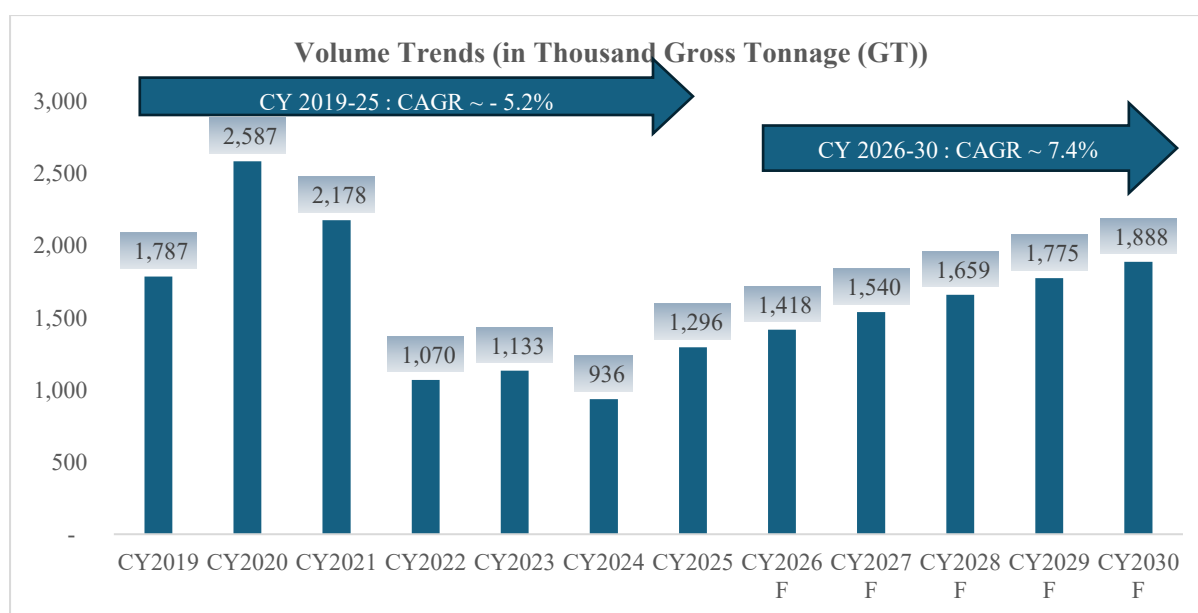
Freight rate movements, commodity trade patterns, and global economic activity are expected to remain key determinants of recycling volumes. Periods of weaker vessel earnings and reduced profitability typically encourage shipowners to retire older and less efficient vessels, thereby increasing recycling activity. Furthermore, the shipping industry's ongoing decarbonization efforts and the adoption of more stringent environmental standards are likely to accelerate the replacement of aging tonnage.

Technological advancements in bulk carrier design are also contributing to this transition. New-generation vessels incorporating larger cargo capacities, optimized hull designs, fuel-efficient propulsion systems, digital navigation technologies, and lower-emission operating capabilities are improving overall fleet economics.

As a result, shipowners are increasingly incentivized to phase out obsolete vessels, supporting growth in the bulk carrier recycling market over the forecast period. Additionally, the implementation of the Hong Kong International Convention (HKC) and the growing preference for environmentally compliant recycling facilities are expected to further strengthen recycling activity across the bulk carrier segment.

3.3.2 Container Ships

Chart: Global: Ship Recycling (Container Ships) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

In CY2025, container ships accounted for 15.5% of the global ship recycling market, making them the second-largest vessel segment by recycling volume. The segment recorded a recycling volume of 1,296 thousand Gross Tonnage (GT), reflecting a CAGR of -5.2% during CY2019-CY2025. Recycling activity during the period was influenced by shipping market dynamics, fleet utilization levels, and the postponement of scrapping decisions during periods of strong container trade and elevated freight rates.

The recycling of container ships has been largely driven by ongoing fleet modernization initiatives, with shipping companies increasingly replacing smaller and aging vessels with larger, more fuel-efficient ships that offer higher cargo capacity, lower operating costs, and better economies of scale. Furthermore, rapid advancements in vessel design, automation, digital technologies, and propulsion systems have accelerated the adoption of next-generation container vessels.

These modern ships provide improved fuel efficiency, enhanced operational performance, and better integration with evolving global supply chain requirements, reducing the commercial competitiveness of older vessels.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the container ship recycling segment is projected to reach a volume of 1,888 thousand GT by CY2030F, expanding at a CAGR of 7.4% during CY2026F-CY2030F. Growth is expected to be supported by continued fleet renewal, evolving trade patterns and increasing regulatory pressure on vessel efficiency and emissions performance.

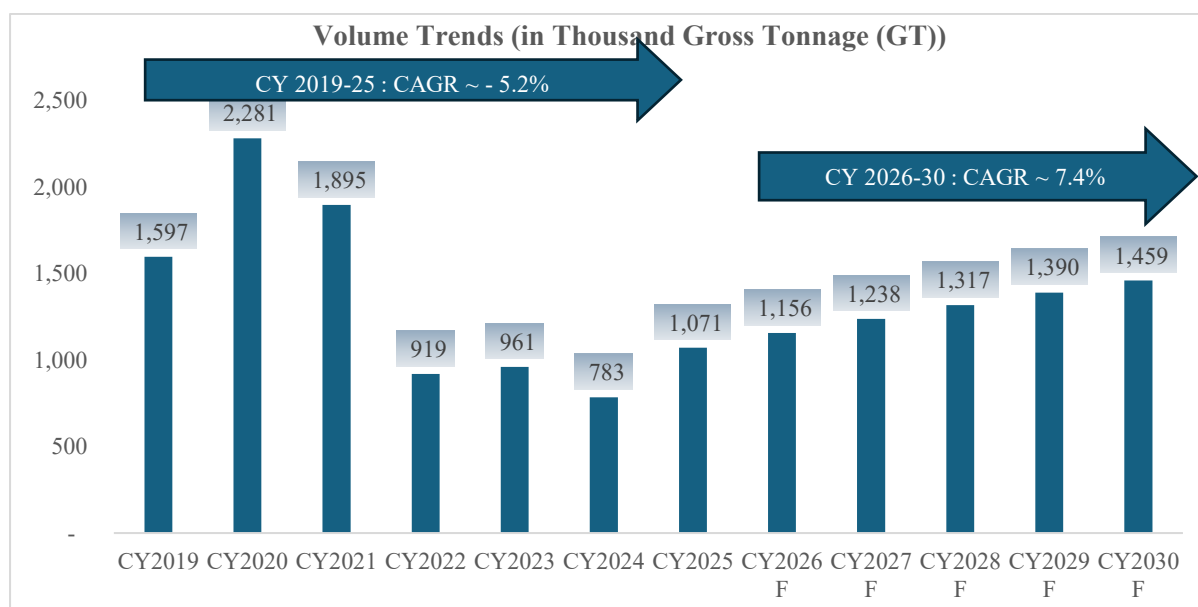
The expansion of global port infrastructure and the widespread deployment of Ultra-Large Container Vessels (ULCVs) have reshaped container shipping networks and intensified economies of scale across major trade routes. As a result, many older and smaller container vessels have become commercially less viable and are increasingly being retired and sent for recycling. Market conditions such as fluctuations in freight rates, vessel charter earnings, and global trade volumes are also expected to influence scrapping activity, particularly for aging ships facing lower profitability.

In addition, the shipping industry's decarbonization agenda is anticipated to accelerate recycling demand. Compliance with tightening carbon-intensity regulations, greenhouse gas reduction targets, and environmental standards is encouraging shipowners to invest in low-emission, alternative-fuel, and energy-efficient vessels. Consequently, conventional ships that require significant retrofit investments or face higher operating costs are increasingly being phased out.

The implementation of the Hong Kong International Convention (HKC) and growing preference for environmentally compliant recycling facilities are expected to further support recycling activity within the container ship segment over the forecast period.

3.3.3 Oil Tankers

Chart: Global: Ship Recycling (Oil Tankers) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

In CY2025, oil tankers accounted for 12.8% of the global ship recycling market, with recycling volumes reaching 1,071 thousand Gross Tonnage (GT). The segment registered a CAGR of -6.4% during CY2019–CY2025, reflecting fluctuations in scrapping activity, tanker market conditions, and periods of extended vessel operation driven by favorable freight earnings and oil trade dynamics.

The recycling of oil tankers has been primarily driven by ongoing fleet renewal initiatives across the global tanker industry. As vessels age, declining fuel efficiency, increasing maintenance requirements, and higher operating costs reduce their commercial attractiveness, prompting shipowners to retire older tankers in favour of modern, technologically advanced vessels.

In addition, stringent international regulations related to emissions, ballast water management, and energy efficiency have significantly increased compliance obligations for aging fleets. The substantial investment required to retrofit older tankers to meet evolving regulatory standards often makes recycling a more economically viable option than continued operation.

- **Forecast Outlook (CY2026F–CY2030F)**

Looking ahead, the oil tanker recycling segment is projected to reach a volume of 1,459 thousand GT by CY2030F, expanding at a CAGR of 6.0% during CY2026F–CY2030F. Growth is expected to be supported by regulatory-driven fleet modernization, changing energy transportation patterns, and increasing adoption of environmentally sustainable vessel technologies.

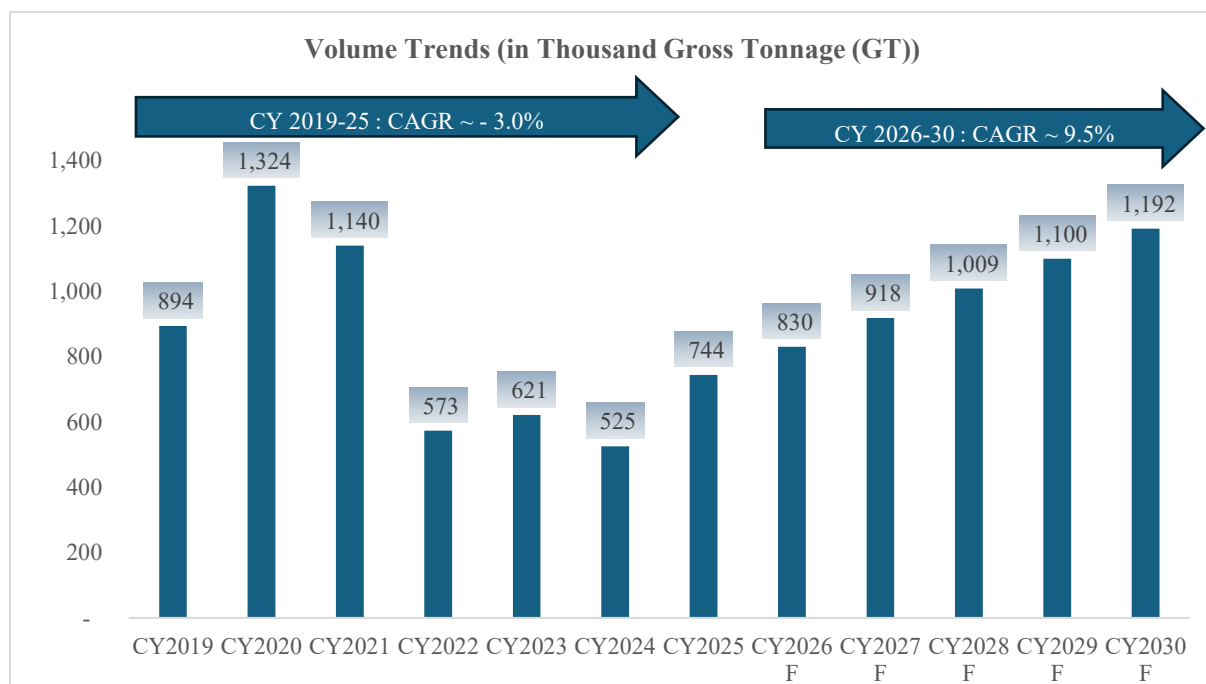
Market conditions within the global crude oil and petroleum products trade will continue to play a critical role in determining recycling activity. Fluctuations in tanker utilization rates, freight earnings, and transportation demand directly influence vessel profitability. During periods of weaker charter rates and reduced trade volumes, shipowners often accelerate the retirement of older and less competitive tankers to optimize fleet efficiency and profitability.

Furthermore, the shipping industry's decarbonization efforts are expected to strengthen recycling demand over the forecast period. The transition toward next-generation tankers powered by cleaner fuels such as LNG, methanol, ammonia, and other low-carbon alternatives is encouraging shipowners to replace conventional vessels with more environmentally sustainable options.

Advances in vessel design, fuel efficiency technologies, and emissions-reduction systems are further supporting fleet renewal. Consequently, older tankers that are costly to retrofit and operate under increasingly stringent environmental regulations are likely to be phased out, contributing to higher recycling volumes globally.

3.3.4 General Cargo

Chart: Global: Ship Recycling (Oil Tankers) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019–CY2025)**

The general cargo segment accounted for 8.9% of the global ship recycling market in CY2025, reaching a volume of 744 thousand Gross Tonnage (GT). The segment registered a CAGR of -3.0% during CY2019–CY2025, reflecting a gradual decline in recycling volumes over the historical period.

The recycling of general cargo vessels is largely driven by the aging profile of regional and short-sea trading fleets. A considerable number of these vessels have surpassed their economic operating life, resulting in higher maintenance and repair expenses, lower fuel efficiency, increased downtime, and reduced operational reliability.

Consequently, recycling older vessels has become a more cost-effective option for shipowners compared to extending their service life.

The market has also been influenced by the implementation of increasingly stringent international environmental regulations related to energy efficiency, carbon intensity, and emissions reduction. To comply with these evolving requirements, shipowners are progressively replacing older general cargo vessels with modern, fuel-efficient ships that offer improved environmental performance, enhanced operational efficiency, and lower lifecycle costs.

- **Forecast Outlook (CY2026F-CY2030F)**

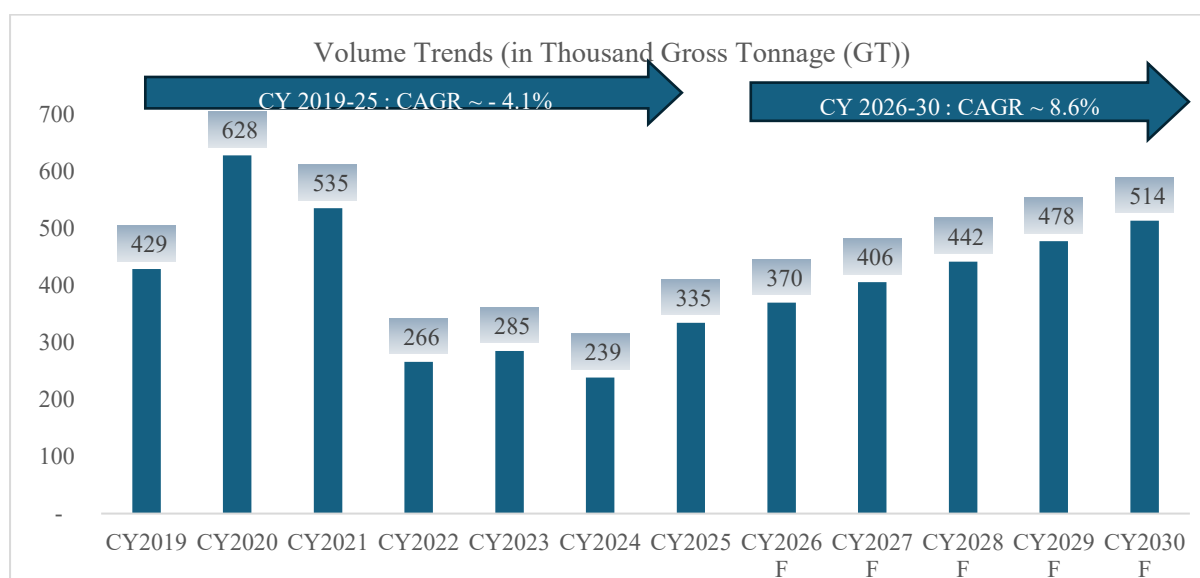
Looking ahead, the general cargo segment is expected to reach a volume of 1,192 thousand Gross Tonnage (GT) by CY2030F, expanding at a CAGR of 9.5% during CY2026F-CY2030F. Growth is anticipated to be supported by ongoing fleet renewal activities, regulatory compliance requirements, and the continued retirement of aging general cargo vessels.

The increasing adoption of containerized shipping across global trade routes is expected to remain a key driver of recycling activity within this segment. As container vessels provide greater cargo handling efficiency, faster turnaround times, and lower transportation costs, the role of conventional general cargo and breakbulk vessels continues to diminish in many markets. This transition is expected to accelerate the phase-out and recycling of older vessels.

In addition, prevailing market conditions are likely to influence recycling decisions among shipowners. Periods of weak freight rates, subdued trade activity, rising maintenance expenditures, and costly dry-docking requirements can significantly reduce the economic viability of aging vessels. As a result, shipowners are increasingly expected to recycle older assets, optimize fleet composition, and improve overall operational efficiency, supporting the growth of the general cargo ship recycling segment over the forecast period.

3.3.5 Ro-Ro (Roll-on/Roll-off)

Chart: Global: Ship Recycling (Oil Tankers) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Ro-Ro (Roll-on/Roll-off) segment accounted for 4.0% of the global ship recycling market in CY2025, reaching a volume of 335 thousand Gross Tonnage (GT). The segment recorded a CAGR of -4.1% during CY2019-CY2025, reflecting fluctuations in vessel retirement activity and fleet replacement cycles across the global Ro-Ro shipping industry.

The recycling of Ro-Ro vessels has been primarily driven by the aging profile of global ferry, vehicle carrier, and cargo Ro-Ro fleets, many of which are approaching the end of their economic service life. Rising maintenance

and repair costs, coupled with declining operational efficiency, have encouraged shipowners to retire older vessels and replace them with newer and more technologically advanced designs.

In addition, increasingly stringent international environmental regulations, including the IMO's decarbonization targets, Energy Efficiency Existing Ship Index (EEXI), and Carbon Intensity Indicator (CII) requirements, have accelerated fleet renewal across the sector. Given the significant costs associated with retrofitting older vessels to comply with evolving environmental standards, recycling has become a more commercially attractive option for many shipowners.

- **Forecast Outlook (CY2026F–CY2030F)**

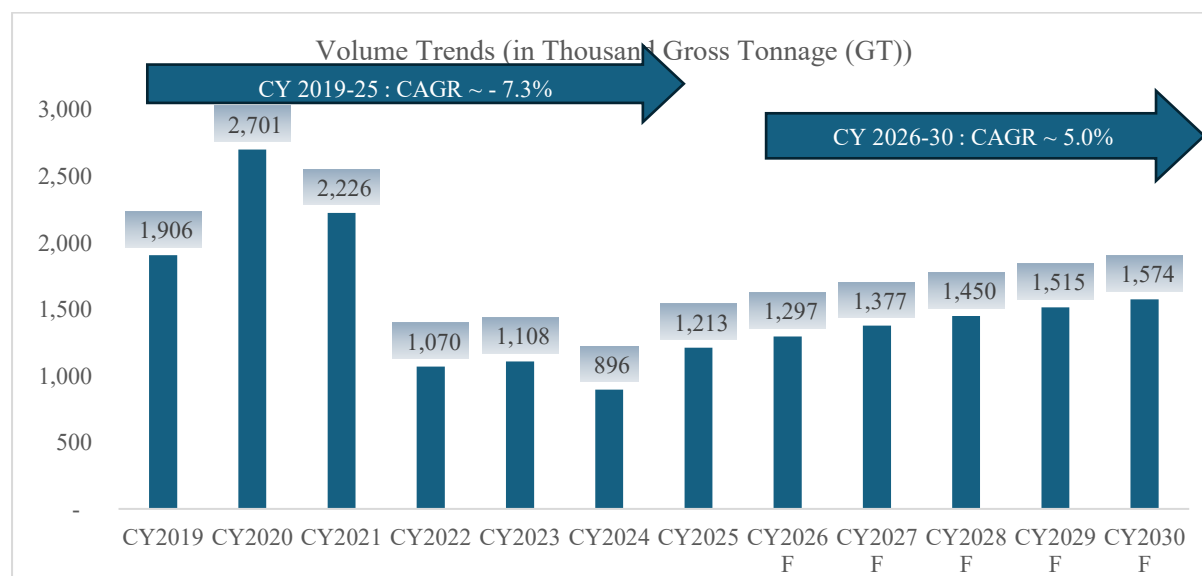
Looking ahead, the Ro-Ro segment is projected to reach a volume of 514 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 8.6% during CY2026F–CY2030F. Growth is expected to be supported by ongoing fleet modernization initiatives, increasing investments in advanced vessel technologies, and stricter environmental compliance requirements.

Growing demand for vehicle transportation and freight logistics worldwide is encouraging shipping companies to invest in larger, fuel-efficient, and dual-fuel Ro-Ro vessels with enhanced cargo capacity and reduced emissions. This transition towards next-generation vessels is expected to increase the number of aging Ro-Ro ships being retired and sent for recycling over the forecast period.

The segment is also expected to benefit from the high recovery value associated with dismantled Ro-Ro vessels. These ships contain substantial quantities of high-quality steel, along with reusable onboard equipment such as ramps, engines, electrical systems, cargo handling machinery, and other marine components. The recovery and reuse of these materials support circular economy objectives while improving the economic viability of ship recycling activities.

3.3.6 Others

Chart: Global: Ship Recycling (Oil Tankers) Market: Volume Trends (in Million Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

In CY2025, the other types accounted for a share of 14.5% of the global ship recycling market. The market reached a volume of 1,213 thousand Gross Tonnage (GT), representing a CAGR of -7.3% during CY2019-CY2025.

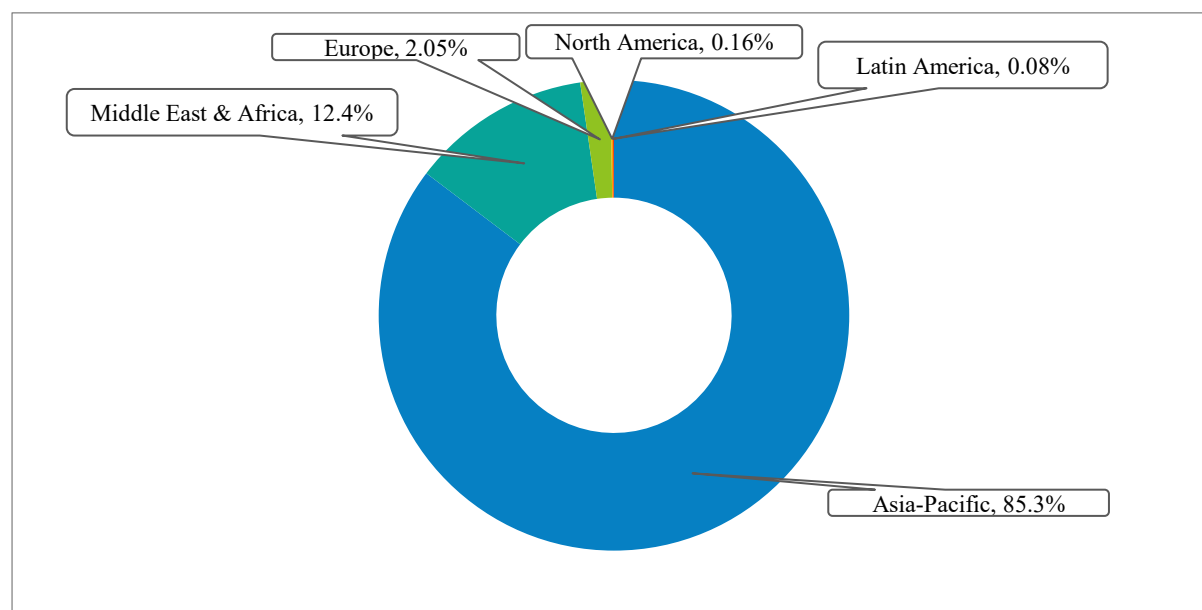
- **Forecast Outlook (CY2026F–CY2030F)**

Looking forward, the market in this segment is expected to reach a volume of 1,574 thousand Gross Tonnage (GT) by CY2030F, representing a CAGR of 5.0% during CY2026F–CY2030F. Growth is anticipated to be supported

by increasing vessel retirements, ongoing fleet modernization efforts and the continued replacement of aging assets with newer and more efficient vessel types.

3.4 Market Breakup by Region

Chart: Global: Ship Recycling Market: Breakup by Region (in %), CY2025



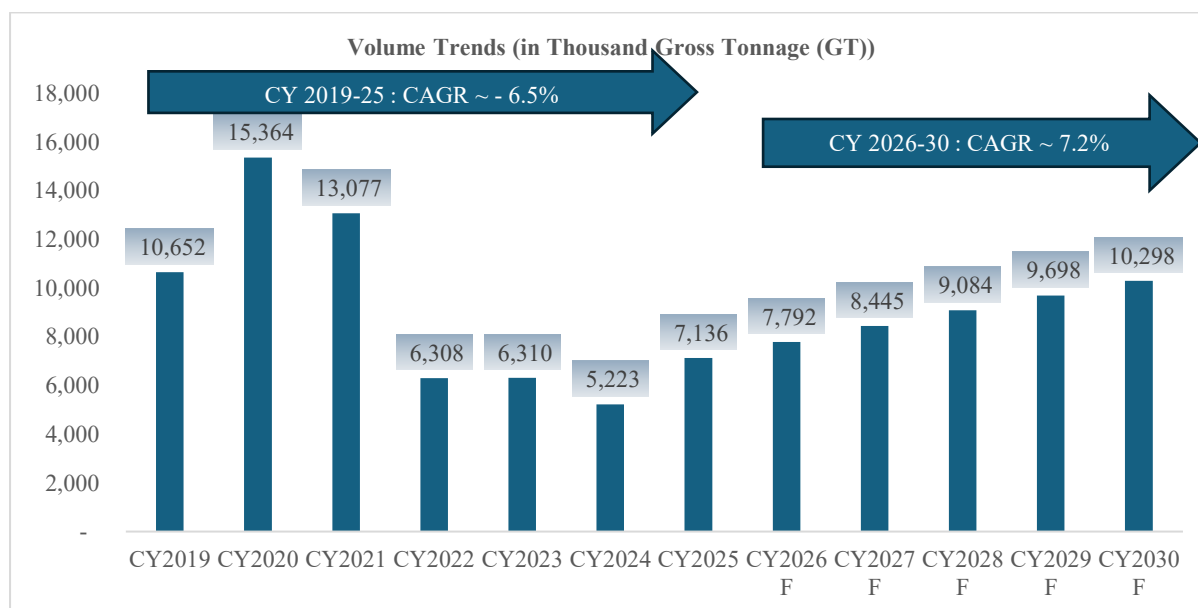
Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

In CY2025, Asia-Pacific accounted for the largest share of the global ship recycling market, representing 85.3% of total recycling volume. The region's dominance is supported by the concentration of leading recycling nations, particularly India, Bangladesh and Pakistan, which collectively handle a substantial portion of the world's end-of-life vessel dismantling activities. Growing investments in compliant recycling infrastructure and increasing adoption of international environmental standards have further strengthened the region's position in the global market.

Asia-Pacific was followed by the Middle East and Africa, which accounted for 12.4% of global recycling volumes. Europe contributed 2.05%, while North America and Latin America represented 0.16% and 0.08%, respectively. Asia-Pacific remains the primary destination for global ship recycling activities.

3.4.1 Asia-Pacific

Chart: Asia-Pacific: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019–CY2025)**

The Asia-Pacific ship recycling market reached a volume of 7,136 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -6.5% during CY2019-CY2025. Despite the contraction over the historical period, the region maintained its position as the dominant global ship recycling hub, supported by its extensive dismantling capacity and well-established recycling ecosystem.

The region's leadership is primarily driven by major recycling nations such as India, Bangladesh, and Pakistan, which collectively account for a significant share of global ship dismantling activity. Asia-Pacific benefits from established recycling infrastructure, a skilled workforce, competitive operating costs, and strong downstream steel re-rolling industries, making it the preferred destination for end-of-life vessels.

In addition, the implementation of the Hong Kong International Convention (HKC) has accelerated the modernization of recycling facilities across the region. Ship recycling yards are increasingly investing in mechanized dismantling systems, hazardous waste management infrastructure, worker safety measures, and environmentally compliant operations to meet international standards and enhance their attractiveness to global shipowners.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the Asia-Pacific ship recycling market is projected to grow at a CAGR of 7.2% during CY2026F-CY2030F, reaching a volume of approximately 10,298 thousand GT by CY2030F. Growth is expected to be supported by strong steel demand, increasing vessel retirements, and continuous improvements in recycling infrastructure and environmental compliance.

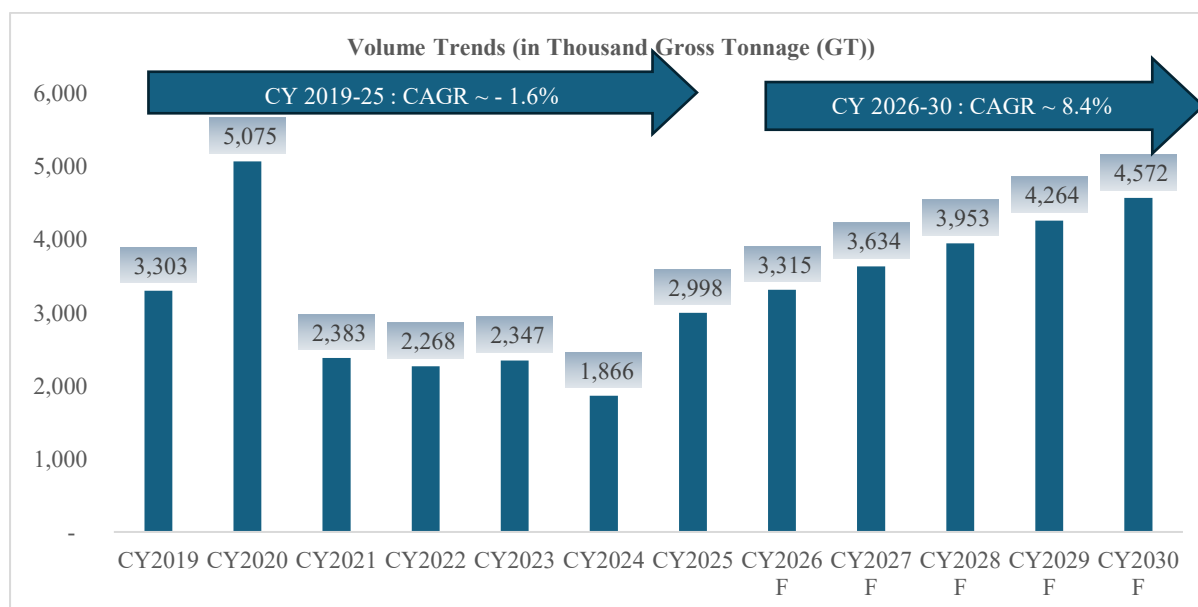
Robust demand for recycled steel from the construction, infrastructure, engineering, manufacturing, and industrial sectors is expected to remain a key growth driver. The recovery of steel and other valuable materials from dismantled vessels promotes resource efficiency, reduces reliance on virgin raw materials, and supports circular economy objectives across the region. Furthermore, the ongoing retirement of aging commercial fleets, coupled with stricter environmental regulations and global decarbonization targets, is expected to accelerate recycling activity as shipowners replace older vessels with modern, fuel-efficient alternatives.

The market is also poised to benefit from continued investments in capacity expansion, digital compliance systems, mechanized dismantling technologies, and sustainable recycling practices. Countries such as India are

further strengthening their competitive position through HKC-compliant facilities and enhanced environmental standards. As a result, Asia-Pacific is expected to maintain its dominance in the global ship recycling industry while reinforcing its role as the primary destination for environmentally responsible vessel recycling over the forecast period.

3.4.1.1 India

Chart: India: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The Indian ship recycling market reached a volume of 2,998 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -1.6% during CY2019–CY2025. Despite moderate contraction during the historical period, India maintained its position as one of the world's leading ship recycling destinations, supported by its well-established recycling ecosystem, large dismantling capacity, and strong domestic demand for recycled materials.

India's leadership in the market is primarily driven by the Alang-Sosiya ship recycling cluster in Gujarat, which is one of the largest ship recycling hubs globally. The country's competitive labor costs, developed downstream steel re-rolling industry, and extensive experience in vessel dismantling continue to attract a significant share of end-of-life vessels. Additionally, growing emphasis on environmentally responsible recycling practices and investments in Hong Kong International Convention (HKC)-compliant facilities have strengthened India's competitiveness in the global ship recycling industry.

Forecast Outlook (CY2026F-CY2030F)

Looking ahead, the Indian ship recycling market is expected to grow at a CAGR of 8.4% during CY2026F-CY2030F, reaching a volume of approximately 4,572 thousand GT by CY2030F. Growth is anticipated to be driven by favorable policy support, increasing vessel retirements, expanding recycling infrastructure, and rising demand for recycled steel across key end-use industries.

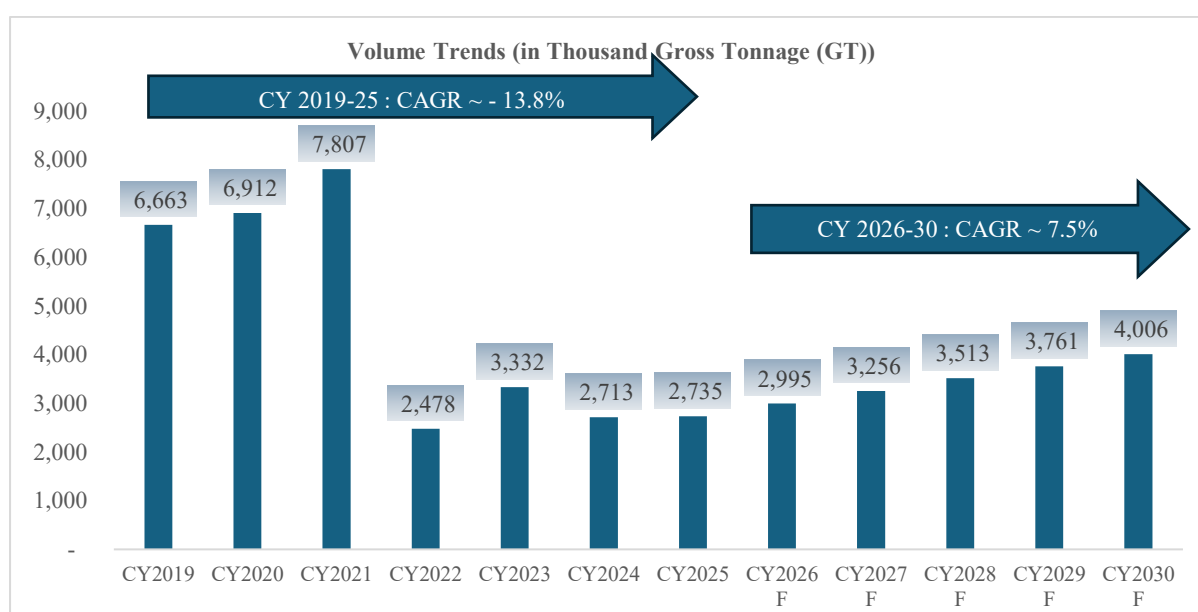
India is expected to benefit significantly from the implementation of the Recycling of Ships Act, 2019, which aligns domestic regulations with international standards and enhances the country's attractiveness as a compliant recycling destination. Furthermore, government initiatives aimed at modernizing recycling infrastructure, digitalizing compliance processes, and encouraging sustainable ship recycling practices are expected to support long-term industry growth.

The planned expansion of Alang's recycling capacity to approximately 9 million Light Displacement Tonnage (LDT), coupled with ongoing investments in HKC-compliant facilities, is expected to strengthen India's ability to handle a larger share of global vessel recycling demand. Efforts to secure broader recognition through inclusion in the European Union's approved list of ship recycling facilities could further enhance India's global competitiveness by attracting vessels from European shipowners seeking environmentally compliant recycling solutions.

Additionally, rising global fleet renewal activity, stricter environmental regulations, and the shipping industry's decarbonization efforts are expected to increase the supply of end-of-life vessels available for dismantling. Supported by robust domestic demand for recycled steel from the construction, infrastructure, and manufacturing sectors, India is well positioned to consolidate its role as a leading global ship recycling hub over the forecast period.

3.4.1.2 Bangladesh

Chart: Bangladesh: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The Bangladesh ship recycling market reached a volume of 2,735 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -13.8% during CY2019–CY2025. Despite the decline during the historical period, Bangladesh remained one of the world's largest ship recycling destinations, supported by its extensive dismantling capacity, strong domestic steel demand, and well-established recycling ecosystem.

The country's ship recycling industry is centered around the Chattogram (Chittagong) Ship Breaking Yard, which plays a critical role in supplying scrap steel to the domestic market. Recycled steel obtained from dismantled vessels is widely utilized across Bangladesh's construction, infrastructure, engineering, and manufacturing sectors, reducing dependence on imported raw materials. The availability of a large workforce, competitive operating costs, and a consistent inflow of aging vessels have further reinforced Bangladesh's position as a major participant in the global ship recycling industry. In addition, ongoing investments in infrastructure modernization and the gradual adoption of Hong Kong International Convention (HKC)-compliant recycling facilities have improved the industry's operational capabilities and environmental performance.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the Bangladesh ship recycling market is expected to grow at a CAGR of 7.5% during CY2026F-CY2030F, reaching a volume of approximately 4,006 thousand GT by CY2030. Growth is anticipated to be

supported by rising vessel retirement volumes, sustained domestic steel demand, and continued efforts to modernize recycling operations in line with international standards.

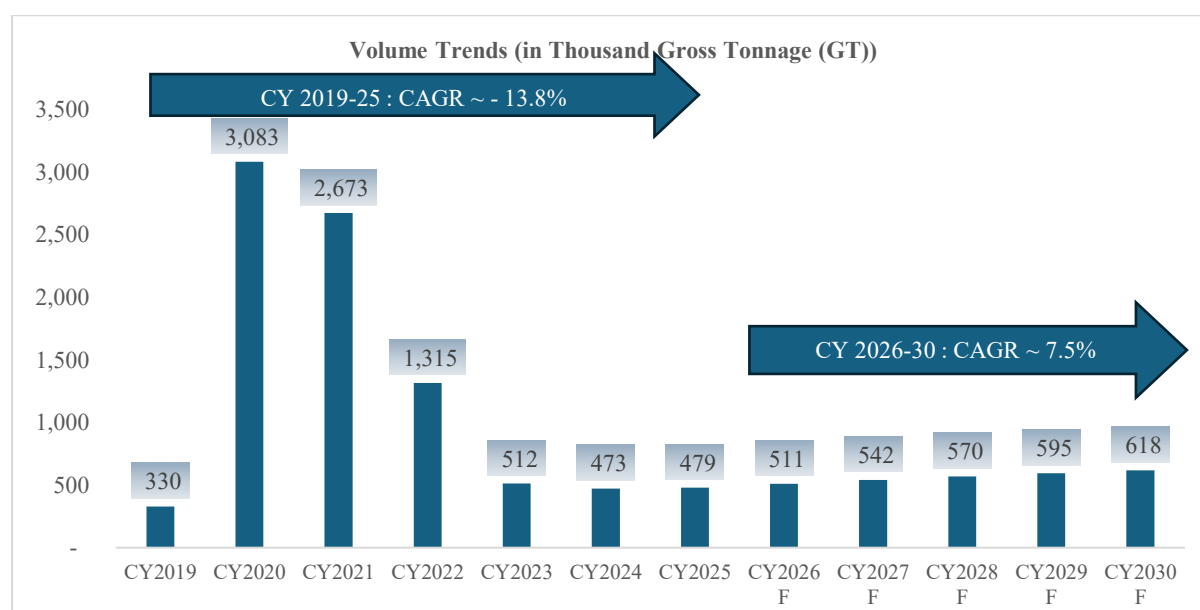
Bangladesh's ship recycling industry is expected to benefit from the growing global requirement for environmentally compliant dismantling facilities following the implementation of the Hong Kong International Convention (HKC). Recycling yards are increasingly investing in upgraded infrastructure, improved hazardous waste management systems, mechanized dismantling processes, and enhanced worker safety measures to align with global best practices and attract a larger share of international vessel recycling demand.

Furthermore, robust demand for affordable steel from the country's rapidly expanding construction and infrastructure sectors is expected to remain a key growth driver. The recovery of steel and other valuable materials from end-of-life vessels supports resource efficiency and provides a cost-effective source of raw materials for domestic industries. At the same time, the global shipping industry's fleet modernization and decarbonization initiatives are likely to increase the number of aging vessels available for recycling, creating additional opportunities for market expansion.

Regulatory reforms focused on strengthening environmental management, improving workplace safety, and enhancing compliance standards are also expected to support the long-term development of the industry. As a result, Bangladesh is well positioned to maintain its status as one of the leading global ship recycling hubs while improving the sustainability and competitiveness of its recycling operations over the forecast period.

3.4.1.3 Pakistan

Chart: Pakistan: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Pakistan ship recycling market reached a volume of 479 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of 6.4% during CY2019-CY2025. Unlike several major ship recycling markets that witnessed contractions during the historical period, Pakistan recorded positive growth, supported by increasing vessel dismantling activity and sustained demand for recycled steel from domestic industries.

The country's ship recycling industry is largely centered around the Gadani Ship Breaking Yard, one of the largest ship recycling facilities in South Asia. The yard benefits from its strategic location along key international shipping routes, providing access to a steady supply of end-of-life vessels.

Pakistan's competitive recycling costs, established dismantling capabilities, and growing steel requirements from the construction and infrastructure sectors have further supported market growth. In addition, ongoing efforts to

modernize recycling operations and enhance environmental and safety practices have strengthened the industry's competitiveness within the regional ship recycling landscape.

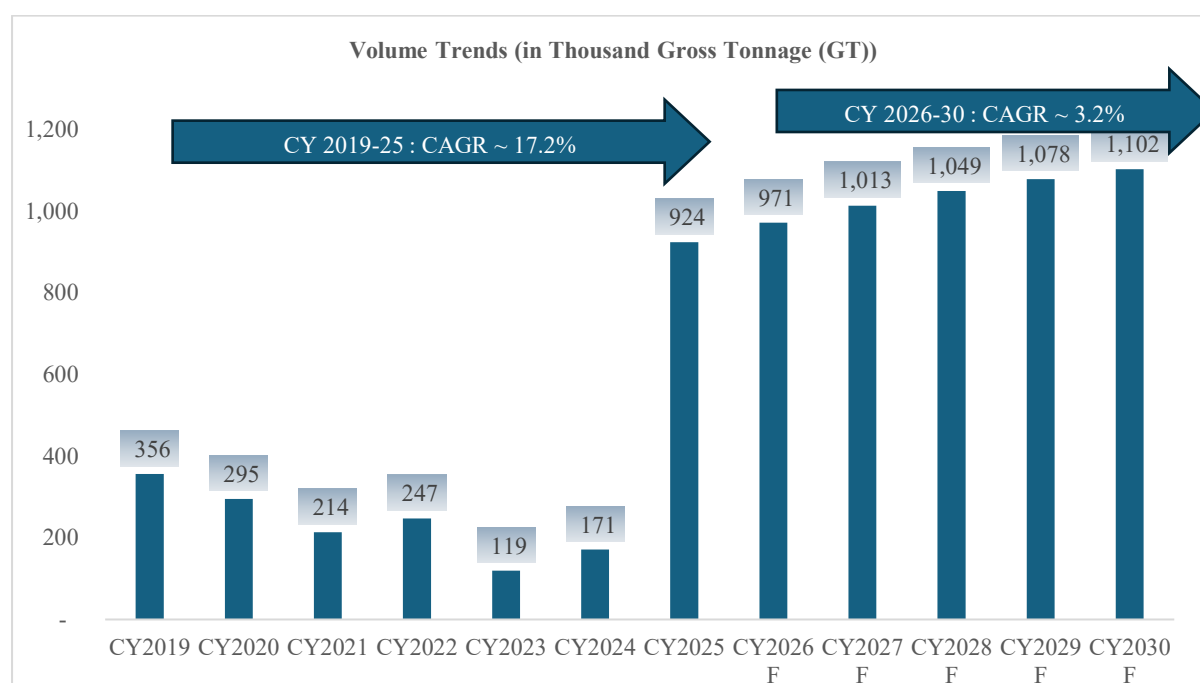
- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the Pakistan ship recycling market is expected to grow at a CAGR of 4.8% during CY2026F-CY2030F, reaching a volume of approximately 618 Thousand GT by CY2030F. Growth is anticipated to be driven by rising vessel retirement volumes, increasing demand for recycled steel, and continued improvements in recycling infrastructure and regulatory compliance.

The Gadani Ship Breaking Yard is expected to remain the primary growth driver for the industry, benefiting from the increasing availability of aging vessels resulting from global fleet renewal and decarbonization initiatives. As shipowners replace older ships with more fuel-efficient and environmentally compliant vessels, the volume of end-of-life vessels available for recycling is likely to increase, creating additional opportunities for the Pakistani market.

3.4.1.4 Others

Chart: Others: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

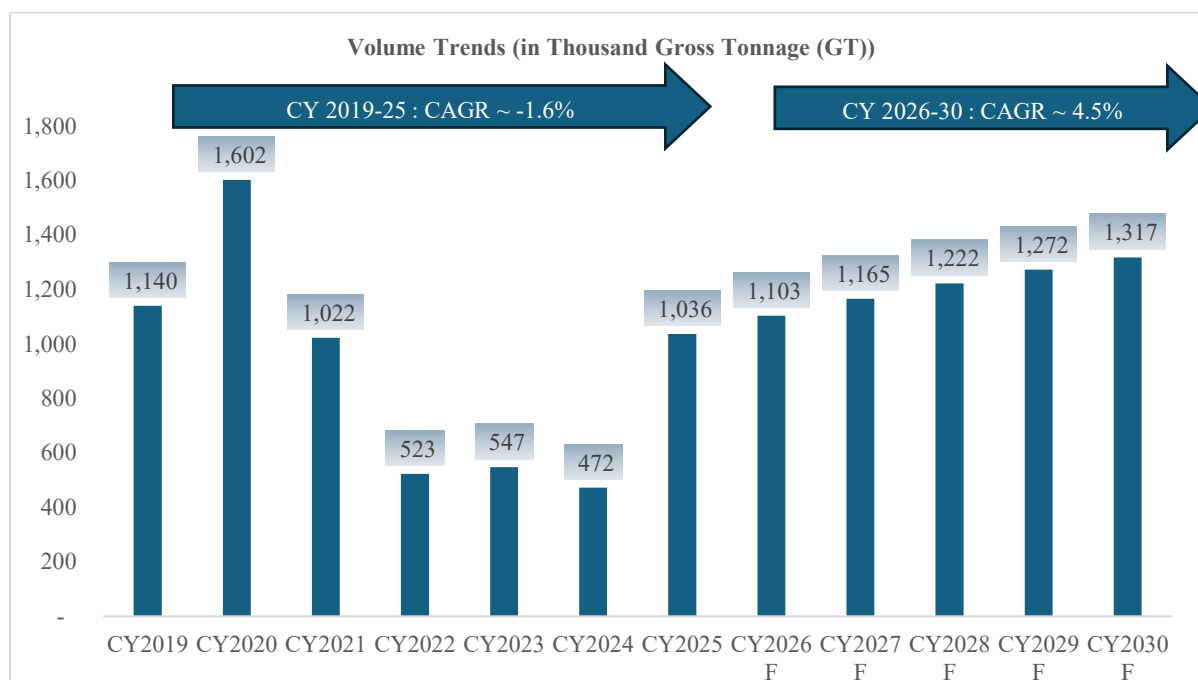
The ship recycling market in other Asia-Pacific countries reached a volume of 924 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of 17.2% during CY2019-CY2025.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the ship recycling market in other Asia-Pacific countries is expected to grow at a CAGR of 3.2% during CY2026F-CY2030F, reaching a volume of approximately 1,102 thousand GT by CY2030.

3.4.2 Middle East & Africa

Chart: Middle East & Africa: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Middle East & Africa ship recycling market reached a volume of 1,036 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -1.6% during CY2019-CY2025. The market was influenced by fluctuations in vessel retirement activity across the region's maritime, shipping, and offshore energy sectors. Despite the presence of a large commercial fleet operating in the region, ship recycling volumes remained moderated by varying market conditions and the operational lifespan of vessels.

The industry is primarily supported by the recycling of aging oil tankers, offshore support vessels (OSVs), cargo ships, and other commercial vessels that serve the region's energy and maritime industries. Increasing maintenance and operating costs associated with older vessels, coupled with ongoing fleet modernization initiatives, have encouraged shipowners and operators to retire and recycle aging assets.

In addition, growing investments in offshore oil and gas exploration, port development and maritime infrastructure expansion have accelerated fleet renewal activities across several countries in the region. The deployment of newer and more technologically advanced vessels has consequently generated a steady flow of end-of-life ships available for recycling, supporting market activity during the historical period.

- **Forecast Outlook (CY2026F-CY2030F)**

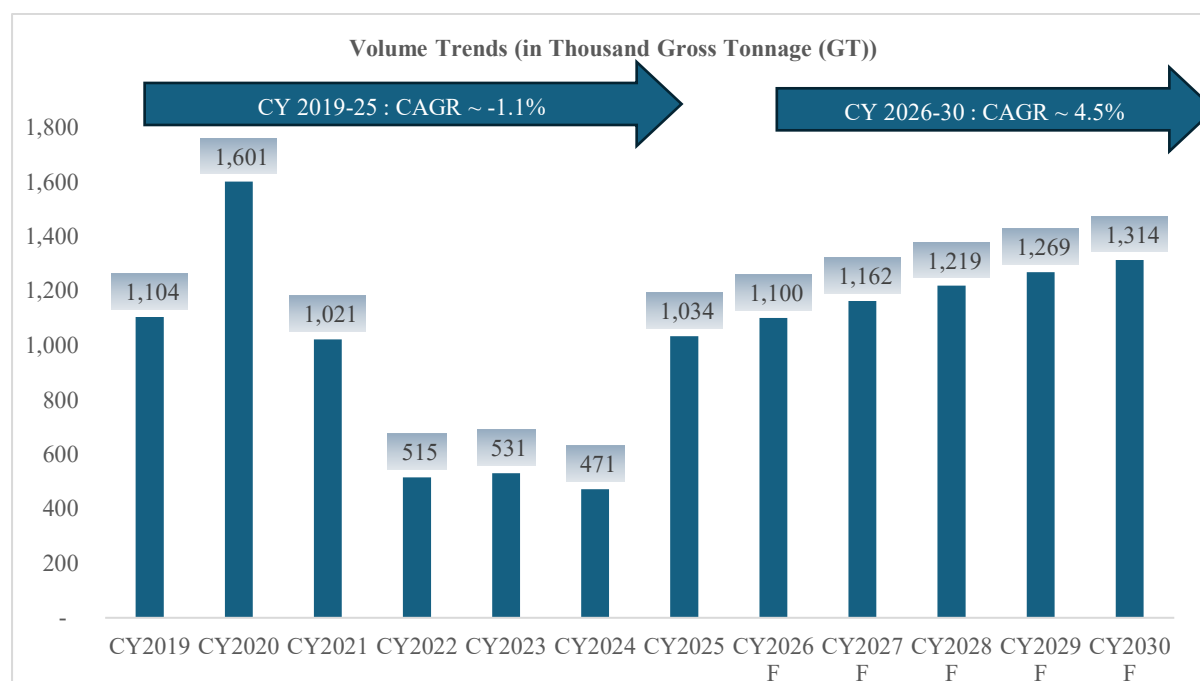
Looking ahead, the Middle East & Africa ship recycling market is expected to grow at a CAGR of 4.5% during CY2026F-CY2030F, reaching a volume of approximately 1,317 thousand GT by CY2030F. Market growth is anticipated to be driven by increasing vessel retirement volumes, continued fleet modernization and rising demand for recycled materials from various industrial sectors across the region.

Several countries are strengthening their regulatory frameworks and promoting environmentally responsible ship recycling practices through improvements in hazardous waste management, worker safety standards, and compliance with international environmental regulations. These initiatives are expected to facilitate the gradual development of more organized and formalized ship recycling capabilities across the region, enhancing the industry's long-term sustainability.

Furthermore, increasing industrialization and infrastructure development are expected to support demand for recycled steel and reusable marine equipment recovered from decommissioned vessels. Combined with growing adoption of circular economy principles and sustainable resource utilization practices, these factors are likely to create favorable conditions for the continued expansion of the Middle East & Africa ship recycling market over the forecast period.

3.4.2.1 Turkey

Chart: Turkey: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Turkey ship recycling market reached a volume of 1,034 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -1.1% during CY2019-CY2025. Market performance during the historical period was influenced by fluctuations in vessel recycling volumes, changes in global shipping activity and the timing of fleet replacement cycles across Europe and the Mediterranean region.

Turkey's ship recycling industry is primarily centered around the Aliaga Ship Recycling Zone, which has established itself as one of the leading ship recycling hubs in the region. The facility specializes in the environmentally compliant dismantling of commercial vessels, offshore assets, and European-flagged ships, supporting a wide range of recycling requirements from regional and international shipowners.

The market has benefited from the implementation of the European Union Ship Recycling Regulation (EU SRR), which has increased demand for approved recycling facilities that meet stringent environmental and safety standards. In addition, Turkey's advanced hazardous waste management capabilities, strong focus on sustainable recycling practices, and strategic location near major European shipping routes have strengthened its position within the global ship recycling industry. Continued investments in modern recycling infrastructure have further enhanced the country's competitiveness and attractiveness as a preferred destination for compliant vessel recycling.

- **Forecast Outlook (CY2026F-CY2030F)**

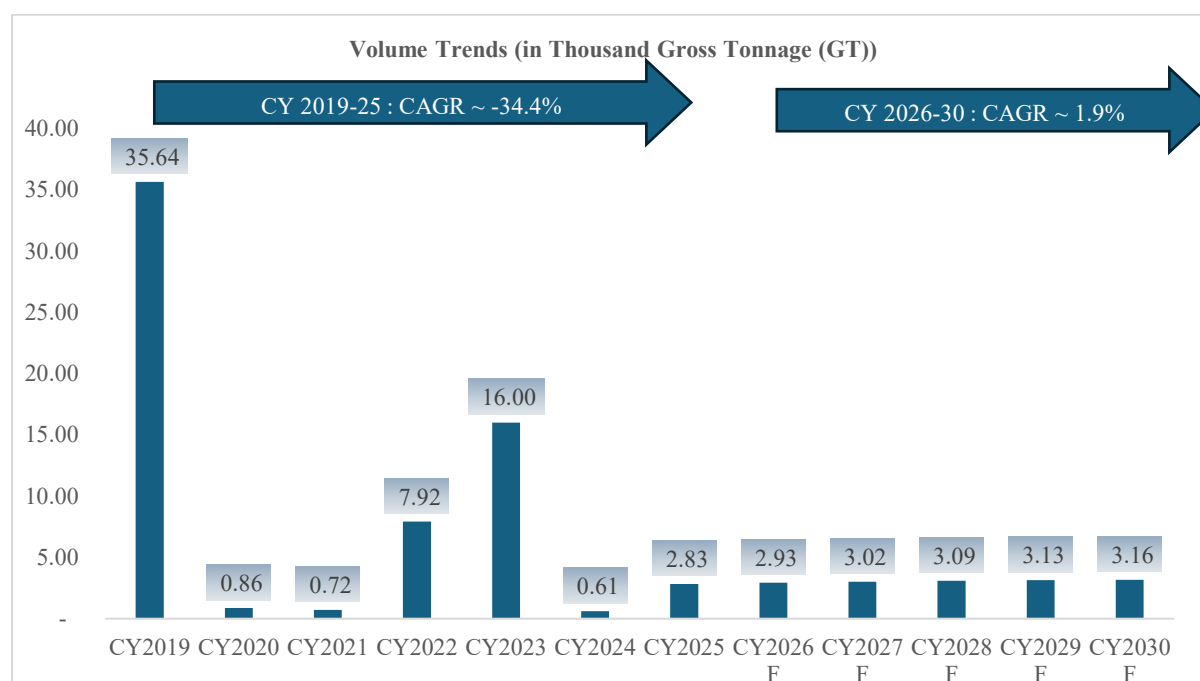
Looking ahead, the Turkey ship recycling market is expected to grow at a CAGR of 4.5% during CY2026F-CY2030F, reaching a volume of approximately 1,314 thousand GT by CY2030F. Growth is expected to be supported by increasing vessel retirement activity, rising demand for environmentally compliant recycling services, and ongoing fleet modernization efforts across Europe and neighboring maritime markets.

The Aliaga Ship Recycling Zone is anticipated to remain the key growth driver for the industry, benefiting from a growing supply of end-of-life vessels generated by fleet renewal initiatives and evolving environmental regulations. As shipowners increasingly replace older vessels with more fuel-efficient and environmentally sustainable alternatives, the volume of ships available for recycling is expected to rise, creating additional opportunities for the Turkish market.

Furthermore, Turkey's established regulatory compliance, advanced recycling capabilities, and continued investments in high-standard ship recycling infrastructure are expected to reinforce its position as a leading destination for responsible vessel dismantling. Supported by increasing emphasis on sustainability, circular economy practices, and resource recovery, the Turkish ship recycling market is expected to maintain steady growth over the forecast period.

3.4.2.2 Others

Chart: Others: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019–CY2025)**

The ship recycling market in the other countries of the Middle East & Africa region reached a volume of 2.8 Thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -34.4% during CY2019–CY2025. The market remained relatively small during the historical period and experienced a significant decline due to limited ship recycling infrastructure, lower vessel dismantling activity, and the absence of large-scale recycling facilities in many countries across the region.

Ship recycling activities in these markets are generally characterized by sporadic vessel disposals and a relatively small volume of end-of-life ships available for dismantling. Compared to established recycling hubs, the sector remains at an early stage of development, with recycling operations often constrained by limited technical capabilities, regulatory frameworks, and investment in dedicated recycling infrastructure.

- **Forecast Outlook (CY2026–CY2030)**

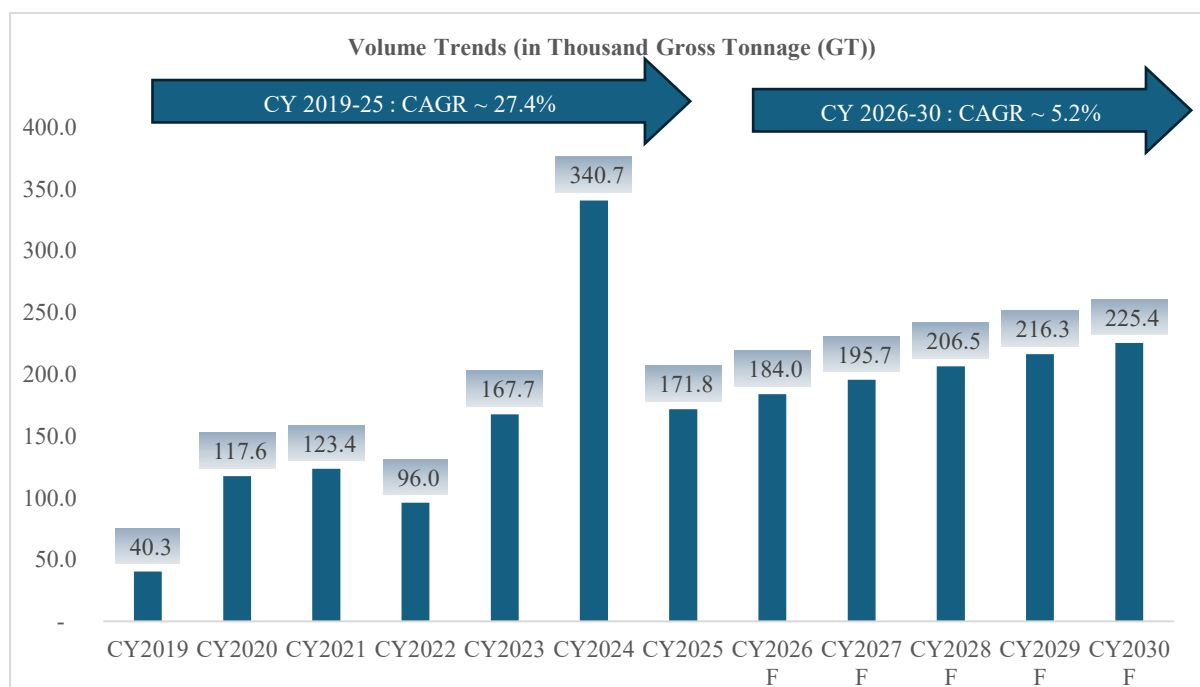
Looking ahead, the ship recycling market in the other countries of the Middle East & Africa region is expected to grow at a CAGR of 1.9% during CY2026F–CY2030F, reaching a volume of approximately 3.2 Thousand GT by CY2030F. Growth is expected to be supported by gradual improvements in maritime infrastructure, increasing awareness of sustainable vessel disposal practices, and rising attention toward resource recovery from decommissioned vessels.

As countries across the region continue to strengthen their maritime and environmental regulations, opportunities for the development of formal ship recycling activities are expected to emerge. In addition, growing industrialization and demand for recycled materials may encourage greater utilization of recycled steel and reusable marine equipment recovered from end-of-life vessels.

Although the market is expected to remain comparatively small relative to major regional recycling hubs, ongoing efforts to improve regulatory compliance, environmental standards, and recycling capabilities are likely to support modest growth in ship recycling activity across these countries over the forecast period.

3.4.3 Europe

Chart: Europe: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The Europe ship recycling market reached a volume of 171.8 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of 27.4% during CY2019-CY2025. The market experienced strong growth during the historical period, supported by the region's stringent regulatory environment, increasing focus on sustainable vessel disposal, and growing fleet renewal activity across the maritime sector.

Europe's ship recycling industry is primarily driven by the European Union Ship Recycling Regulation (EU SRR), which mandates that European-flagged vessels be recycled at approved facilities that adhere to high environmental, health, and safety standards. The implementation of these regulations has reinforced demand for certified and compliant recycling yards while promoting environmentally responsible ship dismantling practices across the region.

The market has also benefited from ongoing fleet modernization initiatives as shipping companies replace aging vessels with more fuel-efficient and lower-emission alternatives to meet decarbonization objectives, carbon intensity requirements, and evolving environmental regulations. This continuous renewal of the regional fleet has increased the number of vessels reaching the end of their operational lifecycle, thereby supporting ship recycling activity throughout the historical period.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the Europe ship recycling market is expected to grow at a CAGR of 5.2% during CY2026F-CY2030F, reaching a volume of approximately 225.4 thousand GT by CY2030F. Growth is expected to be

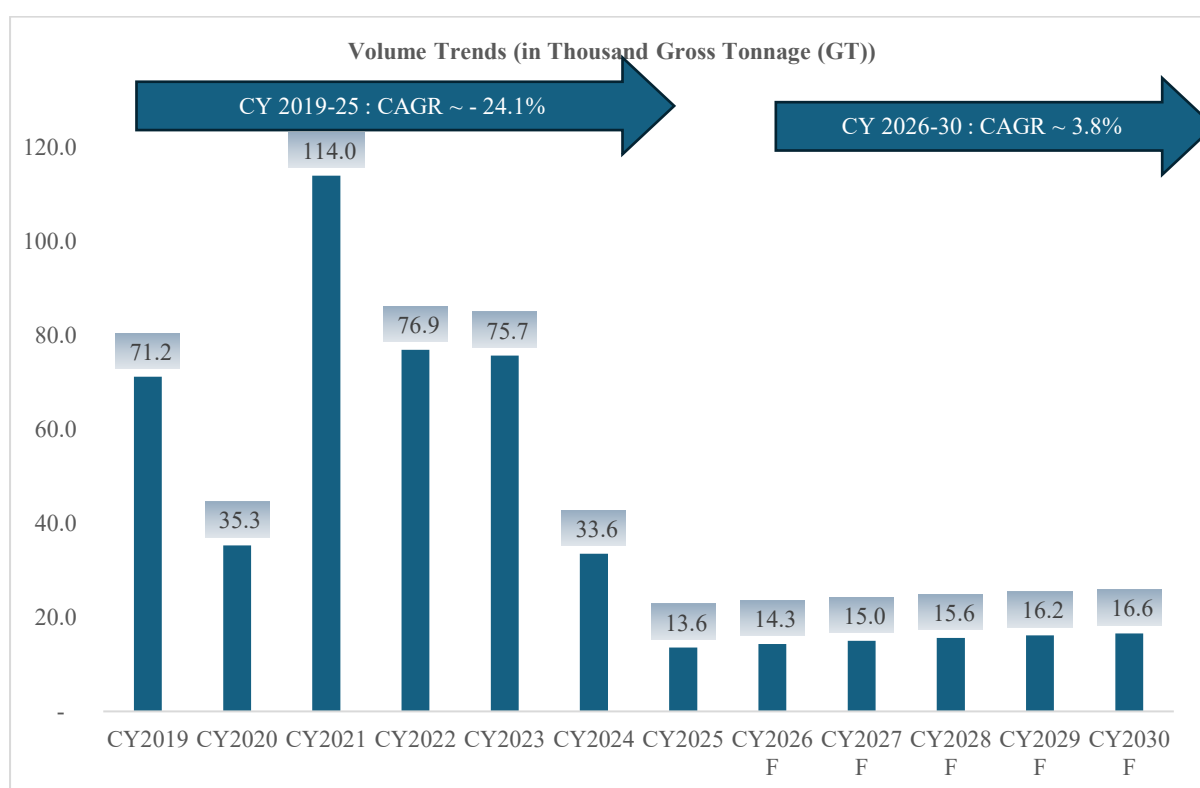
supported by continued regulatory compliance requirements, ongoing fleet replacement programs, and increasing emphasis on sustainable resource recovery from end-of-life vessels.

Europe's strong commitment to sustainability and circular economy principles is expected to remain a key driver of market growth. The recovery and reuse of steel, non-ferrous metals, and other valuable materials from decommissioned vessels contribute to resource efficiency, waste reduction, and lower environmental impact, further strengthening the role of ship recycling within the region's broader sustainability agenda.

In addition, European shipowners are increasingly collaborating with internationally compliant recycling facilities, both within Europe and in approved overseas destinations, to ensure adherence to regulatory and environmental standards. This growing preference for responsible ship recycling practices is expected to encourage further investment in certified recycling infrastructure and support demand for HKC- and EU SRR-compliant ship recycling facilities, contributing to the long-term growth of the European ship recycling market.

3.4.4 North America

Chart: North America: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The North America ship recycling market reached a volume of 13.6 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -24.1% during CY2019–CY2025. The market experienced a decline during the historical period, influenced by fluctuations in vessel retirement activity and the timing of fleet replacement programs across the commercial, offshore, and government maritime sectors.

The region's ship recycling industry is primarily supported by the retirement of aging commercial vessels, offshore support vessels (OSVs), naval ships, and government-owned fleets that have reached the end of their operational life. The ongoing replacement of older vessels with newer, fuel-efficient alternatives has continued to generate a supply of end-of-life ships for recycling activities across the region.

The market is also characterized by a strong regulatory environment governing hazardous waste management, pollution control, and worker safety. Strict environmental and occupational health requirements have encouraged the use of certified ship recycling facilities that comply with high industry standards. As a result, responsible and

environmentally sound ship recycling practices have become an important feature of the North American ship recycling landscape.

- **Forecast Outlook (CY2026F-CY2030F)**

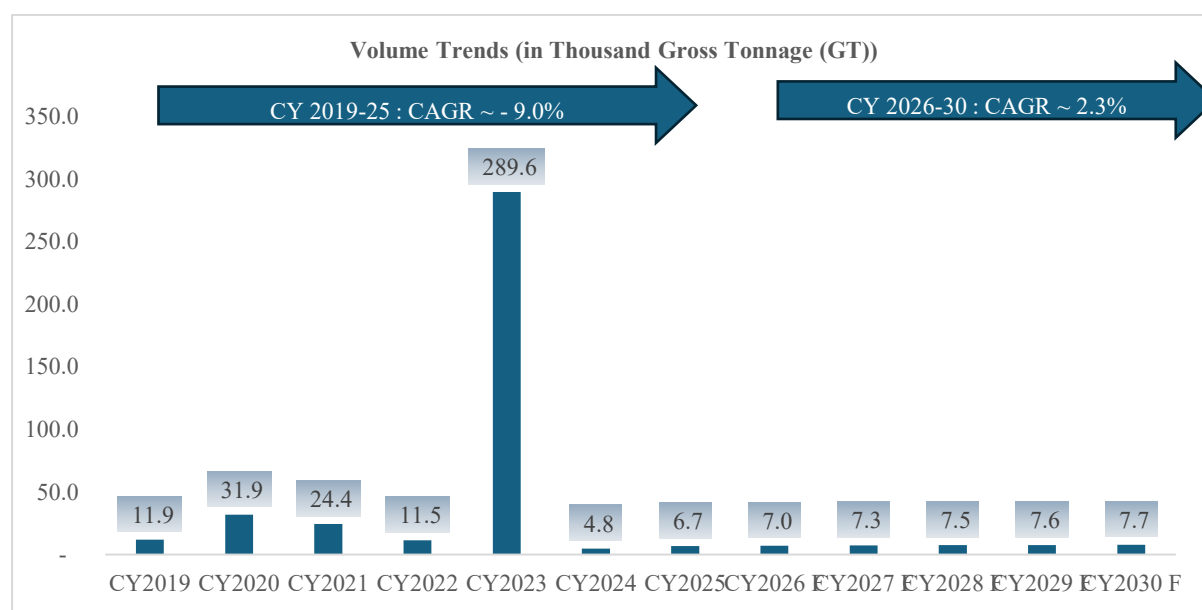
Looking ahead, the North America ship recycling market is expected to grow at a CAGR of 3.8% during CY2026F-CY2030F, reaching a volume of approximately 16.6 thousand GT by CY2030F. Market growth is expected to be supported by increasing vessel retirement volumes, ongoing fleet modernization initiatives, and rising focus on sustainable resource recovery.

Fleet modernization across the commercial shipping, offshore energy, and defense sectors is anticipated to remain a key growth driver. Investments in technologically advanced vessels with improved fuel efficiency, enhanced digital capabilities, and lower emissions are accelerating the retirement of obsolete ships, creating additional opportunities for the ship recycling industry.

Furthermore, growing emphasis on resource recovery and circular economy initiatives is expected to support market expansion over the forecast period. Increasing demand for recycled steel, non-ferrous metals, and reusable marine equipment recovered from end-of-life vessels, coupled with broader sustainability and decarbonization objectives, is likely to encourage continued investment in modern ship recycling infrastructure and advanced dismantling technologies across North America.

3.4.5 Latin America

Chart: Latin America: Ship Recycling Market: Volume Trends (in Thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Latin America ship recycling market reached a volume of 6.7 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -9.0% during CY2019-CY2025. The market experienced a decline during the historical period, reflecting variations in vessel retirement activity and the relatively gradual development of ship recycling capabilities across the region.

The market is primarily supported by the retirement of aging commercial vessels, offshore support vessels (OSVs), fishing fleets, and coastal ships operating across regional maritime routes. Rising maintenance and operating costs, combined with ongoing fleet modernization initiatives, have encouraged shipowners and operators to recycle older vessels and replace them with more efficient alternatives.

In addition, increasing investments in maritime trade, port expansion, offshore oil and gas activities, and coastal infrastructure development have supported fleet renewal across several countries in the region. As operators

continue to upgrade their fleets with newer and more technologically advanced vessels, the supply of end-of-life ships available for recycling has gradually increased, supporting market activity during the historical period.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the Latin America ship recycling market is expected to grow at a CAGR of 2.3% during CY2026F-CY2030F, reaching a volume of approximately 7.7 thousand GT by CY2030F. Growth is expected to be driven by increasing vessel retirement volumes, continued fleet modernization, and the gradual development of formal ship recycling practices across the region.

Countries across Latin America are strengthening environmental regulations and aligning ship recycling activities with international standards to improve worker safety, hazardous waste management, and environmental protection. These efforts are expected to support the gradual formalization of the regional ship recycling industry and encourage greater adoption of environmentally responsible recycling practices.

Furthermore, rising demand for recycled steel and other recovered materials from the construction, manufacturing, and infrastructure sectors is expected to support market growth. Combined with the increasing adoption of circular economy principles and sustainable resource utilization practices, these factors are likely to encourage further investment in responsible ship recycling operations and resource recovery initiatives across Latin America over the forecast period.

3.5 Growth Drivers of the Global Ship Recycling Industry

- **Aging Global Fleet and Increasing Vessel Retirements:** One of the most significant growth drivers of the global ship recycling industry is the aging profile of the world's commercial fleet. A large number of bulk carriers, oil tankers, container ships, general cargo vessels, offshore support vessels (OSVs), and specialized ships are approaching or have exceeded their economic service life. As vessels age, they experience declining operational efficiency, higher fuel consumption, increased maintenance requirements and greater downtime.

Shipowners often face a critical decision between investing substantial capital in vessel upgrades and regulatory compliance or retiring the vessel altogether. In many cases, recycling becomes the more economically attractive option, particularly when maintenance expenses begin to outweigh the revenue generated by the vessel. Consequently, the continuous retirement of aging fleets generates a steady flow of end-of-life vessels for recycling yards worldwide.

- **Maritime Decarbonization and Environmental Compliance:** The shipping industry is undergoing one of the largest transitions in its history as regulators push for lower carbon emissions and greener shipping operations. The International Maritime Organization (IMO) has introduced various measures targeting greenhouse gas (GHG) emissions, carbon intensity reduction, and improved energy efficiency.

To comply with these requirements, shipowners are increasingly replacing older vessels with modern ships equipped with:

- Fuel-efficient engines
- Alternative fuel technologies
- Energy-saving devices
- Advanced digital navigation and monitoring systems
- Lower-emission propulsion systems

Many older vessels cannot economically meet these standards without extensive retrofitting. As a result, fleet renewal programs are accelerating vessel retirements and driving additional ship recycling demand globally.

- **Fleet Modernization Across Commercial Shipping:** The global shipping industry constantly seeks to improve operational efficiency, cargo capacity, and fuel economy. New-generation vessels offer significant advantages over older ships, including:
 - Lower operating costs
 - Higher cargo carrying capacity

- Reduced fuel consumption
- Improved reliability
- Enhanced safety features
- Better environmental performance

Shipping companies are therefore replacing aging vessels with technologically advanced fleets. This trend is particularly evident in container shipping, tanker transportation, LNG carriers, and dry bulk shipping, where operational efficiency directly impacts profitability.

As fleet modernization continues across global shipping markets, recycling volumes are expected to increase substantially.

- **Rising Demand for Recycled Steel and Scrap Materials:** Ship recycling is an important source of secondary raw materials for numerous industries. A dismantled vessel can provide:
 - Recycled steel
 - Non-ferrous metals (copper, aluminium, brass)
 - Machinery and equipment
 - Electrical components
 - Marine fixtures
 - Reusable spare parts

The construction, manufacturing, engineering, and infrastructure sectors increasingly rely on recycled steel due to its cost advantages and lower environmental impact compared to virgin steel production.

In countries such as India, Bangladesh, Pakistan, and Turkey, recycled ship steel plays a vital role in meeting domestic construction and industrial demand. Growing infrastructure investment worldwide is expected to further support demand for recycled materials recovered from ships.

- **Favorable Scrap Steel Prices**
The profitability of ship recycling is heavily influenced by steel scrap prices. When scrap steel values rise, recycling yards are willing to offer higher prices for end-of-life vessels.

This creates a favorable environment for shipowners because:

- Vessel recycling becomes more financially attractive.
- Cash buyers increase acquisition activity.
- Recycling yards expand purchasing volumes.

Periods of strong steel demand and elevated commodity prices therefore tend to stimulate ship recycling activity globally.

3.6 Key Challenges of the Global Ship Recycling Industry

- **Stringent Environmental and Regulatory Compliance Requirements:** One of the most significant challenges facing the global ship recycling industry is compliance with increasingly stringent environmental regulations. International frameworks such as the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC) and the European Union Ship Recycling Regulation (EU SRR) require recycling facilities to adhere to strict standards related to hazardous waste management, environmental protection, worker safety, and operational transparency.

Compliance often requires substantial investments in:

- Impermeable flooring systems
- Hazardous waste treatment facilities
- Pollution control equipment
- Monitoring and reporting systems
- Specialized worker training programs

For many ship recycling yards, particularly in developing countries, meeting these requirements can significantly increase operational costs and reduce competitiveness.

- **Hazardous Materials Management:** Ships contain various hazardous substances accumulated during decades of operation, including:
 - Asbestos
 - Polychlorinated Biphenyls (PCBs)
 - Lead-based paints
 - Heavy metals
 - Residual fuel oils
 - Oily sludge and contaminated water

Safe identification, handling, storage, transportation, and disposal of these materials remain major operational challenges. Improper management can lead to environmental contamination, worker health risks, financial penalties, and reputational damage. As environmental standards continue to tighten, the costs and complexity associated with hazardous waste management are expected to increase.

- **Worker Safety and Occupational Health Risks:** Ship recycling remains one of the most labour-intensive industrial activities globally. Workers are often exposed to:
 - Dangerous cutting operations
 - Falling steel structures
 - Explosions from trapped gases
 - Exposure to hazardous chemicals
 - Heavy machinery accidents
 - Confined-space hazards

Although many recycling facilities have improved safety standards in recent years, maintaining consistent occupational health and safety practices remains an ongoing challenge. Compliance with international labor and safety standards often requires continuous investments in equipment, training, and safety management systems.

- **Cyclical Nature of Shipping Markets:** The supply of vessels available for recycling is heavily influenced by global shipping cycles.

During periods of:

- Strong freight rates
- High vessel earnings
- Robust trade activity

Shipowners often keep older vessels in operation longer than anticipated. Conversely, during shipping downturns, vessel retirements increase sharply. This cyclical nature creates uncertainty for recycling facilities, making capacity planning and investment decisions more difficult.

- **High Capital Requirements for Facility Upgrades:** As environmental and safety requirements become more demanding, recycling facilities must continuously invest in infrastructure upgrades.

Key investments include:

- Advanced dismantling equipment
- Waste collection systems
- Water treatment facilities
- Hazardous material containment areas
- Emergency response systems
- Worker welfare and training facilities

Many smaller operators struggle to secure sufficient financing for these upgrades, potentially limiting their ability to compete with larger and better-capitalized recycling yards.

4. Indian Ship Recycling Industry

4.1 Market Overview

Based on ship recycling volumes during 2025, India emerged as the world's leading ship recycling nation, achieving the Maritime India Vision (MIV) 2030 objective of becoming a global leader in ship recycling ahead of schedule. The industry plays an important role in India's circular economy by recovering steel and other reusable materials from end-of-life vessels, supplying secondary raw materials to downstream industries, generating employment, and promoting resource efficiency. India's growth in ship recycling has been supported by policy reforms, modernization of recycling infrastructure, adoption of international environmental and safety standards, and continued efforts to strengthen the country's maritime ecosystem.

The regulatory framework governing ship recycling in India has evolved significantly following the enactment of the Recycling of Ships Act, 2019, which provides the legal framework for the safe and environmentally sound recycling of ships and aligns India's ship recycling regime with the Hong Kong International Convention (HKC). Investments in compliant recycling facilities, environmental management systems, hazardous waste management infrastructure, and occupational health and safety measures have enhanced the industry's operational standards and strengthened India's global competitiveness in ship recycling.

The Alang-Sosiya Ship Recycling Yard in Gujarat forms the backbone of India's ship recycling industry and is recognized as the world's largest ship recycling cluster. The cluster accounts for approximately 97% of India's ship recycling activity and has an annual recycling capacity of around 6 million gross tonnes (GT). It is supported by an integrated ecosystem comprising authorized recycling yards, steel re-rolling mills, scrap processing units, machinery and equipment resale businesses, hazardous waste treatment and disposal facilities, logistics service providers, oxygen plants, and other ancillary industries. Continuous modernization through mechanized dismantling equipment, engineered recycling plots, impermeable flooring, drainage systems, and improved occupational health and safety practices has enhanced operational efficiency, environmental performance, and compliance with applicable regulatory requirements.

In addition to Alang-Sosiya, ship recycling activities are undertaken at Amar Iron Udyog in Kolkata and Steel Industries Kerala Limited (SILK) in Kerala, although these facilities account for a relatively small share of India's overall ship recycling activity. According to the Ministry of Ports, Shipping and Waterways, a substantial number of Indian ship recycling yards have achieved compliance with the Hong Kong International Convention (HKC), reflecting the industry's progress towards internationally recognized environmental and safety standards.

The long-term outlook for India's ship recycling industry is supported by an aging global fleet, ongoing fleet renewal and modernization, implementation of international environmental regulations, increasing emphasis on environmentally sound ship recycling practices, and continued domestic demand for recycled steel. These factors are expected to support demand for compliant ship recycling facilities over the medium to long term, although industry activity may continue to be influenced by global shipping market conditions, freight rates, scrap steel prices, and vessel availability.

Table: Ship Recycling Facilities and HKC Compliance in India

Ship Recycling Yard	Total Plots	Operational Plots	HKC-Compliant Yards
Alang, Bhavnagar (Gujarat)	150	128	115
Steel Industries Kerala Limited (SILK), Kerala	1	1	0
Amar Iron Udyog, Kolkata (West Bengal)	1	1	0

Source: Ministry of Ports, Shipping and Waterways.

The India ship recycling market reached a volume of 2,998 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -1.6% during CY2019-CY2025. Despite the moderate contraction witnessed over the historical period, the industry continues to be supported by a combination of favorable policy initiatives, increasing global vessel retirements, robust domestic demand for recycled steel, ongoing capacity enhancements, and continuous modernization of ship recycling infrastructure.

One of the most significant developments supporting the industry's growth has been the strengthening of the regulatory framework governing ship recycling. India aligned its domestic regulations with international best practices through the Recycling of Ships Act, 2019, along with subsequent rules and regulations, enabling ship

recycling facilities to comply with the requirements of the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC). The formal entry into force of the HKC on 26 June 2025 has further strengthened India's competitive position, as many Indian recycling yards had already achieved compliance ahead of several competing ship recycling destinations. This early adoption has enhanced global confidence in India's ability to offer environmentally sound and safe recycling services.

At the same time, the global shipping industry is experiencing an increase in vessel retirements driven by stricter environmental regulations, decarbonization targets, rising operating and maintenance costs, and fleet modernization programs. These factors are generating a steady pipeline of end-of-life vessels available for recycling. India is well positioned to capitalize on this opportunity, supported by the presence of the Alang-Sosiya Ship Recycling Yard, the world's largest ship recycling cluster and the center of India's ship recycling ecosystem.

To accommodate future demand, the Government of India and industry stakeholders are pursuing initiatives aimed at expanding and upgrading recycling capacity, particularly at Alang. These efforts include the development of additional HKC-compliant recycling plots, mechanization of dismantling operations, improvements in environmental infrastructure, strengthening hazardous waste management systems, and upgrading occupational health and worker safety standards. Furthermore, increased deployment of digital compliance and monitoring systems is helping improve transparency, operational efficiency, and regulatory oversight across the sector.

India's ship recycling industry also benefits from strong domestic demand for recycled steel from the construction, infrastructure, engineering, automotive, and manufacturing sectors. Steel recovered from dismantled vessels serves as an important secondary raw material, reducing dependence on imported scrap and virgin raw materials while lowering energy consumption and supporting India's circular economy objectives. The sector is also aligned with the country's broader sustainability agenda, including efforts to promote low-carbon steel production and resource efficiency. In addition, government support measures such as the Ship Recycling Credit Note Scheme under the Shipbuilding Financial Assistance Policy (SBFA) 2.0 are expected to further encourage the use of domestic ship recycling facilities and strengthen the overall maritime ecosystem.

Despite its favorable long-term outlook, the industry faces several challenges that may constrain growth. Ship recycling activity remains closely linked to global shipping market cycles. Periods of high freight rates often incentivize shipowners to continue operating older vessels for longer durations, thereby reducing the number of ships available for recycling. Industry profitability is also exposed to foreign exchange risks, as vessels are typically acquired in US dollars while the sale of recycled materials occurs largely in Indian rupees. Consequently, fluctuations in currency exchange rates can materially impact margins.

Additionally, the widespread use of usance Letters of Credit (LCs) creates delays in payment realization, resulting in liquidity and working capital pressures for recycling businesses. The industry also faces rising compliance costs associated with maintaining HKC-compliant facilities, including investments in environmental safeguards, worker welfare measures, hazardous waste treatment systems, and infrastructure upgrades. While these investments improve sustainability and international competitiveness, they also increase operating costs and may affect profitability in periods of weaker market conditions.

Overall, the India ship recycling market is projected to grow at a CAGR of 8.4% during CY2026F-CY2030F, reaching a volume of approximately 4,572 thousand Gross Tonnage (GT) by CY2030. Growth over the forecast period is expected to be supported by increasing global vessel retirements, growing preference for environmentally compliant recycling facilities, continued infrastructure modernization, strong domestic demand for recycled steel, and India's established position as a leading global hub for safe and sustainable ship recycling.

Table: India: Ship Recycling Market: Key Industry Highlights, CY2019 and CY2025

Particulars	CY 2019	CY 2025	CAGR (CY2026-CY2030)
India Ship Recycling Market: Volume Trends (in thousand Gross Tonnage (GT))	3,303.00	2,998.00	-1.60%

Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

Table: India: Ship Recycling Market Forecast: Key Industry Highlights, CY2026 and CY2030

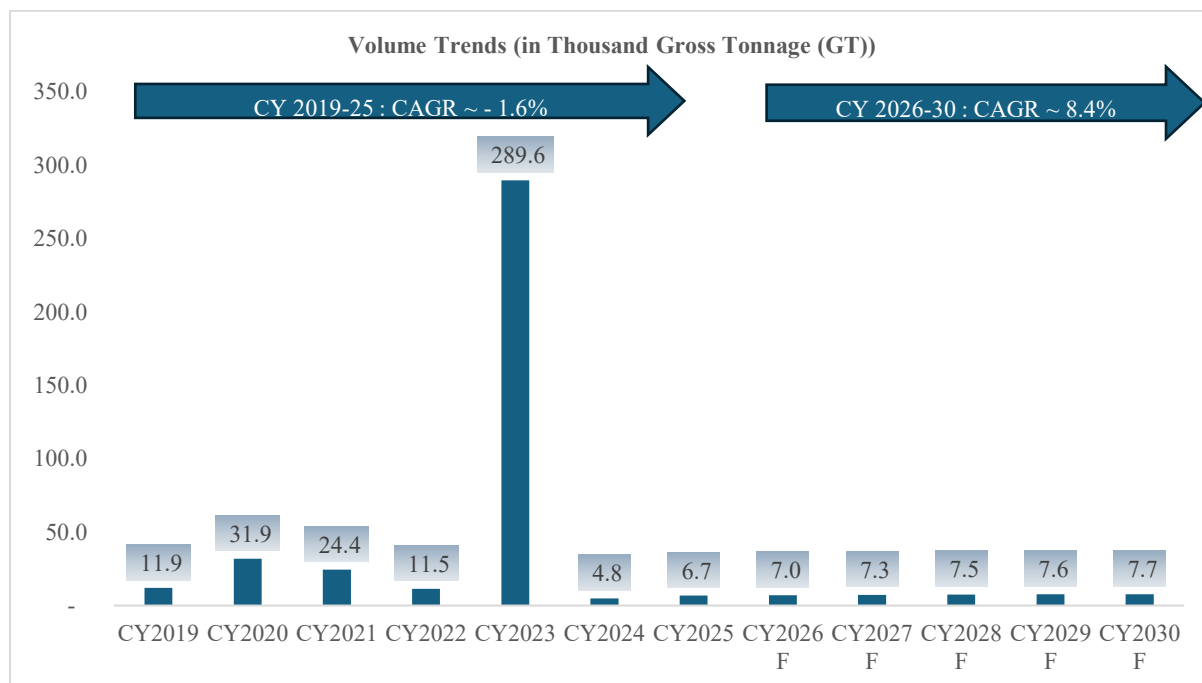
Particulars	CY 2026	CY 2030	CAGR (CY2026-CY2030)
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India Ship Recycling Market: Volume Trends (in thousand Gross Tonnage (GT))	3,315.0 0	4,572.0 0	8.40%
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Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

4.2 Market Performance & Forecast

Chart: India: Ship Recycling Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: UNCTAD (UN Trade and Development), Expert Interactions, IMARC, ICRA Analytics

• Historical Performance (CY2019-CY2025)

The India ship recycling market reached a volume of 2,998 thousand Gross Tonnage (GT) in CY2025, registering a CAGR of -1.6% during CY2019-CY2025. Market performance during the period was influenced by fluctuations in global shipping activity, freight market conditions, and variability in the availability of end-of-life vessels for recycling. Delayed scrapping decisions by shipowners, supported by periods of favorable shipping rates and vessel demand, also affected recycling volumes across the country.

Despite the modest decline, India maintained its position as one of the world's leading ship recycling destinations, supported by its well-established recycling infrastructure and competitive operating environment. A key factor strengthening the market was the implementation of the Hong Kong International Convention (HKC) and India's regulatory framework under the Recycling of Ships Act, 2019, which enhanced environmental compliance, worker safety standards, and transparency across recycling operations. Early alignment with internationally recognized regulations improved the confidence of global shipowners and strengthened India's reputation as a responsible ship recycling hub.

The market also benefited from the gradual retirement of aging vessels as shipping companies modernized their fleets to comply with increasingly stringent environmental regulations and emission reduction requirements. Furthermore, ongoing investments in upgrading and expanding HKC-compliant recycling facilities, particularly at Alang, improved operational efficiency and positioned India to capture a larger share of global ship recycling activity.

• Forecast Outlook (CY2026F-CY2030F)

Looking ahead, the India ship recycling market is projected to grow at a CAGR of 8.4% during CY2026F-CY2030F, reaching approximately 4,572 thousand Gross Tonnage (GT) by CY2030F. Growth is expected to be driven by rising vessel retirement rates, supportive regulatory developments, and continued investments in modern recycling infrastructure.

A major growth catalyst is the increasing enforcement of international environmental standards, including the Hong Kong International Convention (HKC), which is encouraging shipowners to recycle vessels at compliant facilities that meet globally accepted environmental, health, and safety requirements. At the same time, the shipping industry's decarbonization initiatives and transition toward more fuel-efficient vessels are expected to accelerate fleet replacement cycles, generating a larger supply of end-of-life ships for recycling.

The market is also anticipated to benefit from ongoing modernization and capacity expansion of ship recycling facilities across the country. India is actively pursuing the inclusion of its recycling yards on the European Union's approved list of ship recycling facilities, while plans to expand Alang's recycling capacity to approximately 9 million light displacement tonnes (LDT) are expected to further strengthen the country's position in the global market. These developments are likely to enhance India's attractiveness as a preferred destination for environmentally responsible ship recycling.

In addition, India's mature ship recycling ecosystem is expected to provide a strong foundation for future growth. The country benefits from extensive recycling infrastructure, a well-developed steel re-rolling and scrap processing industry, and competitive labor and operational costs. Government initiatives such as the Ship Recycling Credit Note Scheme under the Shipbuilding Financial Assistance Policy (SBFA) 2.0 are further supporting environmentally compliant recycling practices while promoting greater integration between the domestic shipbuilding and ship recycling sectors. Combined with sustained demand for recycled steel and increasing emphasis on circular economy principles, these factors are expected to drive long-term growth in India's ship recycling market.

4.3 Green Ship Recycling in India

Green ship recycling in India refers to the safe and environmentally responsible dismantling of end-of-life vessels in accordance with the Recycling of Ships Act, 2019, the Hong Kong International Convention (HKC), and other relevant national and international regulations. The process focuses on environmentally sound recycling practices, worker health and safety, effective management of hazardous materials, pollution prevention, and the maximization of resource recovery through a comprehensive regulatory and compliance framework.

India has established itself as a leading global hub for green ship recycling, with the Alang-Sosiya Ship Recycling Yard (ASSRY) in Gujarat serving as the country's principal recycling center. Over the past decade, recycling facilities have invested significantly in infrastructure modernization, including the installation of impermeable flooring, advanced drainage networks, dedicated hazardous waste storage facilities, mechanized cargo and material handling systems, emergency response infrastructure, and enhanced occupational safety measures. These developments have enabled Indian ship recycling yards to evolve from traditional ship-breaking operations into facilities that comply with internationally accepted environmental and safety standards.

A significant milestone for the industry was the entry into force of the Hong Kong International Convention on 26 June 2025, which established mandatory global regulations for safe and environmentally sound ship recycling. India had proactively aligned its domestic regulatory framework with HKC requirements through the implementation of the Recycling of Ships Act, 2019 and the corresponding Rules, enabling the country to achieve substantial compliance ahead of the Convention's enforcement. Currently, approximately 115 operational recycling yards at Alang have obtained HKC Statements of Compliance (SoC) issued by recognized classification societies, while the remaining facilities continue to upgrade their infrastructure and operational practices to meet international requirements.

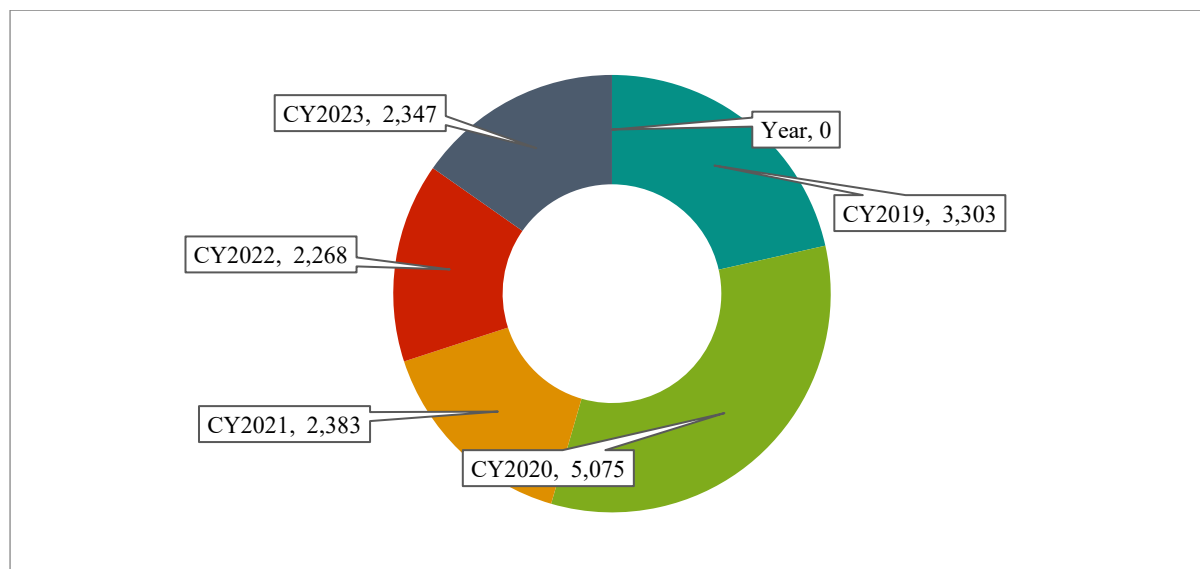
Green ship recycling plays an important role in supporting India's broader sustainability and circular economy goals. The industry facilitates the recovery of high-quality steel and reusable equipment from end-of-life vessels, reducing dependence on virgin raw materials, lowering the carbon footprint associated with primary steel production, and promoting responsible waste management practices. By extending material lifecycles and improving resource efficiency, ship recycling contributes to sustainable industrial development and environmental conservation.

Simultaneously, the Government of India is accelerating the modernization of ship recycling infrastructure by encouraging facilities to meet stringent international environmental standards and pursuing the inclusion of Indian recycling yards in the European Union List of Approved Ship Recycling Facilities. These efforts are expected to enhance India's attractiveness among international shipowners, stimulate private-sector investment, strengthen the

maritime ecosystem, and further consolidate the country's position as a global leader in safe, sustainable, and environmentally compliant ship recycling.

4.4 Indian Ship Recycling Market: Breakup by Type (CY2019-CY2030F)

Chart: India: Ship Recycling Market: Breakup by Type (in %), CY2025

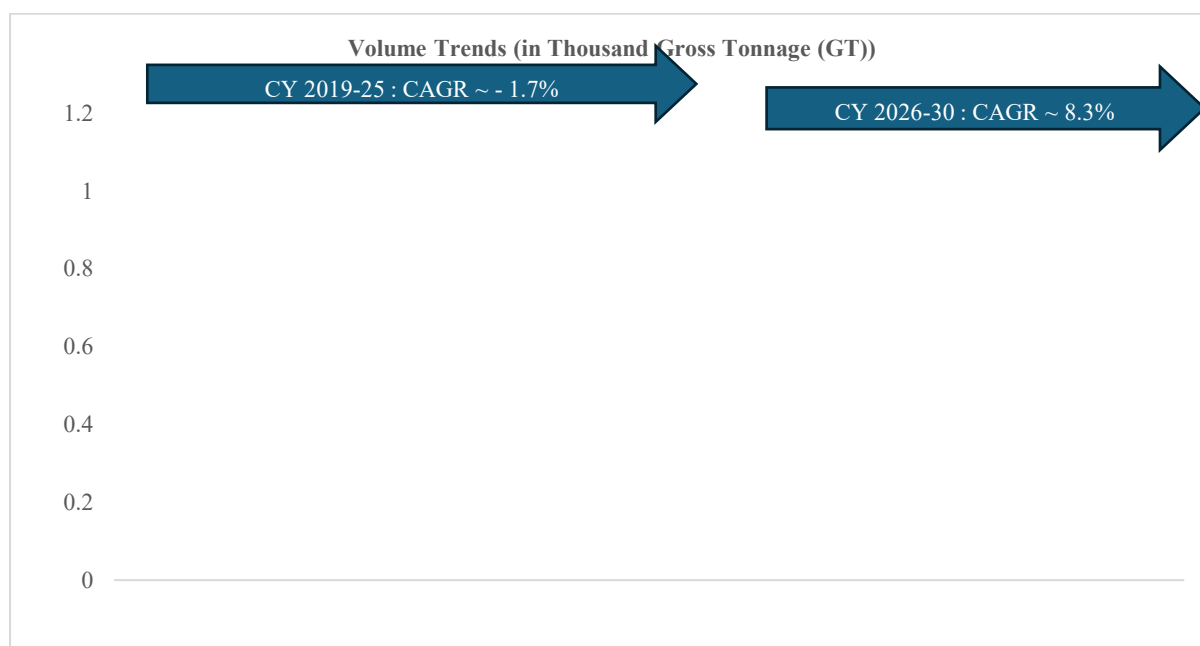


Source: Expert Interactions, IMARC, ICRA Analytics

In CY2025, container ships emerged as the largest vessel category in India's ship recycling market, accounting for 32.4% of the total market volume. The segment was followed by oil tankers, which contributed 20.8% of the market, and bulk carriers with a share of 13.8%. General cargo vessels accounted for 10.2%, while Ro-Ro (Roll-on/Roll-off) vessels represented 4.4% of the market. The remaining 18.4% was comprised of other vessel types.

4.4.1 Container Ships

Chart: Global: Ship Recycling (Container Ships) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The container ships segment accounted for 32.4% of India's ship recycling market in CY2025, reaching a volume of 972 thousand Gross Tonnage (GT). The segment recorded a CAGR of -1.7% during CY2019-CY2025, reflecting fluctuations in global container shipping activity, vessel utilization rates, and scrapping decisions by shipowners.

The recycling of container ships in India has been primarily supported by ongoing global fleet modernization initiatives, whereby older feeder and mid-sized container vessels are being replaced with larger, more fuel-efficient, and higher-capacity ships operating across major international trade routes. As these aging vessels reach the end of their economic life, a significant number are directed to Indian recycling yards for dismantling and resource recovery.

In addition, increasingly stringent environmental regulations, carbon intensity requirements, and operational efficiency standards are encouraging shipping companies to retire older container vessels. The substantial capital expenditure required to retrofit legacy vessels and ensure compliance with evolving emission norms has further accelerated their retirement, thereby supporting recycling activity within this segment.

- **Forecast Outlook (CY2026F-CY2030F)**

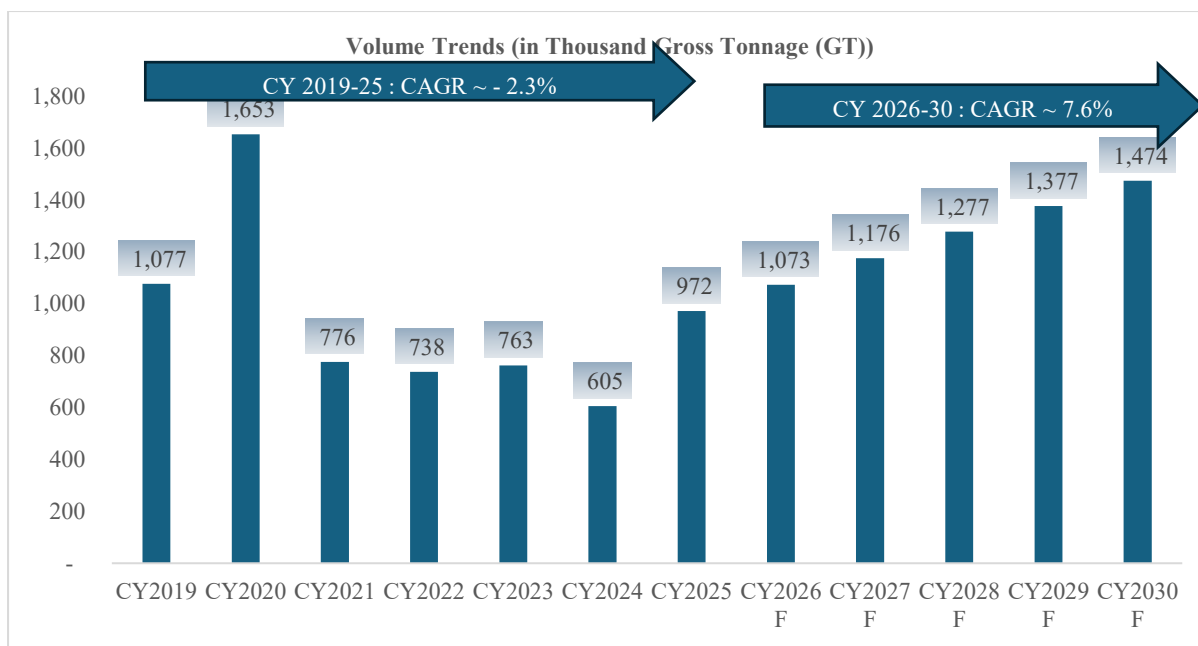
Looking ahead, the container ships segment is expected to reach a volume of 1,474 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 8.3% during CY2026F-CY2030F. Growth is anticipated to be driven by continued fleet renewal programs, evolving global trade dynamics, and increasing regulatory pressure on vessel efficiency and environmental performance.

India's strategic location along major East–West maritime trade corridors enhance its accessibility for vessel recycling activities. The country's proximity to key international shipping routes facilitates the efficient transfer of end-of-life container vessels to Indian recycling facilities, reinforcing the position of Alang as one of the world's leading ship recycling destinations.

The segment is also expected to benefit from continuous investments in HKC-compliant infrastructure, mechanized dismantling systems, and advanced hazardous material management practices. These improvements are enhancing India's capability to safely and efficiently recycle technologically advanced container vessels while maintaining compliance with internationally recognized environmental, health, and safety standards.

4.4.2 Oil Tankers

Chart: Global: Ship Recycling (Oil Tankers) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The oil tankers segment accounted for 20.8% of India's ship recycling market in CY2025, reaching a volume of 624 thousand Gross Tonnage (GT). The segment recorded a CAGR of -2.3% during CY2019-CY2025, reflecting fluctuations in global tanker markets, vessel utilization rates, and the timing of scrapping decisions by shipowners.

The recycling of oil tankers in India has been strongly supported by the presence of the Alang-Sosiya Ship Recycling Yard, which possesses the infrastructure, skilled workforce, and downstream steel processing ecosystem required to dismantle large crude oil and product tankers efficiently. India's strategic location along major international shipping routes further enhances its attractiveness as a preferred destination for the recycling of end-of-life tanker vessels.

In addition, increasingly stringent international regulations related to emissions control, ballast water management, energy efficiency, and decarbonization have significantly increased the operating and compliance costs associated with older tanker fleets. As a result, many shipowners have opted to recycle aging vessels rather than undertake costly retrofitting programs or invest in extending their operational lifespan.

- Forecast Outlook (CY2026F–CY2030F)**

Looking ahead, the oil tankers segment is projected to reach a volume of 917 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 7.6% during CY2026F–CY2030F. Growth is expected to be driven by increasing tanker retirements, tightening environmental regulations, and continued investments in India's ship recycling infrastructure.

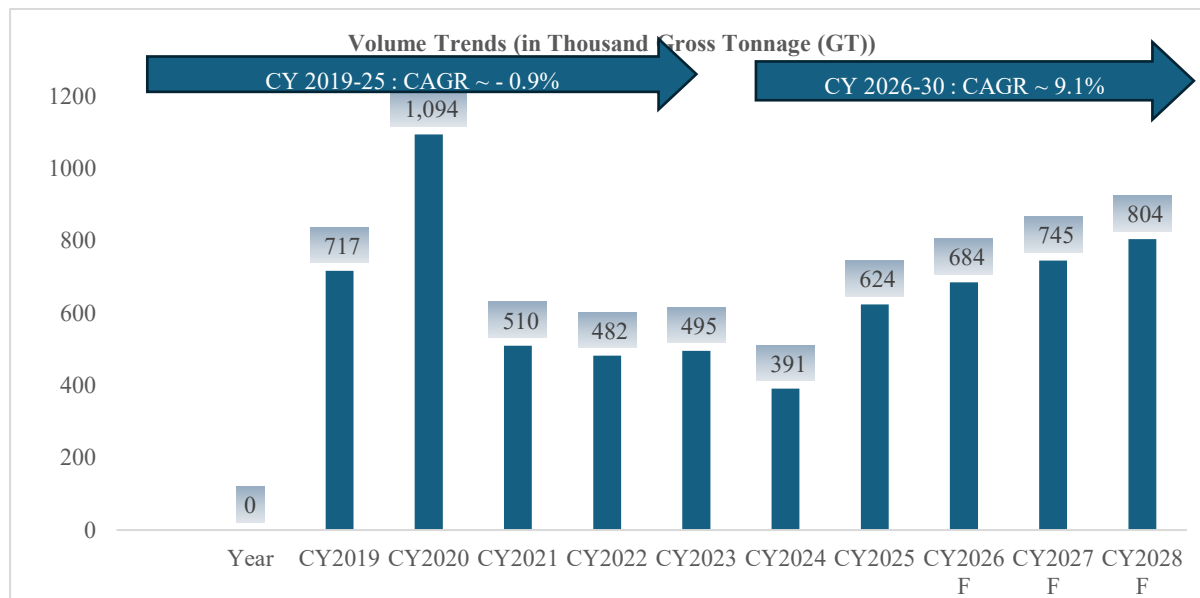
India's position as one of the world's largest crude oil importers ensures substantial tanker traffic along its western coastline, particularly around major ports and offshore energy terminals. This concentration of tanker activity improves accessibility for vessel recycling and supports a steady flow of aging tankers from global shipping markets into Indian recycling facilities.

The segment is also expected to benefit from sustained demand for high-quality recycled steel from India's construction, infrastructure, manufacturing, and engineering sectors. Steel recovered from tanker vessels serves as a valuable secondary raw material, supporting resource efficiency and circular economy objectives. Furthermore, ongoing upgrades to HKC-compliant recycling infrastructure, mechanized dismantling systems, and

hazardous material management capabilities are enhancing material recovery rates, operational efficiency, and environmental performance, thereby strengthening the economic attractiveness of oil tanker recycling in India..

4.4.3 Bulk Carriers

Chart: Global: Ship Recycling (Bulk Carriers) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The bulk carriers segment accounted for 13.8% of India's ship recycling market in CY2025, reaching a volume of 414 thousand Gross Tonnage (GT). The segment recorded a CAGR of -0.9% during CY2019-CY2025, reflecting fluctuations in global dry bulk trade activity, freight market conditions, and vessel retirement patterns.

The recycling of bulk carriers in India has been supported by the large global fleet of aging vessels engaged in the transportation of iron ore, coal, fertilizers, grains, and other dry bulk commodities. As these vessels experience declining operational efficiency and rising maintenance requirements with age, many are directed to Indian recycling facilities, particularly the Alang-Sosiya Ship Recycling Yard, upon reaching the end of their commercial lifespan.

In addition, volatility in global dry bulk freight markets has played an important role in influencing scrapping decisions. Periods of weak freight rates, lower vessel earnings, and subdued trade activity often encourage shipowners to retire older bulk carriers rather than incur significant expenditures on maintenance, repairs, and regulatory compliance upgrades. These factors have continued to support recycling activity within the segment despite cyclical fluctuations in global shipping markets.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the bulk carriers segment is projected to reach a volume of 654 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 9.1% during CY2026F-CY2030F. Growth is expected to be driven by increasing retirement of aging vessels, continued fleet modernization, and sustained demand for recycled steel across India.

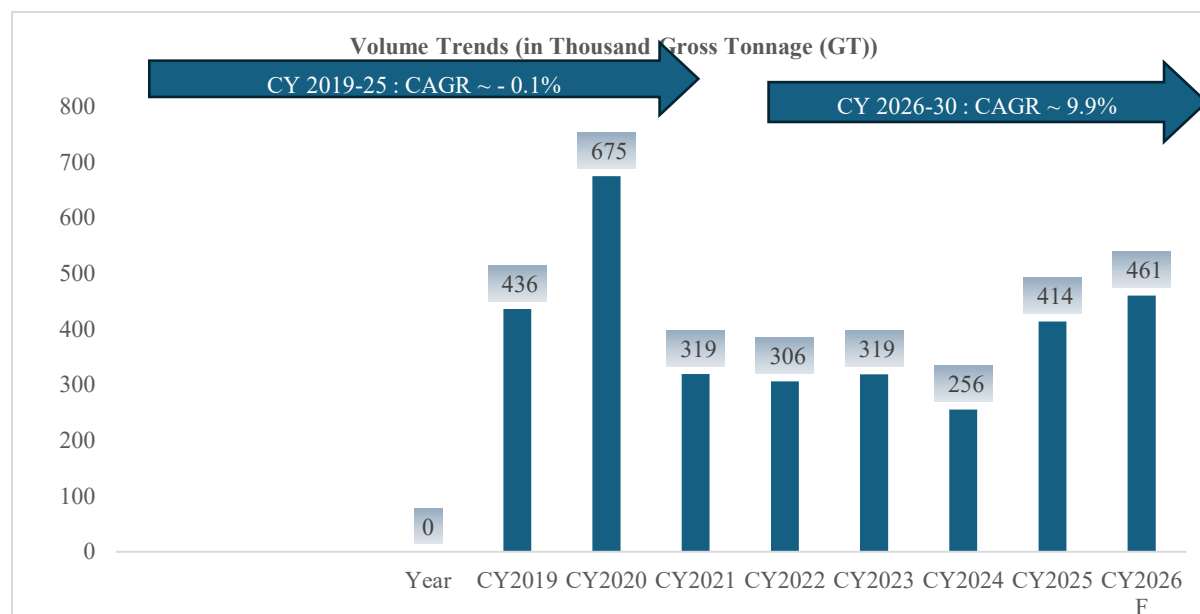
India's rapidly expanding steel, infrastructure, manufacturing, and construction sectors are expected to generate strong demand for recycled ferrous scrap recovered from dismantled bulk carriers. The availability of an established downstream steel processing ecosystem ensures efficient utilization of recovered materials, thereby improving the economic viability of bulk carrier recycling operations and supporting market growth.

The segment is also anticipated to benefit from ongoing modernization initiatives at the Alang-Sosiya ship recycling cluster, including the adoption of mechanized cutting equipment, improved yard infrastructure, enhanced material handling capabilities, and strengthened environmental and safety management systems. These

upgrades have significantly improved India's ability to dismantle large bulk carriers safely, efficiently, and in accordance with internationally recognized environmental, health, and safety standards.

4.4.4 General Cargo

Chart: Global: Ship Recycling (General Cargo) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The general cargo segment accounted for 10.2% of India's ship recycling market in CY2025, reaching a volume of 306 thousand Gross Tonnage (GT). The segment recorded a CAGR of -0.1% during CY2019-CY2025, reflecting relatively stable recycling activity supported by the gradual retirement of aging vessels operating in regional and coastal trade routes.

The recycling of general cargo vessels in India has been primarily driven by the decommissioning of older coastal and regional trading fleets that have become economically unviable due to rising maintenance expenditures, lower fuel efficiency, and increasing repair requirements. Upon reaching the end of their operational life, many of these vessels are sent to the Alang-Sosiya Ship Recycling Yard, where they are dismantled and recycled.

In addition, India's growing emphasis on coastal shipping under initiatives such as the Maritime India Vision 2030 and the Sagarmala Programme has encouraged fleet modernization across the domestic shipping sector. Shipping companies are increasingly replacing aging general cargo vessels with larger, more efficient, and technologically advanced ships, thereby supporting the supply of end-of-life vessels available for recycling.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the general cargo segment is expected to reach a volume of 500 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 9.9% during CY2026F-CY2030F. Growth is anticipated to be driven by accelerating fleet renewal, tightening environmental regulations, and sustained demand for recycled materials across key end-use industries.

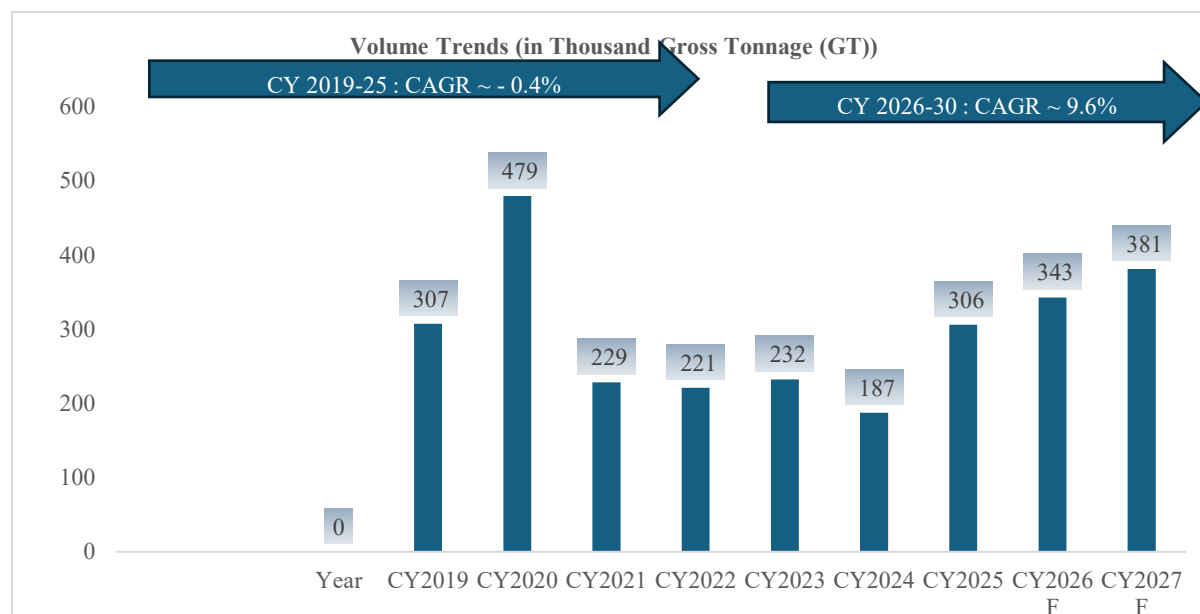
The implementation of stricter international regulations related to energy efficiency, carbon emissions, and environmental performance is expected to encourage shipowners to retire older general cargo vessels rather than undertake costly retrofitting and compliance-related upgrades. As a result, the volume of end-of-life vessels entering recycling facilities is projected to increase over the forecast period.

The segment is also expected to benefit from strong domestic demand for recycled steel from India's construction, infrastructure, engineering, and manufacturing sectors. The presence of a well-established steel re-rolling and

scrap processing ecosystem around the Alang-Sosiya ship recycling cluster facilitates the efficient utilization of recovered materials, improving the overall economics of vessel recycling.

4.4.5 Ro-Ro (Roll-on/Roll-off)

Chart: Global: Ship Recycling (Ro-Ro (Roll-on/Roll-off)) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The Ro-Ro (Roll-on/Roll-off) segment accounted for 4.4% of India's ship recycling market in CY2025, reaching a volume of 132 thousand Gross Tonnage (GT). The segment recorded a CAGR of -0.4% during CY2019-CY2025, reflecting fluctuations in vessel retirement activity and replacement cycles within the global Ro-Ro shipping fleet.

The recycling of Ro-Ro vessels in India has been supported by the country's well-established ship recycling infrastructure, particularly the Alang-Sosiya Ship Recycling Yard, which possesses the capability to dismantle large vehicle carriers, passenger ferries, and Ro-Ro cargo vessels in compliance with internationally recognized environmental, health, and safety standards.

In addition, ongoing fleet modernization initiatives across the global shipping industry have encouraged operators to replace aging Ro-Ro vessels with newer, more fuel-efficient ships that comply with increasingly stringent emission and environmental regulations. As these vessels reach the end of their commercial life, a growing number are being directed to Indian recycling facilities. India's competitive recycling costs, skilled workforce, and expanding network of HKC-compliant ship recycling yards have further strengthened its position as a preferred destination for Ro-Ro vessel recycling.

- **Forecast Outlook (CY2026F-CY2030F)**

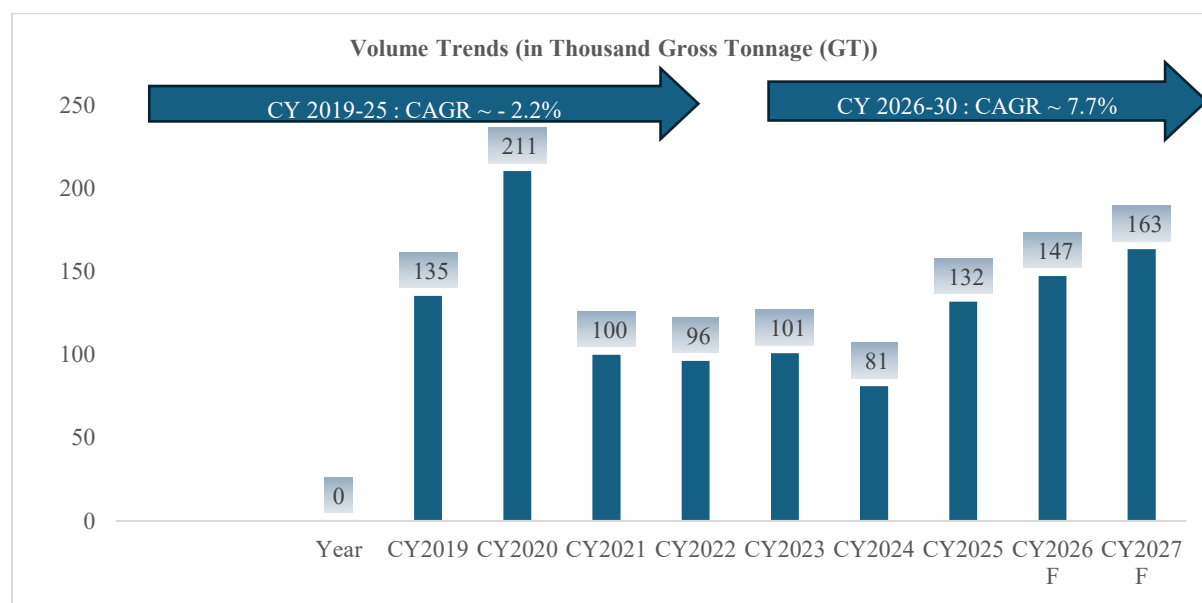
Looking ahead, the Ro-Ro segment is projected to reach a volume of 213 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 9.6% during CY2026F-CY2030F. Growth is expected to be driven by increased retirement of aging vehicle carriers and ferries, continued fleet modernization, and strengthening environmental compliance requirements across the maritime industry.

The dismantling of Ro-Ro vessels generates substantial volumes of high-grade recycled steel along with reusable machinery, cargo handling equipment, electrical systems, and marine components. These recovered materials are extensively utilized across India's steel, engineering, manufacturing, ship repair, and secondary equipment markets, enhancing the value recovery potential and economic attractiveness of recycling operations.

The segment is also expected to benefit from ongoing investments in mechanized dismantling technologies, hazardous waste management systems, and environmentally compliant recycling infrastructure. These developments are improving India's ability to safely and efficiently recycle technologically advanced Ro-Ro vessels while meeting internationally recognized environmental and occupational safety standards.

4.4.6 Others

Chart: Global: Ship Recycling (Others) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

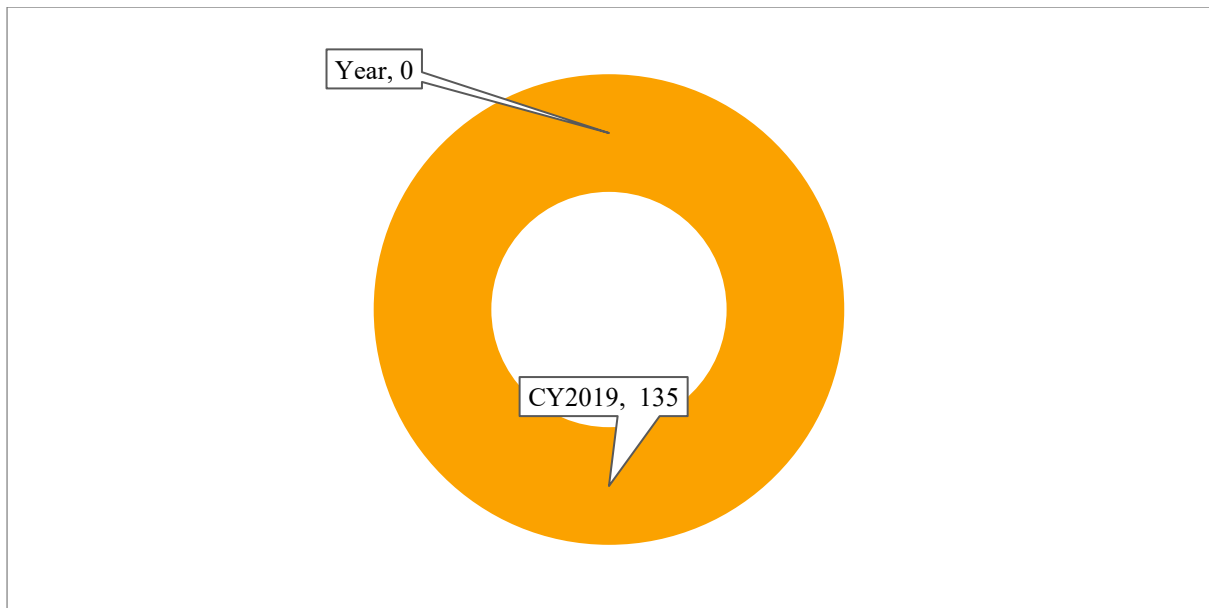
In CY2025, the other types accounted for a share of 18.4% of the ship recycling market in India. The market reached a volume of 552 thousand Gross Tonnage (GT), representing a CAGR of -2.2% during CY2019-CY2025.

- **Forecast Outlook (CY2026F-CY2030F)**

Looking forward, the market in this segment is expected to reach a volume of 815 thousand Gross Tonnage (GT) by CY2030, representing a CAGR of 7.7% during CY2026-CY2030.

4.5 Indian Ship Recycling Market: End-Use Breakdown of Recycled Scrap (CY2019-CY2030)

Chart: India: Ship Recycling Market: End-Use Breakdown of Recycled Scrap (in %), CY2025



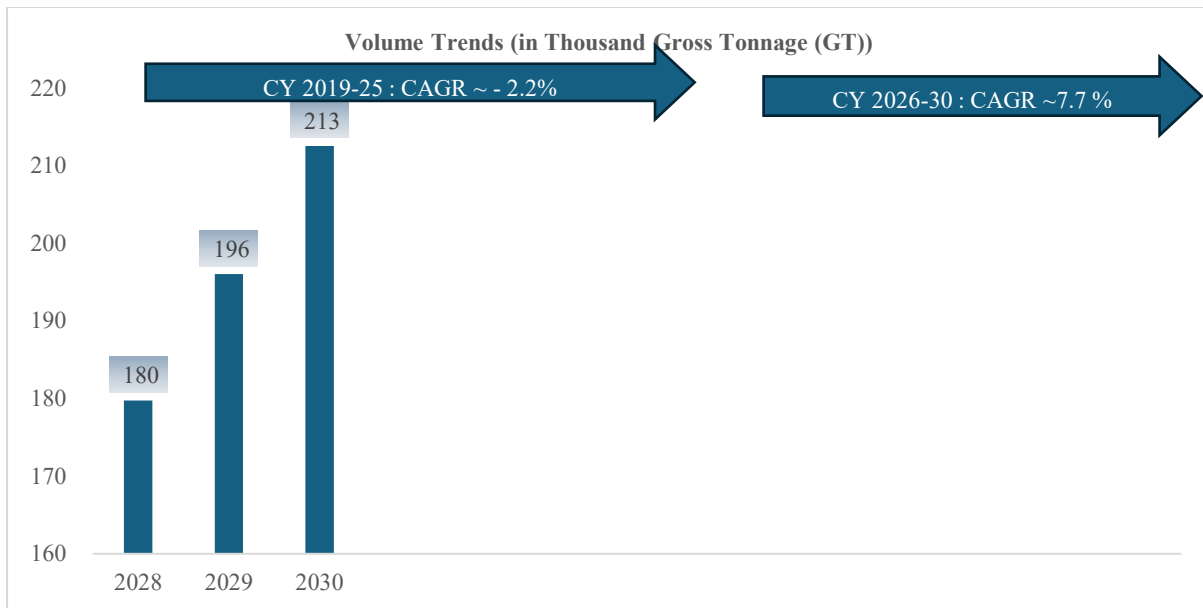
Source: Expert Interactions, IMARC, ICRA Analytics

In CY2025, steel mills accounted for the largest share of recycled ship scrap consumption in India, representing 49.1% of the total market volume.

The segment was followed by fabrication, which contributed 34.4% of the market, while industrial applications accounted for 12.8%. The remaining 3.7% was attributed to other end-use sectors.

4.5.1 Steel Mills

Chart: India: Ship Recycling (Steel Mills) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The steel mills segment accounted for the largest share of India's ship recycling market, reaching a volume of 1,472 thousand Gross Tonnage (GT) in CY2025 and representing 49.1% of the total market volume. The segment recorded a CAGR of -2.2% during CY2019-CY2025, reflecting fluctuations in vessel recycling activity, scrap availability and broader market conditions impacting steel production and raw material demand.

Despite the decline, steel mills remained the primary end-user of recycled ship scrap in India, supported by the country's expanding steel industry and increasing crude steel production. Steel recovered from dismantled vessels serves as an important secondary raw material for induction furnaces, electric arc furnaces (EAFs), integrated steel plants, and re-rolling mills, helping reduce dependence on imported scrap and virgin iron ore. Growing emphasis on resource efficiency, circular economy initiatives, and sustainable manufacturing practices further strengthened the role of recycled ship scrap within the domestic steel value chain.

The segment also benefited from policy support aimed at promoting scrap utilization and reducing the environmental footprint of steel production. Increasing focus on green steel production, decarbonization targets, and lower-carbon manufacturing processes encouraged steel producers to incorporate higher volumes of recycled ferrous scrap into their operations. In addition, the availability of quality ship-recycled steel from major recycling hubs such as Alang continued to support raw material requirements across the industry.

- Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the steel mills segment is projected to reach a volume of 2,173 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 7.7% during CY2026F-CY2030F. Growth is expected to be supported by rising steel demand, increasing scrap-based steel production and the growing adoption of sustainable manufacturing practices across the industry.

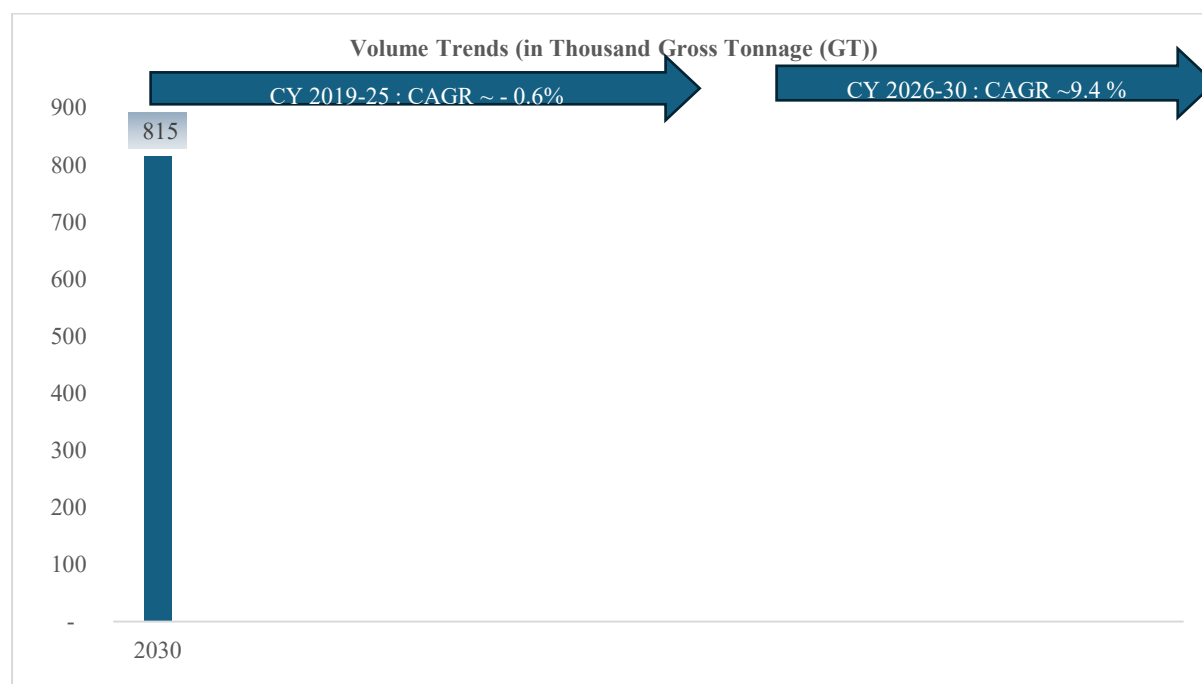
Robust demand from infrastructure development, construction, railways, automotive, engineering, renewable energy, and capital goods sectors is expected to drive the consumption of recycled steel over the forecast period. Ship-recycled scrap offers a cost-effective and high-quality source of raw material while helping steel manufacturers reduce energy consumption and greenhouse gas emissions compared with primary steel production, making it an increasingly attractive input for steelmaking operations.

The market is also expected to benefit from India's efforts to promote resource efficiency and accelerate the transition toward low-carbon industrial production. As environmental regulations become more stringent and sustainability commitments gain importance, steel manufacturers are likely to increase the use of recycled feedstock to improve operational efficiency and reduce carbon emissions. These trends are expected to strengthen long-term demand for ship-recycled steel within the domestic market.

Furthermore, the presence of a well-established ecosystem of steel re-rolling mills and secondary steel producers in proximity to the Alang-Sosiya ship recycling cluster is expected to enhance market growth. This integrated value chain facilitates the efficient processing, distribution, and utilization of recovered steel, reducing transportation costs, improving material recovery rates, and supporting the overall economic competitiveness of India's ship recycling industry. Combined with sustained investments in recycling infrastructure and growing demand for sustainable raw materials, these factors are expected to support continued expansion of the steel mills segment through CY2030F.

4.5.2 Fabrication

Chart: India: Ship Recycling (Fabrication) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019-CY2025)**

The steel mills segment accounted for the largest share of India's ship recycling market, reaching a volume of 1,472 thousand Gross Tonnage (GT) in CY2025 and representing 49.1% of the total market volume. The segment recorded a CAGR of -2.2% during CY2019-CY2025, reflecting fluctuations in vessel recycling activity, scrap availability, and broader market conditions impacting steel production and raw material demand.

Despite the decline, steel mills remained the primary end-user of recycled ship scrap in India, supported by the country's expanding steel industry and increasing crude steel production. Steel recovered from dismantled vessels serves as an important secondary raw material for induction furnaces, electric arc furnaces (EAFs), integrated steel plants, and re-rolling mills, helping reduce dependence on imported scrap and virgin iron ore. Growing emphasis on resource efficiency, circular economy initiatives, and sustainable manufacturing practices further strengthened the role of recycled ship scrap within the domestic steel value chain.

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steel demand, increasing scrap-based steel production, and the growing adoption of sustainable manufacturing practices across the industry.

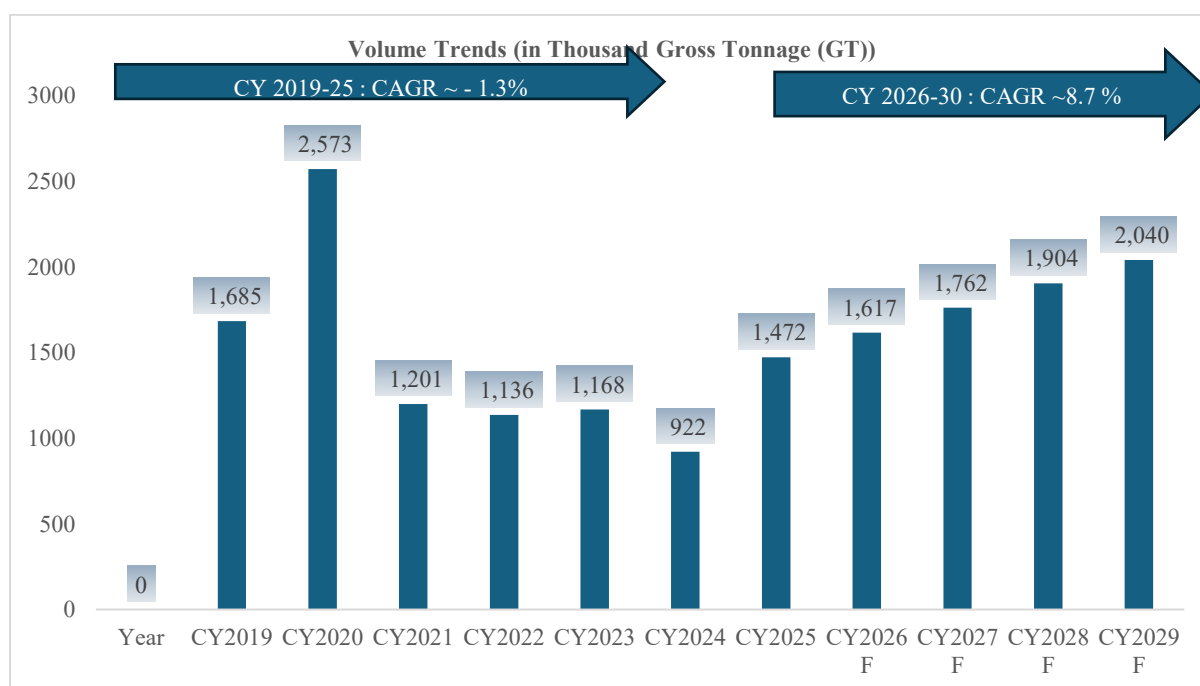
Robust demand from infrastructure development, construction, railways, automotive, engineering, renewable energy, and capital goods sectors is expected to drive the consumption of recycled steel over the forecast period. Ship-recycled scrap offers a cost-effective and high-quality source of raw material while helping steel manufacturers reduce energy consumption and greenhouse gas emissions compared with primary steel production, making it an increasingly attractive input for steelmaking operations.

The market is also expected to benefit from India's efforts to promote resource efficiency and accelerate the transition toward low-carbon industrial production. As environmental regulations become more stringent and sustainability commitments gain importance, steel manufacturers are likely to increase the use of recycled feedstock to improve operational efficiency and reduce carbon emissions. These trends are expected to strengthen long-term demand for ship-recycled steel within the domestic market.

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4.5.3 Industrial Use

Chart: India: Ship Recycling (Industrial Use) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- Historical Performance (CY2019-CY2025)**

The industrial use segment accounted for 12.8% of India's ship recycling market in CY2025, reaching a volume of 384 thousand Gross Tonnage (GT). The segment recorded a CAGR of -1.3% during CY2019-CY2025, reflecting fluctuations in ship recycling activity and the availability of recoverable materials from end-of-life vessels.

Demand within the segment was supported by the continued expansion of India's manufacturing, engineering, and heavy industrial sectors, which utilize recovered steel plates, pipes, machinery, structural sections, and other reusable components obtained from dismantled ships. These materials are widely used across industrial operations due to their cost-effectiveness and availability.

The segment also benefited from growing investments in industrial infrastructure, power generation facilities, oil and gas projects, refineries, cement plants, shipbuilding activities, and large-scale engineering developments. The increasing use of recycled materials in these sectors helped reduce procurement costs while providing a reliable alternative to newly manufactured steel and equipment.

- **Forecast Outlook (CY2026F-CY2030F)**

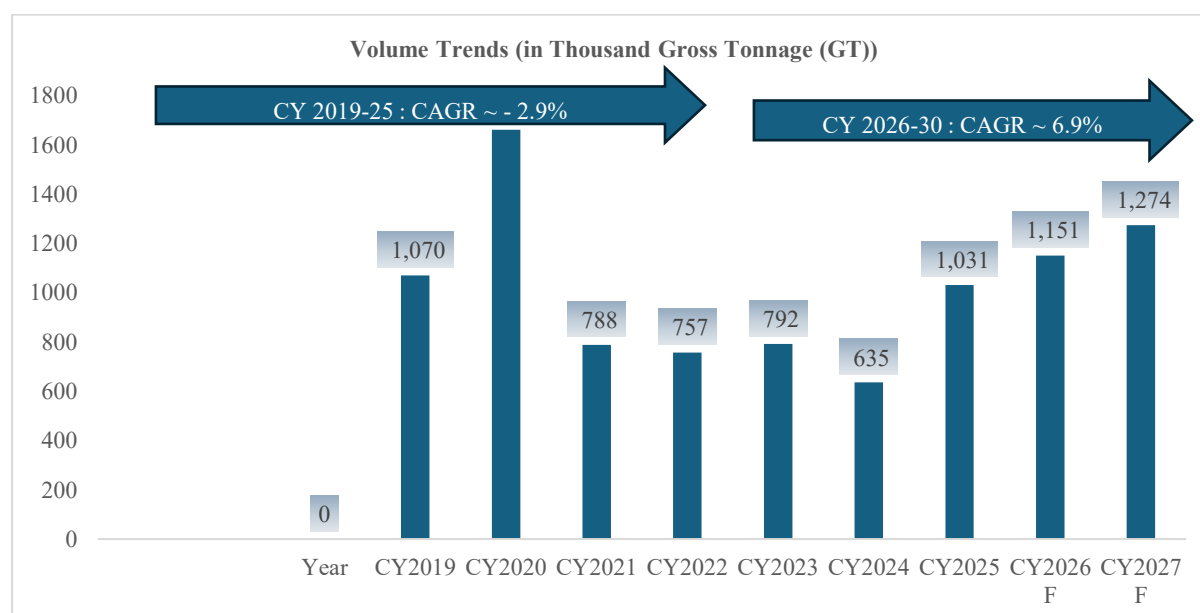
Looking ahead, the industrial use segment is expected to reach a volume of 593 Thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 8.7% during CY2026F-CY2030F. Growth is expected to be driven by rising industrial activity, increasing investments in infrastructure and manufacturing projects, and greater adoption of recycled materials across end-user industries.

Increasing adoption of circular economy principles, sustainable manufacturing practices, and resource conservation initiatives is also expected to support long-term demand growth. As industries place greater emphasis on environmental sustainability and efficient resource utilization, the use of recycled steel and reusable components sourced from end-of-life vessels is likely to increase.

Furthermore, growing awareness regarding carbon reduction, waste minimization, and responsible material sourcing is expected to strengthen the role of recycled ship scrap in industrial applications. Combined with ongoing investments in recycling infrastructure and improved material recovery processes, these factors are expected to support sustained growth in the industrial use segment through CY2030F.

4.5.4 Others

Chart: India: Ship Recycling (Other: End Uses) Market: Volume Trends (in thousand Gross Tonnage (GT)), CY2019-CY2030F



Source: Expert Interactions, IMARC, ICRA Analytics

- **Historical Performance (CY2019–CY2025)**

The other end-use segment accounted for 3.7% of India's ship recycling market in CY2025, reaching a volume of 111 thousand Gross Tonnage (GT). The segment recorded a CAGR of -2.9% during CY2019–CY2025, reflecting

fluctuations in ship recycling activity, scrap availability, and demand from niche applications utilizing materials recovered from end-of-life vessels.

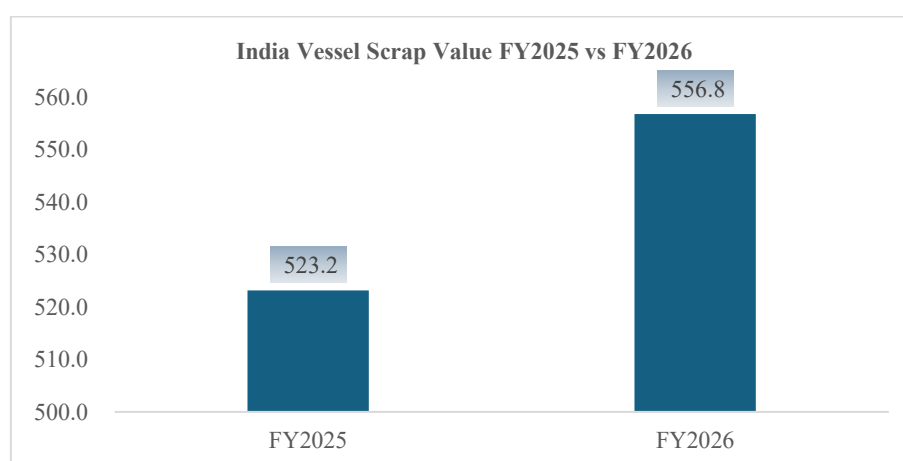
- **Forecast Outlook (CY2026F-CY2030F)**

Looking ahead, the other end-use segment is projected to reach a volume of 158 thousand Gross Tonnage (GT) by CY2030, registering a CAGR of 6.9% during CY2026F-CY2030F. Growth is expected to be driven by increasing awareness of resource efficiency, rising adoption of recycled materials across smaller industries, and growing emphasis on circular economy practices.

4.6 Export–Import Dynamics and Domestic Scrap Movement

4.6.1 Export-Import Dynamics

Chart: India Vessel Scrap Value FY2025 vs FY2026 (in Million USD)



Source: Ministry of Commerce and India, ICRA Analytics

India's ship recycling industry operates predominantly on an import-driven model, wherein end-of-life vessels acquired from international shipowners are dismantled at authorized recycling facilities, with the Alang-Sosiya Ship Recycling Cluster in Gujarat serving as the country's principal recycling hub.

Imports under HS Code 8908 (vessels and other floating structures for breaking up) have demonstrated steady growth, increasing from USD 523.2 million in FY2024-25 to USD 556.8 million in FY2025-26, representing a 6.4% year-on-year increase. This growth reflects improving operational throughput at Alang-Sosiya as well as India's increasing competitiveness compared to other major ship recycling destinations such as Bangladesh and Turkey, supported by advantages in both cost efficiency and regulatory compliance.

India's extensive network of Hong Kong International Convention (HKC)-compliant recycling facilities has become a significant competitive advantage. Growing regulatory requirements and ESG expectations are prompting European and international shipowners to prioritize environmentally responsible vessel disposal. Consequently, the increasing number of HKC-certified yards in India is attracting a larger share of high-value vessels, contributing to growth in both import volumes and average import values.

On the export side, shipments of recovered materials remain minimal. Most outputs generated through ship recycling including steel scrap, non-ferrous metals, marine machinery, equipment, and reusable components are consumed within the domestic market. As a result, the industry functions largely as a closed-loop domestic value chain, where imported end-of-life vessels are converted into raw materials and reusable assets that support India's steel, manufacturing, engineering, and infrastructure sectors rather than being re-exported.

4.6.2 Domestic Utilization and Value Chain Integration of Recovered Ship Recycling Materials

Ship recycling in India is supported by a well-established domestic value chain that enables the efficient utilization of materials recovered from end-of-life vessels. Steel, which typically constitutes around 75–85% of a vessel's total recoverable weight, represents the primary output of the recycling process. The recovered steel is supplied

to a broad network of downstream consumers, including steel mills, re-rolling mills, fabrication units, engineering companies, and construction material processors across the country. This diversified end-user base ensures steady material absorption, reduces inventory accumulation at recycling yards, and supports stable operational throughput, particularly at the Alang-Sosiya Ship Recycling Cluster.

The industry has also fostered a comprehensive ecosystem of ancillary businesses comprising scrap processors, metal traders, transport operators, steel re-rollers, equipment refurbishers, and recycling intermediaries. This ecosystem facilitates the seamless movement of recovered materials into the broader economy and maximizes value realization from each recycled vessel. Beyond ferrous materials, ship recycling generates significant volumes of non-ferrous metals such as copper, aluminium, and brass, along with reusable marine machinery, navigation equipment, electrical systems, furniture, and fittings. These materials are distributed through specialized secondary markets, further enhancing resource recovery and economic returns.

Recovered materials are transported to major industrial and manufacturing centers across states such as Gujarat, Maharashtra, Rajasthan, Punjab, Haryana, and Tamil Nadu. Gujarat serves as a key processing and re-rolling hub, while Maharashtra has significant demand from steel and fabrication industries. Rajasthan and Punjab are major consumers through induction furnace and re-rolling mill operations, whereas Haryana and Tamil Nadu utilize recovered materials across automotive, engineering, and manufacturing applications. This widespread geographic distribution demonstrates the deep integration of ship recycling within India's industrial supply chain and extends its economic contribution far beyond the maritime sector.

The industry also plays an important role in supporting India's raw material security. Ship recycling currently contributes an estimated 20–25% of the country's total ferrous scrap requirement, making it a strategically important source of feedstock for electric arc furnaces (EAFs), induction furnaces, and secondary steel producers. By supplying domestically generated recycled steel, the industry helps reduce dependence on imported scrap, improves resource efficiency, supports circular economy objectives, and contributes to the decarbonization of steel production.

As India's infrastructure, construction, manufacturing, and engineering sectors continue to expand, the importance of ship recycling as a source of sustainable raw materials is expected to increase further. The sector is gradually transitioning from a traditional scrapping activity to a critical component of India's circular economy and low-carbon industrial strategy, supported by growing demand for recycled materials, supportive policy initiatives, and increasing compliance with international environmental and safety standards.

4.7 Adoption of Advanced Techniques in Indian Ship Recycling

4.7.1 Mechanized Dismantling

India has established itself as a global leader in responsible ship recycling through continuous modernization of its recycling infrastructure and the adoption of advanced dismantling technologies. Over the past decade, the sector has evolved from traditional, labour-intensive ship-breaking operations into a highly regulated and increasingly mechanized industry supported by a strong legislative framework, institutional oversight, and rigorous compliance mechanisms. Significant investments in modern equipment, environmental protection systems, and worker safety infrastructure particularly at the Alang-Sosiya Ship Recycling Yard have enhanced operational efficiency while minimizing environmental and occupational risks. As a result, India has strengthened its position as one of the world's most trusted destinations for safe, environmentally sound, and compliant ship recycling.

Indian ship recycling facilities have substantially upgraded their operational capabilities and now incorporate:

- Engineered and clearly demarcated ship recycling plots.
- A structured three-stage dismantling methodology comprising primary structural dismantling, secondary segregation of components, and tertiary material recovery, enabling improved efficiency and higher resource utilization.
- Mechanized material handling and heavy-lifting equipment, including hydraulic cranes, excavators, forklifts, winches, and gas-cutting systems for the safe handling of large vessel sections.
- Impermeable work surfaces, dedicated drainage networks, and specialized hazardous waste storage facilities designed to prevent soil and marine contamination.

- Authorized downstream linkages for the treatment, recycling, recovery, and disposal of various waste streams.
- Comprehensive occupational health and safety systems, emergency response mechanisms, welfare facilities, and worker protection programs.

Most authorized ship recycling facilities operate in compliance with the Hong Kong International Convention (HKC) and have implemented internationally recognized management and safety standards. Safety practices are monitored by qualified professionals, with regular training programs covering enclosed-space operations, firefighting, working at heights, emergency evacuation, and rescue procedures. Several facilities have also begun integrating virtual reality (VR)-based training modules to simulate hazardous work environments and improve workforce preparedness.

The increasing adoption of mechanized dismantling methods has facilitated more controlled cutting operations, safer handling of heavy ship structures, faster recovery of recyclable materials, improved environmental management, and lower dependence on manual labor. Collectively, these advancements have enhanced the competitiveness and sustainability of India's ship recycling industry, reinforcing its reputation as a preferred global destination for responsible ship recycling.

4.7.2 Hazardous Material Controls and Handling

- **Inventory of Hazardous Materials (IHM) and Lifecycle Compliance**

The Inventory of Hazardous Materials (IHM) serves as a critical element in ensuring safe and environmentally responsible ship recycling practices. It provides a comprehensive record of hazardous substances present on board a vessel and facilitates their identification, documentation, and monitoring throughout the ship's operational life and prior to recycling. In line with the requirements of the Hong Kong International Convention (HKC), India has made the IHM mandatory for ships entering authorized recycling facilities. The inventory enables recycling yards to identify, remove, handle, store, transport, and dispose of hazardous materials in a controlled manner, while enhancing transparency, traceability, and risk management across the ship recycling process.

- **Safe Handling and Disposal of Hazardous Materials**

Authorized ship recycling facilities in India follow established procedures for the identification, segregation, removal, storage, transportation, treatment, and final disposal of hazardous substances, including asbestos, polychlorinated biphenyls (PCBs), ozone-depleting substances, heavy metals, oils, chemicals, and radioactive materials. The availability of dedicated hazardous waste storage areas, spill containment systems, impermeable work surfaces, and authorized downstream treatment and disposal facilities helps prevent environmental contamination and supports compliance with both domestic and international hazardous waste management regulations.

- **Worker Protection and Emergency Preparedness**

Indian ship recycling facilities are required to implement robust occupational health and safety measures to safeguard workers engaged in hazardous material handling activities. These measures include the provision of personal protective equipment (PPE), periodic safety training, medical monitoring, emergency response protocols, fire protection systems, gas detection and monitoring mechanisms, and standardized operating procedures. Together, these initiatives help reduce occupational health risks, enhance workplace safety, and strengthen environmental protection during ship dismantling operations.

4.7.3 Compliance with International Conventions

- **Hong Kong International Convention (HKC)**

The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC) is the primary international framework governing ship recycling activities worldwide. The Convention establishes

legally binding obligations for ships, shipowners, flag States, ship recycling States, and recycling facilities to ensure that vessels are dismantled in a safe and environmentally responsible manner.

The Convention entered into force on 26 June 2025, making compliance mandatory for all Parties. India aligned its domestic regulatory framework with HKC requirements through the Recycling of Ships Act, 2019, the Ship Recycling Rules, 2021, and related regulations, enabling Indian recycling facilities to adopt HKC-compliant practices well before the Convention became legally effective.

Under the HKC framework, India performs responsibilities both as a flag State and as a ship recycling State, including:

- Lifecycle certification and compliance monitoring of ships.
- Authorization, inspection, regulation, and supervision of ship recycling facilities.
- Monitoring and management of hazardous materials through the Inventory of Hazardous Materials (IHM).
- Review, preparation, and approval of Ship Recycling Plans (SRPs).
- Maintenance of records, certification processes, and international reporting to ensure adherence to Convention requirements.

• **EU Ship Recycling Regulation (EU SRR)**

At present, no ship recycling facilities in India are included on the European List of Approved Ship Recycling Facilities under the European Union Ship Recycling Regulation (EU SRR). Although more than 110 ship recycling yards in Alang possess HKC compliance certifications, they are not currently recognized under the EU framework. Consequently, ships flying EU or EEA member-state flags may only be recycled at facilities approved under the European List.

The Government of India continues to engage with relevant European authorities to facilitate the inclusion of Indian recycling facilities on the approved list. Ongoing efforts are focused on demonstrating compliance with the environmental, safety, and operational requirements prescribed under the EU SRR.

The EU Ship Recycling Regulation was adopted on 20 November 2013 and became fully applicable on 31 December 2018. The regulation was introduced to prevent and minimize accidents, injuries, and adverse environmental impacts associated with ship recycling activities.

The key objectives of the EU SRR include:

- Ensuring that ships are dismantled at facilities that operate in a safe and environmentally sound manner.
- Protecting human health and marine ecosystems from harmful impacts.
- Promoting sustainable resource utilization and waste management practices.
- Improving transparency and accountability in ship recycling operations.

The regulation mandates that vessels flying the flag of an EU member state must be recycled only at facilities listed on the European List of Approved Ship Recycling Facilities.

• **Adoption of International Management Standards**

In addition to compliance with HKC requirements, many Indian ship recycling facilities have adopted internationally recognized management systems to strengthen operational quality, environmental performance, and occupational health and safety standards. These include:

- **ISO 9001** – Quality Management Systems
- **ISO 14001** – Environmental Management Systems
- **ISO 45001** – Occupational Health and Safety Management Systems
- **ISO 30000** – Ship Recycling Management Systems

The adoption of these standards supports continuous improvement in operational efficiency, regulatory compliance, worker welfare, environmental stewardship, and overall management performance across India's ship recycling industry.

4.8 Regulatory Policies and Programs

4.8.1 Key National Regulations

India has established a robust regulatory framework to ensure that ship recycling activities are conducted in a safe, environmentally sustainable, and internationally compliant manner. The framework covers the entire ship recycling lifecycle, including vessel preparation, certification, authorization of recycling facilities, operational procedures, worker welfare, environmental protection, inspections, and regulatory oversight. The **Directorate General of Shipping (DG Shipping)** serves as the designated National Authority responsible for implementing these regulations and monitoring compliance across the sector.

- **Recycling of Ships Act, 2019:** The Recycling of Ships Act, 2019 constitutes the principal legislation governing ship recycling activities in India. The Act aligns India's domestic legal framework with the requirements of the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC) by establishing provisions relating to the authorization and regulation of ship recycling facilities, certification of ships, preparation of Ship Recycling Plans (SRPs), maintenance of the Inventory of Hazardous Materials (IHM), inspections, audits, and enforcement procedures.
- **Ship Recycling Rules, 2021:** The Ship Recycling Rules, 2021 provide the operational and procedural framework for implementing the provisions of the Act. These Rules prescribe detailed technical requirements covering facility authorization, hazardous material management, occupational health and safety standards, environmental safeguards, emergency preparedness measures, worker training, reporting obligations, documentation requirements, and compliance procedures throughout the ship dismantling process.
- **Ship Recycling Regulations, 2026:** The Ship Recycling Regulations, 2026 further strengthen the regulatory regime by introducing enhanced operational standards, inspection methodologies, environmental protection measures, safety requirements, and digital compliance systems. The Regulations are intended to promote uniform implementation of regulatory requirements across authorized ship recycling facilities while improving transparency, monitoring capabilities, and enforcement effectiveness.

4.8.2 Environmental and Safety Frameworks

India's environmental and occupational safety framework for ship recycling is designed to minimize environmental risks while ensuring the protection, health, and safety of workers engaged in dismantling activities. The framework is closely aligned with the provisions of the Hong Kong International Convention (HKC), which entered into force on 26 June 2025. India proactively harmonized its regulatory system with HKC requirements prior to its implementation, enabling domestic recycling facilities to achieve compliance with internationally recognized environmental and safety standards.

- **Inventory of Hazardous Materials (IHM):** A key element of the environmental management framework is the mandatory Inventory of Hazardous Materials (IHM). The IHM identifies and documents hazardous substances present on board a ship, including asbestos, polychlorinated biphenyls (PCBs), ozone-depleting substances, heavy metals, and other regulated materials. Maintained throughout a vessel's operational life, the inventory enables ship recycling facilities to safely identify, remove, handle, transport, and dispose of hazardous materials before dismantling activities commence.
- **Ship Recycling Facility Plans and Worker Safety Measures:** Authorized recycling facilities are required to develop comprehensive Ship Recycling Facility Plans (SRFPs) and vessel-specific Ship Recycling Plans (SRPs) that outline dismantling procedures, hazardous waste management protocols, environmental protection measures, emergency response arrangements, and occupational safety practices. Facilities must also implement pollution control measures, maintain impermeable working areas and drainage systems, provide suitable personal protective equipment (PPE), conduct regular worker training, and establish emergency medical response mechanisms to protect both workers and the surrounding environment.

- **Adoption of International Management Standards:** To improve operational performance and strengthen global competitiveness, many Indian ship recycling facilities have adopted internationally recognized management systems, including:
 - **ISO 9001** – Quality Management Systems
 - **ISO 14001** – Environmental Management Systems
 - **ISO 45001** – Occupational Health and Safety Management Systems
 - **ISO 30000** – Ship Recycling Management Systems

The adoption of these standards supports improved environmental stewardship, enhanced workplace safety, standardized operational practices, and greater international acceptance of India's ship recycling industry.

4.8.3 Government Initiatives and Incentives

The Government of India has introduced a range of policy measures and incentive programs aimed at modernizing the ship recycling sector, encouraging environmentally responsible recycling practices, and strengthening linkages between ship recycling and domestic shipbuilding. These initiatives contribute to the development of a sustainable maritime circular economy while reinforcing India's position as a leading global ship recycling destination.

- **Ship Recycling Credit Note Scheme (SBFA 2.0):** One of the key initiatives is the Ship Recycling Credit Note Scheme, introduced under the Shipbuilding Financial Assistance Policy (SBFA) 2.0. Under this scheme, shipowners who recycle vessels at authorized HKC-compliant Indian recycling facilities are eligible to receive a Credit Note equivalent to 40% of the scrap value of the recycled ship. The Credit Note can subsequently be utilized when placing orders for new vessels at Indian shipyards. The scheme is designed to encourage environmentally compliant ship recycling, promote domestic shipbuilding, improve resource utilization, and enhance the competitiveness of India's maritime manufacturing ecosystem.
- **Digitalization of Regulatory Governance:** The Government is also advancing the digital transformation of ship recycling governance through the deployment of electronic platforms for facility authorization, ship certification, inspection management, compliance monitoring, reporting, and record maintenance. These digital initiatives enhance transparency, streamline administrative processes, facilitate real-time regulatory oversight, and strengthen enforcement of environmental and safety requirements across authorized recycling facilities.
- **Support for Green Steel and Circular Economy Initiatives:** Ship recycling has become an important component of India's broader sustainability agenda through its contribution to Green Steel and circular economy objectives. Steel recovered from end-of-life vessels significantly reduces energy consumption, greenhouse gas emissions, and dependence on virgin raw materials when compared with primary steel production. By encouraging the utilization of recycled steel, the Government supports resource conservation, industrial sustainability, material efficiency, and national decarbonization goals while strengthening the role of ship recycling within India's circular economy framework.

4.9 Key Growth Drivers and Challenges

4.9.1 Key Growth Drivers

- **HKC Implementation and Regulatory Reforms:** The implementation of the Hong Kong International Convention (HKC) and India's Recycling of Ships Act, 2019 has significantly strengthened the credibility of the Indian ship recycling industry. Compliance with international environmental, occupational health, and safety standards has enhanced the confidence of global shipowners, many of whom now prefer recycling vessels at certified facilities. The regulatory framework has also improved transparency, hazardous waste management, worker protection, and operational governance, helping India emerge as a preferred destination for responsible ship recycling.

Impact on growth:

- Increased inflow of vessels from international shipowners.
- Improved global acceptance of Indian recycling yards.

- Better environmental and safety performance.
- Enhanced competitiveness against regional recycling hubs.
- **Government Incentives and Digitalization:** The Government of India has introduced several initiatives to promote sustainable ship recycling. The Ship Recycling Credit Note Scheme under SBFA 2.0 incentivizes shipowners to recycle vessels at authorized Indian facilities. Simultaneously, digitalization of approvals, inspections, certification, and compliance monitoring is improving regulatory efficiency and reducing administrative delays.

Impact on growth:

- Encourages environmentally compliant recycling.
- Supports integration between ship recycling and domestic shipbuilding.
- Improves ease of doing business.
- Strengthens regulatory oversight and transparency.
- **Expansion of Recycling Capacity:** India is continuously investing in the expansion of recycling infrastructure, particularly at Alang-Sosiya Ship Recycling Yard, the country's largest recycling hub. Capacity expansion through modernization of plots, deeper waterfront access, and improved infrastructure enables the handling of larger and more complex vessels.

Impact on growth:

- Higher annual recycling throughput.
- Ability to process larger vessels.
- Increased foreign vessel inflow.
- Stronger position in the global recycling market.
- **Modernization and Mechanization of Ship Recycling Yards:** Indian yards have transitioned from traditional labour-intensive dismantling to increasingly mechanized operations. The adoption of hydraulic cranes, excavators, forklifts, heavy-lifting equipment, impermeable flooring, drainage systems, and hazardous waste management facilities has significantly improved productivity and safety.

Impact on growth:

- Higher operational efficiency.
- Faster dismantling cycles.
- Improved worker safety.
- Greater environmental compliance.
- Better recovery rates of steel and reusable materials.
- **Rising Global Vessel Retirements:** The global shipping industry is undergoing fleet renewal driven by decarbonization targets, fuel-efficiency requirements, and stricter emission regulations. As older vessels become uneconomical to operate, the number of end-of-life ships available for recycling is expected to increase substantially.

Impact on growth:

- Larger supply of vessels available for recycling.
- Higher recycling volumes over the forecast period.
- Increased opportunities for Indian recycling facilities.
- **Adoption of International Standards (ISO):** Many Indian recycling facilities have adopted internationally recognized standards such as ISO 9001, ISO 14001, ISO 45001, and ISO 30000. These certifications strengthen environmental performance, operational quality, and occupational health and safety management.

Impact on growth:

- Improves international reputation.
- Facilitates business with global shipowners.

- Enhances compliance and risk management.
- Supports long-term industry sustainability.
- **Growing Demand for Recycled Steel:** Steel recovered from dismantled vessels serves as a valuable raw material for steel mills, re-rolling mills, fabrication industries, and engineering applications. India's rapid infrastructure development, construction activity, manufacturing expansion, and urbanization continue to drive demand for recycled steel.

Impact on growth:

- Supports strong demand for ship scrap.
- Reduces dependence on imported scrap and iron ore.
- Promotes circular economy principles.
- Supports low-carbon steel production.

4.9.2 Key Challenges

- **Rising Compliance and Infrastructure Costs:** Meeting HKC requirements and other environmental standards requires significant investment in infrastructure upgrades, worker safety measures, hazardous waste management systems, and monitoring technologies. Smaller yards often face financial challenges in complying with these requirements.

Impact on the industry:

- Increased operating expenses.
- Reduced profitability for smaller facilities.
- Delays in achieving full compliance.
- Higher capital expenditure requirements.
- **Foreign Exchange Volatility:** Ships are generally purchased in U.S. dollars, while much of the recycling activity and steel sales occur in Indian rupees. Fluctuations in exchange rates can significantly affect acquisition costs and profit margins.

Impact on the industry:

- Uncertainty in transaction profitability.
- Pressure on working capital.
- Increased financial risk during vessel procurement.
- **Steel Price Volatility:** The value of a recycled vessel is heavily dependent on prevailing steel scrap prices. Changes in domestic and global steel markets directly impact recycling yard profitability and vessel acquisition decisions.

Impact on the industry:

- Unpredictable revenue streams.
- Margin fluctuations.
- Conservative purchasing behaviour during market downturns.
- Increased exposure to commodity cycles.
- **Working Capital and Liquidity Constraints:** Ship recycling is capital intensive. Yard operators must make substantial upfront payments to acquire vessels while revenues are realized gradually through the sale of recovered steel, machinery, and equipment.

Impact on the industry:

- Cash flow pressure.
- Dependence on external financing.
- Higher financing costs.
- Limited ability to scale operations rapidly.

- **Cyclical Ship Availability:** The availability of vessels for recycling is highly dependent on global shipping cycles. During periods of strong freight rates and shipping demand, shipowners tend to extend vessel lifespans instead of scrapping them, reducing recycling volumes.

Impact on the industry:

- Irregular vessel supply.
- Capacity utilization fluctuations.
- Revenue uncertainty.
- Difficulty in long-term planning.

4.10 Importance of Offshore Support Vessel Services in India & Turkey

4.10.1 Importance of Offshore Support Vessel (OSV) Services in India

Offshore Support Vessel (OSV) services form a critical part of India's offshore energy and maritime ecosystem by providing logistics, operational, construction, maintenance, and emergency support for offshore oil and gas exploration, offshore wind developments, and subsea infrastructure projects. With the Government of India accelerating domestic hydrocarbon exploration under the Open Acreage Licensing Policy (OALP), expanding offshore natural gas production, and promoting renewable energy projects in offshore locations, demand for specialized vessels such as Anchor Handling Tug Supply (AHTS) vessels, Platform Supply Vessels (PSVs), Multi-Purpose Support Vessels (MPSVs), Construction Support Vessels (CSOVs), and Crew Transfer Vessels (CTVs) continues to increase.

- **Supporting Expansion of Offshore Oil and Gas Exploration:** India's offshore exploration activities are expanding across key hydrocarbon-producing regions, including the Mumbai Offshore Basin, Krishna-Godavari Basin, and Kerala-Konkan Basin, creating growing demand for reliable offshore logistics and support services. In September 2025, the Government approved three offshore exploration blocks in the Kerala-Konkan Basin, with exploration activities being undertaken by Oil India Limited (OIL) and ONGC. Supporting infrastructure, including a dedicated shore base at Cochin Port, has been established to provide berthing facilities, refueling services, warehousing, dry bulk handling, potable water supply, shore power, and round-the-clock operational support for offshore vessels. These developments are increasing demand for OSVs responsible for transporting equipment, fuel, supplies, personnel, and emergency resources to offshore drilling and production installations.
- **Enhancing Production from Mature Offshore Fields:** Offshore support vessels also play a vital role in sustaining and enhancing production from India's mature offshore oil and gas assets. In May 2026, ONGC appointed BP Exploration Services India Limited as its Technical Services Provider (TSP) to improve production from offshore fields in the Mumbai Offshore Basin. The collaboration aims to increase crude oil production by approximately 10.8% and natural gas production by 31.5% over the next decade through enhanced reservoir management, well optimization, and infrastructure modernization. Achieving these objectives will require continued deployment of platform supply vessels, anchor handling vessels, maintenance support vessels, and subsea intervention vessels to support drilling operations, inspections, equipment replacement, and field maintenance activities.
- **Supporting India's Offshore Wind Industry:** India's emerging offshore wind sector is creating significant opportunities for specialized offshore support vessels. The launch of India's first hybrid-electric, methanol-ready Commissioning Service Operation Vessel (CSOV) by Cochin Shipyard Limited (CSL) in October 2025 highlights the country's growing capability to support offshore renewable energy projects. Equipped with dynamic positioning (DP2) systems, battery-based energy storage, motion-compensated gangways, and onboard accommodation facilities, these vessels are designed to safely transport technicians, support turbine installation activities, and undertake long-term operation and maintenance services for offshore wind farms along India's coastline.
- **Strengthening Maritime Infrastructure and Indigenous Shipbuilding:** The rising demand for offshore support vessels is also driving investments in domestic shipbuilding under initiatives such as Maritime India Vision 2030 and Atmanirbhar Bharat. Indian shipyards, particularly Cochin Shipyard

Limited, are expanding their capabilities to design and construct technologically advanced offshore vessels, including hybrid-powered service vessels, offshore wind support vessels, dredgers, and specialized maritime assets. The development of indigenous OSV manufacturing capabilities helps reduce import dependence, strengthens India's maritime industrial ecosystem, and creates opportunities for exporting high-value specialized vessels to international markets.

- **Enhancing Offshore Safety and Operational Efficiency:** Offshore support vessels perform a wide range of mission-critical functions, including anchor handling, towing, platform supply, subsea inspection, firefighting, standby rescue operations, medical evacuation, oil spill response, and offshore construction support. These services improve operational efficiency, minimize downtime, ensure uninterrupted offshore production, and strengthen safety across drilling rigs, production platforms, subsea pipelines, and offshore renewable energy projects. As offshore activities increasingly move into deeper waters and more technically complex environments, advanced OSVs equipped with dynamic positioning systems, digital navigation technologies, and low-emission propulsion systems are becoming essential for ensuring safe, efficient, and environmentally sustainable offshore operations.

4.10.2 Importance of Offshore Support Vessel (OSV) Services in Turkey

Offshore Support Vessel (OSV) services are becoming increasingly important within Turkey's offshore energy sector as the country intensifies domestic hydrocarbon production, expands offshore exploration activities, and advances the development of its offshore renewable energy industry. The growing scale and complexity of offshore operations are driving demand for specialized vessels capable of supporting exploration, production, construction, logistics, maintenance, and emergency response activities across a variety of offshore environments.

- **Supporting Expansion of Black Sea Natural Gas Production:** Offshore support vessels play a crucial role in the development and operation of Turkey's flagship Sakarya Gas Field in the Black Sea. As of April 2025, daily natural gas production from the field had reached approximately 9.5 million cubic metres, with the government continuing efforts to expand offshore production and reduce dependence on imported energy sources. OSVs provide essential services to drilling rigs and offshore production facilities, including the transportation of personnel, fuel, equipment, and operational supplies. In addition, they perform anchor handling, towing, subsea intervention, maintenance support, logistics management, and emergency response functions, ensuring the safe and uninterrupted operation of offshore assets.
- **Facilitating New Offshore Exploration Activities:** Turkey is actively expanding its offshore oil and gas exploration footprint both domestically and internationally through the activities of Türkiye Petrolleri Anonim Ortaklığı (TPAO). The country is pursuing exploration opportunities in Bulgarian waters, offshore Somalia, offshore Pakistan, and the Caspian region as part of its strategy to strengthen long-term energy security and diversify energy sources. Offshore support vessels are indispensable to these operations, supporting seismic surveys, exploratory drilling campaigns, offshore logistics, supply chain management, anchor handling, crew transportation, and marine safety throughout the exploration lifecycle.
- **Supporting Development of Newly Discovered Gas Reserves:** In May 2025, Turkey announced the discovery of a new 75 billion cubic metre (bcm) natural gas reserve at the Göktepe-3 well in the Black Sea, with an estimated economic value of approximately US\$30 billion. The development and commercialization of this offshore resource are expected to generate substantial demand for offshore support services. A wide range of specialized vessels will be required to support drilling activities, subsea construction, platform operations, equipment transportation, inspection and maintenance programs, and emergency response capabilities, thereby driving long-term growth in the OSV market.
- **Enabling Offshore Wind Development:** Turkey is laying the foundation for a domestic offshore wind industry as part of its broader energy transition strategy. In collaboration with the World Bank, the Ministry of Energy and Natural Resources launched the Offshore Wind Roadmap for Türkiye in November 2024, targeting 5 GW of installed offshore wind capacity by 2035. The development of offshore wind projects is expected to create significant demand for specialized vessels such as Commissioning Service Operation Vessels (CSOVs), Crew Transfer Vessels (CTVs), survey vessels,

cable-laying vessels, and maintenance support vessels. These vessels will play a critical role in seabed surveys, turbine installation, cable deployment, commissioning activities, and long-term operation and maintenance of offshore wind infrastructure.

- **Strengthening Energy Security and Maritime Capabilities:** Turkey currently imports more than 90% of its energy requirements, making the development of domestic offshore energy resources a strategic national priority. Offshore support vessels contribute significantly to improving operational efficiency, minimizing downtime, and ensuring the safe execution of offshore exploration and production activities. At the same time, increasing investments in offshore energy projects are creating opportunities for Turkey's domestic shipbuilding sector to design and construct technologically advanced OSVs equipped with dynamic positioning systems, hybrid propulsion technologies, digital navigation systems, and enhanced operational capabilities. The expansion of domestic OSV construction capacity is expected to strengthen Turkey's maritime industry while contributing to the country's long-term energy security objectives.

4.11 Scrap Steel Generated

Ship recycling serves as a significant source of high-quality ferrous scrap, with steel accounting for approximately 75–85% of a vessel's total weight. During the recycling process, steel plates, structural sections, pipes, and other metallic components are recovered, segregated based on thickness and grade, and supplied to downstream industries including steel mills, re-rolling mills, and fabrication units. Owing to its high tensile strength and favorable metallurgical characteristics, ship scrap is regarded as one of the most valuable forms of recycled steel and requires comparatively limited processing before being reintroduced into industrial applications.

The Alang-Sosiya Ship Recycling Yard (ASSRY) in India, recognized as the world's largest ship recycling cluster, plays a central role in supplying scrap steel to the domestic market. The cluster processes millions of tonnes of end-of-life vessels and provides recovered steel to a broad range of sectors, including construction, infrastructure, automotive, engineering, and manufacturing. In addition, the extensive concentration of steel re-rolling mills and scrap-processing facilities in the vicinity of Alang enables efficient processing, value addition, and distribution of recycled steel across the country.

The growing availability of ship-generated scrap steel is becoming increasingly important as the global steel industry transitions toward lower-carbon production pathways. Steel manufacturing is one of the most carbon-intensive industrial activities, prompting steel producers to increase the use of recycled scrap as part of their decarbonization strategies. Compared with conventional blast furnace-based steelmaking, the use of scrap steel recovered from ships can substantially reduce greenhouse gas emissions while lowering energy consumption and raw material requirements. Furthermore, the increasing adoption of Electric Arc Furnace (EAF) technology, which can utilize significantly higher proportions of recycled steel feedstock, is strengthening demand for quality scrap generated through ship recycling.

As a result, ship recycling is emerging as a critical contributor to green steel production, resource efficiency, and circular economy objectives. By extending the lifecycle of steel and reducing dependence on virgin raw materials, the industry supports sustainable industrial development while helping steel producers meet evolving environmental and sustainability targets.

Looking ahead, global demand for ship scrap is expected to increase as an aging world fleet approaches the end of its operational life. The retirement of a large number of older vessels over the coming decade is anticipated to substantially increase the availability of high-grade recycled steel. Combined with growing demand for sustainable raw materials, expanding investments in scrap-based steelmaking technologies, and increasing emphasis on decarbonization across the steel sector, these trends are expected to further enhance the strategic importance of ship recycling within the global industrial value chain.

5. Competitive Landscape

5.1 Company profiling and benchmarking

1. Rudra Green Ship Recycling Limited (RGSR): The company started as a partnership firm with the name Hari Krishna Steel Corporation in 1994 and engaged in steel trading and ship recycling activities. Subsequently, the business and operations of the partnership firm were taken over by and continued through our Company, which was incorporated as Rudra Green Ship Recycling Private Limited in 2018 which was later converted into a public limited company and renamed as Rudra Green Ship Recycling Limited in May 2026.

RGSR is an India based ship recycling company operating a Hong Kong Convention (HKC) compliant green ship recycling yard at Alang, Gujarat. The company is engaged in the ship recycling business and the trading of ferrous materials and steel products. Its operations involve environmentally compliant dismantling of end-of-life vessels and recover recyclable materials, primarily ferrous scrap and steel, for further processing and sale. In addition, the company purchases and sells ferrous materials and steel products, serving customers across the steel value chain. Through its integrated business model, RGSR maintains a presence across multiple stages of the ship recycling and ferrous materials ecosystem, generating revenue from both the recovery of materials from decommissioned vessels and the trading of steel-related products.

2. VMS Industries Ltd: VMS Industries incorporated in 1991, is engaged in the recycling of end-of-life vessels at its facility in Alang, Gujarat. The company recovers and sells recyclable materials such as ferrous scrap, non-ferrous metals, machinery, and other reusable components obtained from ship dismantling operations. Its operations cater primarily to steel re-rolling mills and other industrial consumers, while adhering to internationally recognized environmental and safety standards.

5.2 Financial benchmarking of key peers in the sector

Table: Financial benchmarking of key peer companies for the Full Year 2026

Comparison with industry peers

For the period ending March 31, 2026		
Particulars	Rudra Green Ship Recycling Limited	VMS Industries Limited
(₹ in Lakhs unless stated otherwise)		
Revenue from Operations	20,451.53	15,743.05
Total Income	20,500.20	16,449.28
Gross Profit	2,637.68	641.47
Gross Margin %	12.90%	4.07%
EBITDA	2,178.61	(31.41)
EBITDA Margin %	10.65%	(0.20%)
EBIT	2,182.03	604.09
EBIT Margin %	10.67%	3.84%
Profit After Tax	1,264.51	146.29
PAT Margin %	6.17%	0.89%
Net Worth	5,247.45	9,715.83
Return on Equity %	27.40%	1.51%
Return on Capital Employed %	19.03%	3.74%
Return on Assets %	7.44%	0.91%
Current Ratio (x)	1.42x	2.46x
Quick Ratio (x)	0.57x	2.22x
Debt / Equity (x)	1.19x	0.30x
Interest Coverage (x)	4.42x	1.54x
Basic EPS (₹) – as reported	4.22	0.60
Book Value per Share (₹)	17.49	39.70

Source: Company Financial Statements, ICRA Analytics

Note: All metrics have been calculated or derived, to the extent ascertainable, based on publicly disclosed information available as of the date of this Draft Red Herring Prospectus

Table: Financial benchmarking of key peer companies for the Full Year 2025**Comparison with industry peers**

For the period ending March 31, 2025		
Particulars	Rudra Green Ship Recycling Limited	VMS Industries Limited
(₹ in Lakhs unless stated otherwise)		
Revenue from Operations	17,919.54	28,855.88
Total Income	17,970.24	29,330.30
Gross Profit	2,144.44	1,828.96
Gross Margin %	11.97%	6.34%
EBITDA	1,672.65	791.97
EBITDA Margin %	9.33%	2.74%
EBIT	1,685.08	1,202.84
EBIT Margin %	9.40%	4.17%
Profit After Tax	981.51	680.45
PAT Margin %	5.46%	2.32%
Net Worth	3,981.90	9,622.53
Return on Equity %	28.11%	8.50%
Return on Capital Employed %	18.25%	8.04%
Return on Assets %	5.91%	4.55%
Current Ratio (x)	1.29x	2.77x
Quick Ratio (x)	0.41x	1.89x
Debt / Equity (x)	1.32x	0.15x
Interest Coverage (x)	4.65x	4.15x
Basic EPS (₹) – as reported	3.27	2.91
Book Value per Share (₹)	13.27	39.32

Source: Company Financial Statements, ICRA Analytics

Note: All metrics have been calculated or derived, to the extent ascertainable, based on publicly disclosed information available as of the date of this Draft Red Herring Prospectus

Table: Financial benchmarking of key peer companies for the Full Year 2024

Comparison with industry peers

For the period ending March 31, 2024		
Particulars	Rudra Green Ship Recycling Limited	VMS Industries Limited
(₹ in Lakhs unless stated otherwise)		
Revenue from Operations	17,950.69	26,637.28
Total Income	18,063.62	27,084.31
Gross Profit	1,294.20	1,763.21
Gross Margin %	7.21%	6.62%
EBITDA	842.61	607.17
EBITDA Margin %	4.69%	2.28%
EBIT	920.43	1,009.01
EBIT Margin %	5.13%	3.79%
Profit After Tax	537.27	631.53
PAT Margin %	2.97%	2.33%
Net Worth	3,000.70	6,393.34
Return on Equity %	19.64%	10.39%
Return on Capital Employed %	14.09%	5.29%
Return on Assets %	5.66%	3.68%
Current Ratio (x)	1.41x	1.58x
Quick Ratio (x)	0.61x	1.19x
Debt / Equity (x)	1.18x	0.35x
Interest Coverage (x)	4.68x	6.14x
Basic EPS (₹) – as reported	1.79	3.82
Book Value per Share (₹)	10.00	38.81

Source: Company Financial Statements, ICRA Analytics

Note: All metrics have been calculated or derived, to the extent ascertainable, based on publicly disclosed information available as of the date of this Draft Red Herring Prospectus.

Table: List of Formulas used for the key peer comparison

SR. No.	Formula	
1	Revenue from Operations	Revenue from operations as per restated financial statements.
2	Total Income	Revenue from operations plus Other Income
3	Gross Profit	Revenue from operations minus cost of materials consumed, purchases of stock-in-trade and changes in inventories.
4	Gross Margin %	Gross Profit as a percentage of revenue from operations
5	EBITDA	EBITDA refers to earnings before interest and taxes plus depreciation and amortisation less other income
6	EBITDA Margin %	EBITDA Margin refers to EBITDA during a given period as a percentage of revenue from operations during that period.
7	EBIT	EBIT refers to restated profit before tax plus finance costs
8	EBIT Margin %	EBIT Margin refers to EBIT during a given period as a percentage of revenue from operations during that period.
9	Profit After Tax	Profit After Tax as per restated financial statements.
10	PAT Margin %	PAT Margin refers to PAT during a given period as a percentage of total income during that period.
11	Net Worth	Total Equity
12	Return on Equity %	Return on equity (RoE) is equal to PAT divided by average net worth (average of opening and closing net worth)
13	Return on Capital Employed %	Return on Capital Employed (RoCE) (%) is calculated as EBIT divided by Capital Employed, where Capital Employed is net worth plus total borrowings (non-current and current)
14	Return on Assets %	Return on Assets is calculated as PAT divided by Total Assets as at the end of the year
15	Current Ratio (x)	Current Ratio refers to current assets divided by current liabilities
16	Quick Ratio (x)	Quick Ratio refers to current assets less inventories divided by current liabilities
17	Debt / Equity (x)	Debt-to-Equity ratio is calculated as total borrowings (non-current and current) divided by total equity.
18	Interest Coverage (x)	EBIT divided by Finance Cost
19	Basic EPS (₹)	Restated profit for the year divided by the weighted average number of equity shares (for our Company, face value of ₹5 each, as adjusted for sub-division); for the listed peer, as reported
20	Book Value per Share (₹)	Net worth divided by the number of equity shares outstanding at the end of the year (for our Company, face value of ₹5 each, as adjusted for sub-division)

Table: List of Abbreviations

Abbreviation	Full Form
IMF	International Monetary Fund
WEO	World Economic Outlook
GDP	Gross Domestic Product
GVA	Gross Value Added

GFCF	Gross Fixed Capital Formation
PFCE	Private Final Consumption Expenditure
IIP	Index of Industrial Production
CPI	Consumer Price Index
MPC	Monetary Policy Committee
RBI	Reserve Bank of India
NSO	National Statistical Office
MoSPI	Ministry of Statistics and Programme Implementation
PMI	Purchasing Managers' Index
CAGR	Compound Annual Growth Rate
CY	Calendar Year
FY	Financial Year
PE	Provisional Estimates
FRE	First Revised Estimates
GT	Gross Tonnage
LDT	Light Displacement Tonnage
HKC	Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships
EU SRR	European Union Ship Recycling Regulation
BIMCO	Baltic and International Maritime Council
UNCTAD	United Nations Trade and Development
IHM	Inventory of Hazardous Materials
SRP	Ship Recycling Plan
SRFP	Ship Recycling Facility Plan
DG Shipping	Directorate General of Shipping
SBFA	Shipbuilding Financial Assistance Policy
ISO	International Organization for Standardization
ESG	Environmental, Social and Governance
PCB	Polychlorinated Biphenyl
PPE	Personal Protective Equipment
VR	Virtual Reality
EAF	Electric Arc Furnace
OSV	Offshore Support Vessel
AHTS	Anchor Handling Tug Supply Vessel
PSV	Platform Supply Vessel
MPSV	Multi-Purpose Support Vessel
CSOV	Construction/Commissioning Service Operation Vessel
CTV	Crew Transfer Vessel
OALP	Open Acreage Licensing Policy
TSP	Technical Services Provider
DP2	Dynamic Positioning Class 2
LNG	Liquefied Natural Gas
EEXI	Energy Efficiency Existing Ship Index
CII	Carbon Intensity Indicator
GHG	Greenhouse Gas
LC	Letter of Credit
MIV 2030	Maritime India Vision 2030
ASSRY	Alang-Sosiya Ship Recycling Yard

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